

Bismillah. UIU - Seniors' Workbook Topics Explained

Ahmmad Nur Swapnil

- UIU — Seniors' Workbook Topics Explained
 - Workbook-to-chapter map
- 1. Structured programming: parameter passing
 - 7.1 Prototype, call, definition
 - 11.1 Address and dereference
 - 11.3 Arrays are not pointers
 - 11.4 Pointer parameters
 - 20.2 Pointer versus ordinary parameter
- 2. Object-oriented programming
 - Part I — OOP foundations
 - Constructors
 - Lecture 3 — Assignment, copying, parameters, returns, and friends
 - References
 - Lecture 5 — Function/constructor overloading, defaults, and ambiguity
 - Lecture 7 — Inheritance, access, construction order, and the diamond
 - Lecture 10 — Virtual functions and runtime polymorphism
 - The pass-by-value correction
 - Overriding, hiding, and dispatch
 - Abstract classes and anonymous subclasses
 - Interfaces
 - Abstract class versus interface in Java
 - Workbook board supplement — friends, constructors, and Java's diamond
- 3. Discrete mathematics and number theory
 - 106. Graph definitions and types
 - 107. Handshaking lemma and degree sequences
 - 110. Walks, trails, paths, and cycles
 - 111. Connectedness and components
 - 112. Trees and forests
 - 113. Core tree proofs from the slides
 - Seniors' workbook supplement — number theory for the board
 - 15.1 The pigeonhole principle – a solid answer
- 4. Data structures, searching, sorting and dynamic programming
 - 1. Algorithms, ADTs, correctness, and analysis
 - 2.1 Recursion mental model
 - 2.2 Recursion versus iteration
 - 2.5 Solving recurrences
 - 3.2 Array-based list
 - 3.3 Singly linked list

- 3.6 Array versus linked representation
- 4.1 Stack ADT and representations
- 5.1 Queue ADT
- 5.2 Why a naive array queue fails
- 5.3 Circular queue: one-slot-empty design
- 5.4 Circular queue with a count
- 6. Binary trees and traversals
- 7. Binary search trees
- 8. Binary heaps, priority queues, build-heap, and heapsort
- 9. Graph language and representations
- 10. BFS, DFS, bipartite testing, cycles, and topological ordering
- 11. Searching and sorting
- 12.1 Pattern and proof method
- 12.3 Maximum subarray by divide and conquer
- 14.1 When DP applies
- 14.2 Fibonacci: the smallest overlap example
- 14.3 Longest common subsequence (LCS)
- 14.5 0/1 knapsack
- 14.12 Memoization versus tabulation
- 17. Seniors' workbook supplement — search, sorting, and board simulations
- 5. Graph algorithms, hashing and amortized analysis
 - 3.1 Disjoint-set union (union–find)
 - 3.2 MST definitions and facts
 - 3.3 Kruskal
 - 3.4 Prim
 - 3.6 MST comparison viva table
 - 4. Single-source shortest paths
 - 5.2 Floyd–Warshall
 - 5.3 Johnson's algorithm
 - 13.1 What it is—and is not
 - 13.2 The three methods
 - 13.3 Binary counter
 - 13.4 Stack with multipop
 - 13.5 Dynamic array/table
 - 17. Hashing
 - 21. Coping with hardness: special cases, DP, approximation, and exact exponential algorithms
 - 21.5 Approximation versus heuristic
 - 24. Workbook board drill — negative edges, negative cycles, and runtime
- 6. Database topics from the interviews
 - 1.6 Relational versus non-relational
 - 2.1 Relation vocabulary
 - 2.2 Keys—say these precisely
 - 18.1 Transaction and ACID
 - 18.2 Transaction states

- 26. Workbook interview case — migrate a document database to SQL
- 11.1 SQL injection
- 19.4 SQL injection, prepared statements, and identifiers
- 7. Software engineering
 - 2. What software engineering is
 - 3. SDLC and process models
 - 4. Agile, XP, and Scrum
 - 17.4 Software engineering ethics principles
- 8. Operating systems
 - 1. The operating-system mental model
 - 2. Programs, processes, and threads
 - 3. CPU scheduling
 - 4. Concurrency and synchronization
 - 5. Deadlock, starvation, and livelock
 - 6. Address spaces and virtual memory
 - Workbook supplement — single-core threads, process states, and RR cost
 - 15.1 Scheduling metrics and one complete trace
 - 15.2 MLFQ as an explicit algorithm
 - 15.5 Race condition trace
 - 15.6 Semaphore bounded buffer
 - 15.8 Deadlock detection and Banker safety
 - 15.10 Paging numeric example
 - 15.11 TLB effective access time
 - 15.12 Page-fault service sequence
 - 15.13 Replacement algorithms and Belady anomaly
- 9. Artificial intelligence
 - 1. Four views of AI
 - 3. Agent, percept, action, rationality
 - 4. PEAS
 - 5. Environment dimensions
 - 6. Agent architectures
 - 7. Search-problem formulation
 - 8. Tree search versus graph search
 - 9. Generic best-first graph search
 - 10. Criteria and notation
 - 11. Breadth-first search
 - 12. Uniform-cost search
 - 13. Depth-first search
 - 14. Depth-limited and iterative deepening search
 - 15. Bidirectional search
 - 16. Comparison
 - 17. Greedy and A*
 - 18. Admissibility, consistency, reopening
 - 19. A* optimality proof and the reported $h = 0$ question

- 20. Building heuristics
- 25. Game model and minimax
- 26. Reported infinite-depth question
- 27. Alpha-beta pruning
- 28. Cutoff, evaluation, horizon, chance
- Workbook supplement — completeness, rationality, and game assumptions
- 10. Machine learning
 - 1. What is machine learning?
 - 2. End-to-end ML workflow
 - 3. Preprocessing and feature engineering
 - 4. Loss, risk, and model selection
 - 5. Classification metrics
 - 6. Regression metrics
 - 7. Bias-variance: the precise answer
 - 8. Regularization
 - 9. Linear regression
 - 10. Perceptron
 - 11. Logistic regression
 - 12. Multiclass softmax and cross-entropy
 - 15. Decision trees
 - 16. Bagging and random forest
 - 17. AdaBoost
 - 18. Gradient boosting
 - 19. Stacking
 - 24. Optimization algorithms
 - 34. Maximum likelihood, MAP, and Bayesian learning
 - 37. Gaussian mixture model
 - 38. Expectation-maximization for GMM
 - 39. K-means and Lloyd's algorithm
 - 40. Dimensionality reduction recall
 - 41. MDP definition
 - 42. Value functions and Bellman equations
 - 43. Dynamic programming: value and policy iteration
 - 44. Passive learning, ADP, TD, and active learning
 - 45. Q-learning
 - 46. Bag-of-words to contextual embeddings
 - 51. Autoregressive language modeling
 - 52. Transformer overview
 - 53. Scaled dot-product self-attention
 - 54. Multi-head, multi-query, and grouped-query attention
 - 55. Positional information and RoPE
 - 56. Transformer block details
 - 57. KV cache
 - 58. Encoder-only, decoder-only, and encoder-decoder models

Bismillah.

UIU — Seniors' Workbook Topics Explained

Focused companion to [Varsity_Topics_List_In Memory of Masud.xlsx](#). This book **copies existing explanations**, including their worked examples, equations, code and diagrams. It is not another copy of the entire course pack. The workbook's substantive topics across its subject tabs, interview questions, embedded screenshots and hidden Niche Topics tab determine the selection.

This edition filters those workbook topics to UIU-listed courses: structured programming, OOP, discrete mathematics, data structures, algorithms, DBMS, SWE, OS, AI and ML. SQL injection is retained as a database topic; standalone networking/security/hardware/graphics/TOC and personal research chapters are excluded.

Sections are copied whole so definitions and code prerequisites are not reduced to isolated answers. Original section numbers, source labels, and cross-references are retained; a reference to a section not reproduced here points back to the named full book in [BRACU_Viva_Prep](#). These are corrected explanations, not an endorsement of inaccurate shorthand in the spreadsheet.

The DB tab says only "Work in progress"; database prompts elsewhere in the workbook are included. Resource links, dedication artwork, and unidentified questions have no technical explanation to copy; linked external collections are not silently treated as read or solved. See the existing [20_SENIORS_WORKBOOK_COVERAGE_AUDIT.md](#) for the cell-by-cell audit and screenshot ledger. The original reference books and PDFs remain unchanged.

Workbook-to-chapter map

Chapter	Workbook topic routing
1. Structured programming: parameter passing	DSA B42; BRAC Ques Bank C9
2. Object-oriented programming	DSA B40/B42/B62–B72; Random Screenshots constructors, abstract classes, friends and inheritance; BRAC C37/C45
3. Discrete mathematics and number theory	Runtime Analysis number-theory screenshot; BRAC C5/C37; graph and hashing prerequisites
4. Data structures, searching, sorting and dynamic programming	DSA B7–B38/B58; Board B35/B37; Runtime Analysis sorting screenshot; BRAC array/list, circular array, DP, stock-profit and graph questions
5. Graph algorithms, hashing and amortized analysis	DSA B5/B44–B56; Board B9–B15; Runtime Analysis MST/shortest-path screenshots; hidden Niche Topics C4 (Johnson)
6. Database topics from the interviews	DB tab is work-in-progress; BRAC C19 (ACID), C41 (NoSQL to SQL); Board B41 (SQL injection)
7. Software engineering	BRAC C5 (Agile versus waterfall); BRAC C11 professional ethics

Chapter	Workbook topic routing
8. Operating systems	OS B3–B15; BRAC C11/C13/C31/C37/C49: scheduling, deadlock, semaphores, page tables, replacement and context switching
9. Artificial intelligence	AI B4–B34; DSA BFS/DFS completeness claims; BRAC rationality, search, minimax and alpha-beta; ML horizon effect
10. Machine learning	Entire ML tab; classification/regression/ensemble screenshots; BRAC bias–variance, k-means, MLE/EM, XGBoost, AdaBoost, GD, LSTM, Transformers and LM/LLM

1. Structured programming: parameter passing

Workbook coverage: DSA B42; BRAC Ques Bank C9.

7.1 Prototype, call, definition

```
int add(int x, int y);           /* declaration/prototype */

int main(void) {
    int result = add(2, 3);      /* call */
    printf("%d\n", result);
}

int add(int x, int y) {          /* definition */
    return x + y;
}
```

- Arguments are expressions at the call.
- Parameters are local objects in the called function.
- C passes arguments **by value**.
- To let a function modify a caller object, pass its address by value.

```
void swap(int *a, int *b) {
    int t = *a;
    *a = *b;
    *b = t;
}
```

Excerpt source: `04_C_PROGRAMMING_SLIDE_COMPLETE.md`, original lines 356–383. Original section numbers are retained.

11.1 Address and dereference

```
int x = 10;
int *p = &x;
*p = 20;      /* modifies x */
```

- `&x`: address of `x`;
- `int *p`: pointer intended to point to `int`;
- `*p`: referenced object, only valid if `p` points to a live suitable object;
- `NULL`: null pointer constant/conventional invalid “no object” state;
- uninitialized/dangling/out-of-bounds pointers must not be dereferenced.

Excerpt source: `04_C_PROGRAMMING_SLIDE_COMPLETE.md`, original lines 655–668. Original section numbers are retained.

11.3 Arrays are not pointers

An array is an object containing elements. In most expressions its name converts to a pointer to the first element, but differences remain:

- `sizeof array` is whole array size; `sizeof pointer` is pointer size;
- array assignment is not permitted;
- `&array` has pointer-to-array type;
- string arrays own writable storage, pointer-to-literal variables do not.

Excerpt source: `04_C_PROGRAMMING_SLIDE_COMPLETE.md`, original lines 678–686. Original section numbers are retained.

11.4 Pointer parameters

```
int divide(int num, int den, int *q, int *r) {
    if (den == 0 || q == NULL || r == NULL) return 0;
    *q = num / den;
    *r = num % den;
    return 1;
}
```

C is still pass-by-value: copies of the pointer values are passed, and those copies refer to caller objects.

Excerpt source: `04_C_PROGRAMMING_SLIDE_COMPLETE.md`, original lines 687–699. Original section numbers are retained.

20.2 Pointer versus ordinary parameter

Draw two boxes:

```
caller x=5
ordinary call f(x): parameter receives copy 5
pointer call g(&x): parameter receives copy of x's address -> *p names caller x
```

This is not pass-by-reference at the language level; it is pass-by-value of a pointer.

Excerpt source: `04_C_PROGRAMMING_SLIDE_COMPLETE.md`, original lines 1159–1170. Original section numbers are retained.

2. Object-oriented programming

Workbook coverage: DSA B40/B42/B62–B72; Random Screenshots constructors, abstract classes, friends and inheritance; BRAC C37/C45.

Part I — OOP foundations

Object, class, state, behavior, and identity

- A **class** is a user-defined type or blueprint that declares representation (fields/data members), behavior (methods/member functions), and invariants.
- An **object** is a runtime instance of a class.
- **State** is the current value of its fields.
- **Behavior** is what its methods do.
- **Identity** distinguishes two objects even when their visible state is equal.

```
Class: BankAccount
State: owner, balance
Behavior: deposit(), withdraw()
Invariant: balance must not become negative
Objects: accountA and accountB have separate identities
```

An object is not merely “data plus functions.” Good class design protects a valid state and exposes operations meaningful in the problem domain.

The four central ideas

Encapsulation

Encapsulation bundles state and behavior and controls access to the representation. Data hiding is one consequence of encapsulation.

```
final class Account {
    private long balance;

    Account(long opening) {
        if (opening < 0) throw new IllegalArgumentException();
        balance = opening;
    }

    public void deposit(long amount) {
        if (amount <= 0) throw new IllegalArgumentException();
        balance += amount;
    }

    public long balance() { return balance; }
}
```

Why not make `balance` public? A caller could set it to an invalid value and bypass validation.

Abstraction

Abstraction presents essential behavior and hides irrelevant implementation detail. An interface such as `Shape.area()` says **what** is available; subclasses decide **how** it is computed.

Inheritance

Inheritance creates an **is-a** relationship: a `Rectangle` is a `Shape`. It supports reuse and substitutability, but it also creates coupling. Use it when the subtype truly satisfies the supertype contract, not merely to reuse a few lines.

Polymorphism

Polymorphism means “one interface, multiple implementations.”

- **Compile-time/ad-hoc polymorphism:** overload resolution and C++ operator overloading.
- **Runtime/subtype polymorphism:** an overridden method is selected from the actual object at runtime.
- **Parametric polymorphism:** C++ templates and Java generics apply an abstraction to many types.

Do not say that overloading and overriding are the same. Overloading selects among different parameter lists at compile time; overriding replaces inherited instance behavior and enables runtime dispatch.

Relationships between objects

- **Association:** one object knows or uses another; e.g., `Teacher` teaches `Course`.
- **Aggregation:** weak whole–part relation; parts may outlive the whole, e.g., `Department` and `Teacher`.
- **Composition:** strong ownership; the whole controls the part’s lifetime, e.g., a `House` owns its `Room` objects.
- **Dependency:** temporary use, often through a parameter or local variable.
- **Inheritance:** an is-a/type relationship.

Viva line: “Prefer composition for has-a behavior because it reduces coupling; choose inheritance only when substitutability is valid.”

Small design principles worth saying aloud

- Keep representation private and expose intention-revealing operations.
- Aim for **high cohesion** inside a class and **low coupling** between classes.
- **Single responsibility:** one main reason for a class to change.
- **Open/closed:** extend behavior through abstractions without repeatedly modifying stable code.
- **Liskov substitution:** code expecting a base type must remain correct for every subtype.
- **Interface segregation:** prefer focused interfaces over one large interface.
- **Dependency inversion:** high-level policy should depend on abstractions, not concrete details.
- Prefer immutability when state does not need to change.
- In C++, prefer RAII/value types and the Rule of Zero; in Java, close external resources deterministically with try-with-resources.

C++ versus Java at a glance

Issue	C++	Java
Compilation	Native code is typical	Source → bytecode → JVM/JIT machine code

Issue	C++	Java
Object variables	Can hold an object value directly	Class variable holds a reference value
Memory	Automatic storage, RAII, and explicit dynamic allocation	Heap objects plus garbage collection
Destruction	Deterministic destructor at lifetime end	GC is nondeterministic; use <code>close()</code> for resources
Inheritance	Multiple class inheritance is allowed	One superclass; multiple interfaces
Runtime dispatch	Base function must normally be <code>virtual</code>	Ordinary overridable instance methods dispatch dynamically
Operator overloading	User-defined operators supported	No general user-defined operator overloading
Generic programming	Templates instantiate code, including primitives	Generics are mainly erased and require reference types
Parameters	Value, pointer, or reference	Always pass-by-value, including copied reference values
Exceptions	No checked-exception category	Checked and unchecked exceptions
Root class	No universal mandatory root	Every class ultimately extends <code>Object</code>

Excerpt source: `03_OOP_CPP_JAVA_SLIDE_COMPLETE.md`, original lines 20–123. Original section numbers are retained.

Constructors

A constructor establishes the initial valid state. It has the class name and no return type. It is called when an object is created.

```
class Point {
    int x, y;
public:
    Point() : Point(0, 0) {}           // delegating constructor
    Point(int x, int y) : x(x), y(y) {} // member-initializer list
};
```

Prefer an initializer list: members are initialized directly rather than default-initialized and then assigned. `const` and reference members must be initialized there.

Excerpt source: `03_OOP_CPP_JAVA_SLIDE_COMPLETE.md`, original lines 202–216. Original section numbers are retained.

Lecture 3 — Assignment, copying, parameters, returns, and friends

Copy initialization versus assignment

```
T b = a;    // initialization: invokes copy constructor
T c(a);     // initialization: invokes copy constructor
c = a;      // existing object: invokes copy-assignment operator
```

Compiler-generated copying performs memberwise copying. For a raw owning pointer, that is a shallow copy: two objects point to one allocation, leading to aliasing, double deletion, or dangling pointers.

Rule of Three, Five, and Zero

- If a class manually owns a resource and defines any of destructor, copy constructor, or copy assignment, it usually needs all three: **Rule of Three**.
- In C++11+, also consider move constructor and move assignment: **Rule of Five**.
- Best default: store resources in standard RAII members so none of those five needs custom code: **Rule of Zero**.

Complete runnable ownership/copying example

```
#include <algorithm>
#include <cstdint>
#include <iostream>
#include <utility>

class Buffer {
    std::size_t size_{};
    int* data_{};
public:
    explicit Buffer(std::size_t n)
        : size_(n), data_(n ? new int[n]{} : nullptr) {}

    ~Buffer() { delete[] data_; }

    Buffer(const Buffer& other)
        : Buffer(other.size_) {
        if (size_ != 0)
            std::copy(other.data_, other.data_ + size_, data_);
    }

    Buffer& operator=(const Buffer& other) {
        if (this != &other) {
            Buffer copy(other);    // deep copy
            swap(copy);           // old resource dies with copy
        }
        return *this;
    }
}
```

```

Buffer(Buffer&& other) noexcept
    : size_(std::exchange(other.size_, 0)),
      data_(std::exchange(other.data_, nullptr)) {}

Buffer& operator=(Buffer&& other) noexcept {
    if (this != &other) {
        delete[] data_;
        size_ = std::exchange(other.size_, 0);
        data_ = std::exchange(other.data_, nullptr);
    }
    return *this;
}

void swap(Buffer& other) noexcept {
    std::swap(size_, other.size_);
    std::swap(data_, other.data_);
}

int& operator[](std::size_t i) { return data_[i]; }
const int& operator[](std::size_t i) const { return data_[i]; }
};

int main() {
    Buffer a(2);
    a[0] = 7;
    Buffer b = a;           // deep copy
    b[0] = 99;
    std::cout << a[0] << ' ' << b[0] << '\n'; // 7 99
}

```

In real code, `std::vector<int>` would give Rule-of-Zero behavior with much less risk.

Parameter passing

- `void f(T x)` : copy/move a value; changes affect only `x`.
- `void f(T& x)` : modifiable reference; no copy.
- `void f(const T& x)` : read-only reference; avoids copying and accepts temporaries.
- `void f(T* x)` : pointer may be null and uses `->`.

The old slide wording that a by-value copy “does not call a constructor” is historical/oversimplified. Semantically, the parameter is initialized; copy elision may remove some physical copies.

Returning an object by value is normal modern C++. Copy elision/NRVO and moves make it efficient. Never return a reference or pointer to a destroyed local object.

Friend

A `friend` is not a class member, but it is granted access to private/protected members. Typical cases are symmetric operators, stream insertion/extraction, or a function coordinating two classes.

Friendship is explicit, not inherited, not transitive, and not reciprocal.

Excerpt source: `03_OOP_CPP_JAVA_SLIDE_COMPLETE.md`, original lines 251–349. Original section numbers are retained.

References

A reference is an alias and normally must be initialized. It is not reseatable.

```
void square(int& x) { x *= x; }

int main() {
    int n = 5;
    int& alias = n;
    square(alias);           // n becomes 25
}
```

Do not return a reference to a local variable. A `const T&` may bind to a temporary and extend its lifetime in specific initialization contexts.

Excerpt source: `03_OOP_CPP_JAVA_SLIDE_COMPLETE.md`, original lines 391–406. Original section numbers are retained.

Lecture 5 — Function/constructor overloading, defaults, and ambiguity

Overloading rules

Functions may share a name when their parameter lists differ sufficiently. Return type alone cannot distinguish overloads.

Overload resolution considers exact matches, promotions, standard conversions, user-defined conversions, and ellipsis. If no unique best viable function exists, compilation fails as ambiguous.

```
void print(int);
void print(double);
// int print(int); // return type alone cannot overload void print(int)
```

Constructors are commonly overloaded for flexible creation and array support. A destructor cannot be overloaded.

Copy constructor

Canonical form:

```
T(const T& other);
```

The `const` reference avoids recursively copying the parameter and accepts const objects. It is used when a new object is initialized from an existing object; assignment uses `operator=`.

Default arguments

```
double area(double length, double width = 0.0);
```

- After the first defaulted parameter, every parameter to its right must also have a default.
- Specify the default in one visible declaration, not both declaration and definition.
- Defaults are substituted at the call site and are statically bound.
- Combining overloads and defaults can create ambiguity.

Common ambiguities

- `f(float)` and `f(double)`, called with `int`: neither conversion may be better.
- `f(int)` and `f(int, int = 0)`, called with one argument.
- Value and reference overloads that are equally viable.
- Two user-defined conversions of equal rank.

Address of an overloaded function

The target function-pointer type selects the overload:

```
void space(int);
void space(int, char);

void (*p1)(int) = space;
void (*p2)(int, char) = space;
```

Excerpt source: `03_OOP_CPP_JAVA_SLIDE_COMPLETE.md`, original lines 407–462. Original section numbers are retained.

Lecture 7 — Inheritance, access, construction order, and the diamond

Inheritance access matrix

Base member	Public inheritance	Protected inheritance	Private inheritance
<code>public</code>	public in derived	protected in derived	private in derived
<code>protected</code>	protected in derived	protected in derived	private in derived
<code>private</code>	inaccessible directly	inaccessible directly	inaccessible directly

Private base state still exists inside the derived object; derived code accesses it through accessible base operations.

Default inheritance is private for `class Derived : Base` and public for `struct Derived : Base`.

Construction/destruction order

1. Virtual base subobjects.
2. Direct base subobjects in declaration order.
3. Data members in declaration order.
4. Derived constructor body.

Destruction is the exact reverse. Initializer-list textual order does not change declaration order.

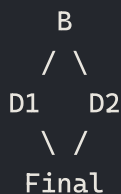
```
class Derived : public Base {
    Member m;
public:
    Derived(int x) : Base(x), m(x) {}
};
```

Multiple and multilevel inheritance

- Multilevel: `A <- B <- C`.
- Multiple: `D : public B1, public B2`.
- Hierarchical: multiple derived classes share one base.

In the diamond, non-virtual inheritance gives the most-derived object two copies of the common base. Virtual inheritance makes it share one common base subobject:

Type relationship:



Object subobjects:

```
non-virtual Final = [D1 [B]] + [D2 [B]]    -> two B objects; B member is ambiguous
virtual Final     = [shared B] + [D1] + [D2] -> one B object
```

```
struct B { int value{}; };
struct D1 : virtual B {};
struct D2 : virtual B {};
struct Final : D1, D2 {};           // one B inside Final
```

The most-derived class constructs a virtual base.

Excerpt source: `03_OOP_CPP_JAVA_SLIDE_COMPLETE.md`, original lines 531–593. Original section numbers are retained.

Lecture 10 — Virtual functions and runtime polymorphism

Early versus late binding

- **Early/static binding:** target resolved at compile time; examples include overloads, non-virtual C++ methods, and static functions.
- **Late/dynamic binding:** target resolved at runtime from the actual object; implemented in C++ with virtual functions.

A base pointer/reference can point/refer to a derived object. The reverse conversion is not automatically safe.

The dispatch picture:

source expression:	Shape* p = pointer to a Triangle object
call:	p->area()
compile time:	verify Shape declares area(); choose virtual slot
runtime object/vptr:	select Triangle's final overrider
executed function:	Triangle::area()

If area() were non-virtual, the static type Shape* would select Shape::area().

Dynamic dispatch therefore needs both facts: the call is through a pointer/reference (not a sliced base value), and the member is virtual. Java ordinary overridable instance methods follow the same runtime-object idea by default; overload resolution and static-method hiding remain compile-time.

Complete runnable runtime-polymorphism example

```
#include <iostream>
#include <memory>
#include <vector>

class Shape {
public:
    virtual double area() const = 0;    // pure virtual
    virtual const char* name() const = 0;
    virtual ~Shape() = default;        // essential for polymorphic deletion
};

class Rectangle final : public Shape {
    double w_, h_;
public:
    Rectangle(double w, double h) : w_(w), h_(h) {}
    double area() const override { return w_ * h_; }
    const char* name() const override { return "rectangle"; }
};

class Triangle final : public Shape {
    double b_, h_;
public:
    Triangle(double b, double h) : b_(b), h_(h) {}
    double area() const override { return 0.5 * b_ * h_; }
    const char* name() const override { return "triangle"; }
};

int main() {
    std::vector<std::unique_ptr<Shape>> shapes;
    shapes.push_back(std::make_unique<Rectangle>(4, 5));
```

```

    shapes.push_back(std::make_unique<Triangle>(4, 3));
    for (const auto& s : shapes)
        std::cout << s->name() << ": " << s->area() << '\n';
}

```

Output:

```

rectangle: 20
triangle: 6

```

An abstract class has at least one pure virtual function and cannot be instantiated, but pointers/references to it are allowed. A derived class remains abstract until it implements all required pure virtual functions.

Virtual-destructor rule

If an object may be deleted through a base pointer, the base destructor must be virtual. Otherwise behavior is undefined and derived cleanup may not run.

Use **override** so the compiler catches signature mismatches; use **final** to prevent further overriding or derivation.

Object slicing

```

Derived d;
Base b = d;           // derived portion is sliced away

```

Pass/store polymorphic objects through references, pointers, or smart pointers—not base values.

Excerpt source: `03_OOP_CPP_JAVA_SLIDE_COMPLETE.md`, original lines 680–769. Original section numbers are retained.

The pass-by-value correction

Java is always pass-by-value. For an object argument, the value copied into the parameter is a reference. Therefore:

- mutating the referenced object is visible to the caller;
- reassigning the parameter to another object is not visible to the caller.

```

static void change(int[] a) {
    a[0] = 99;           // changes caller's array object
    a = new int[3];      // changes only local copied reference
    a[0] = 7;
}

public static void main(String[] args) {
    int[] x = {1};
    change(x);
    System.out.println(x[0]); // 99
}

```

Viva answer: "The object itself is not copied, but Java still passes the reference value by value."

Excerpt source: `03_OOP_CPP_JAVA_SLIDE_COMPLETE.md`, original lines 1055–1077. Original section numbers are retained.

Overriding, hiding, and dispatch

An override has the same signature and a compatible covariant return. It cannot reduce access and cannot throw broader checked exceptions. Use `@Override`.

- Instance method: runtime dispatch based on actual object.
- Static method: hidden; selected by compile-time reference/type.
- Field: hidden; selected by compile-time reference type.
- `private`, `static`, `final` methods and constructors are not overridden.

```
class Base { void speak() { System.out.println("Base"); } }
class Child extends Base {
    @Override void speak() { System.out.println("Child"); }
}

Base x = new Child();
x.speak();                                // Child
```

The reference type controls which members are *accessible* at compile time; the actual type controls the selected override at runtime.

Excerpt source: `03_OOP_CPP_JAVA_SLIDE_COMPLETE.md`, original lines 1264–1284. Original section numbers are retained.

Abstract classes and anonymous subclasses

An abstract class cannot be instantiated and may contain state, constructors, concrete methods, and abstract methods. A concrete subclass must implement all abstract methods. An anonymous subclass can provide a one-off implementation at the creation site.

Excerpt source: `03_OOP_CPP_JAVA_SLIDE_COMPLETE.md`, original lines 1285–1288. Original section numbers are retained.

Interfaces

An interface specifies a role/capability. A class can implement several interfaces, enabling multiple inheritance of type.

- Ordinary interface methods are implicitly `public abstract`.
- Fields are implicitly `public static final`.
- Default methods have implementations and are inherited.
- Static interface methods belong to the interface and are not inherited as instance methods.
- Private interface methods (Java 9+) share code among default/static interface methods.
- Interfaces may extend multiple interfaces.

An implementing method must be public. If a class inherits conflicting unrelated default methods, it must override and resolve the conflict; it may explicitly call `InterfaceName.super.method()` .

```
interface Printable { void print(); }
interface Resettable { default void reset() { System.out.println("reset"); } }

class Report implements Printable, Resettable {
    @Override public void print() { System.out.println("report"); }
}
```

Abstract class versus interface: use an abstract class for shared state/implementation in a close family; use an interface for a capability that unrelated types can implement. Modern interfaces are not literally “pure abstract classes” because they can contain default, static, and private methods.

Excerpt source: `03_OOP_CPP_JAVA_SLIDE_COMPLETE.md` , original lines 1380–1403. Original section numbers are retained.

Abstract class versus interface in Java

Abstract class	Interface
May hold instance state and constructors	No instance state; constants only
Single class inheritance	A class implements many interfaces
Any access for concrete members	Contract methods are public
Shared base implementation/family	Cross-cutting role/capability

Excerpt source: `03_OOP_CPP_JAVA_SLIDE_COMPLETE.md` , original lines 1870–1878. Original section numbers are retained.

Workbook board supplement — friends, constructors, and Java’s diamond

Source: DSA B40/B42/B62–B72 and Random Screenshots images 1/3/6/12. The screenshots mix C++ and Java rules; identify the language first.

A friend class is not an inherited parent

```
#include <iostream>
class Second;
class First {
    int a;
    friend class Second;
public:
    explicit First(int value) : a(value) {}
};
class Second {
    int b;
public:
    explicit Second(int value) : b(value) {}
    int sum(const First& other) const { return other.a + b; }
```

```
};
int main() {
    First x(4);
    Second y(7);
    std::cout << y.sum(x) << '\n'; // 11
}
```

First grants **Second** access. There is no is-a relationship, and First does not thereby gain access to Second's private **b**. The workbook's picture omits initialization and a declared result; the program above is complete. Friendship can support a tightly coupled abstraction, but overuse weakens encapsulation.

Constructor categories are a teaching map, not an exhaustive taxonomy

```
Widget a;           -> default construction
Widget b(5);        -> construction from parameters
Widget c(b);        -> copy construction (when applicable)
Widget d(move(b)) -> move construction (if a suitable move constructor exists)
c = a;              -> assignment, NOT construction
```

C++ also has converting/delegating/inherited constructors; "default, parameterized, copy" is not the complete language taxonomy. A default constructor can have parameters if all have defaults. Java does not synthesize a C++-style copy constructor; copying an object reference does not copy its referent. See the ownership example in C++ Lecture 3 for shallow/deep copy.

Java interfaces can create a method conflict, not two superclass objects

```
interface Left { default String label() { return "L"; } }
interface Right { default String label() { return "R"; } }
class Both implements Left, Right {
    @Override public String label() {
        return Left.super.label() + Right.super.label();
    }
}
// new Both().label() -> "LR"
```

Without the override, unrelated inherited defaults conflict. Java permits one direct superclass and multiple interfaces; C++ virtual base inheritance solves a different duplicated-base-subobject problem. Java abstract classes can have constructors, fields, static/final **concrete** methods and abstract methods. An abstract method itself cannot be final or static. C++ has no **abstract** class keyword: a pure virtual function makes the class abstract.

Finally, C passes arguments **by value**, including pointers. Passing **&x** copies an address that lets the callee modify **x**; it does not add C++ reference parameter semantics. Java likewise passes a copy of a primitive/reference value. Trace both object mutation and parameter reassignment before saying "objects are passed by reference." See C §11.4 and Java current unit 2.

Excerpt source: 03_OOP_CPP_JAVA_SLIDE_COMPLETE.md, original lines 1905–1980. Original section numbers are retained.

3. Discrete mathematics and number theory

Workbook coverage: Runtime Analysis number-theory screenshot; BRAC C5/C37; graph and hashing prerequisites.

106. Graph definitions and types

A finite undirected graph is $G=(V,E)$, where V is the vertex set and each edge joins two vertices.

- In a **simple graph**, an edge is an unordered pair $\{u,v\}$ with $u \neq v$; there are no loops or parallel edges.
- A **multigraph** may have multiple edges between the same endpoints; depending on convention it may also allow loops.
- A **directed graph** or digraph is $G=(V,A)$, where an arc (u,v) has tail u and head v .
- A **weighted graph** associates a cost, distance, capacity, or other value with edges.

Vertices u,v are **adjacent** if uv is an edge. An edge is **incident** with each of its endpoints. The open neighborhood is

$$N(v)=\{u \text{ in } V : uv \text{ in } E\}.$$

For a simple graph, $\deg(v)=|N(v)|$. With parallel edges, degree counts incidences, not distinct neighbors. A loop contributes 2 to undirected degree.

Maximum and minimum degree are

$$\begin{aligned}\Delta(G) &= \max_v \deg(v), \\ \delta(G) &= \min_v \deg(v).\end{aligned}$$

Excerpt source: `10_DISCRETE_MATHEMATICS_SLIDE_COMPLETE.md`, original lines 2719–2742. Original section numbers are retained.

107. Handshaking lemma and degree sequences

Every undirected edge has two endpoints, so double-counting vertex-edge incidences gives

$$\sum_{v \text{ in } V} \deg(v) = 2|E|.$$

Consequences:

1. the degree sum is even;
2. the number of odd-degree vertices is even.

Example: $(2,2,1)$ and $(2,2,2,2,1)$ cannot be degree sequences of any undirected graph because their sums are odd.

An even sum is necessary but not sufficient for a **simple** graph. The slide's $(3,3,3,1)$ has even sum 10 but is impossible on four vertices: if three vertices have degree 3, each connects to every other vertex, forcing the fourth vertex to have degree at least 3, not 1.

Havel-Hakimi test for a simple degree sequence

```
isGraphical(degrees):
    repeatedly:
        remove all zeros
        if list is empty: return true
        sort nonincreasing
        remove first value d
        if d < 0 or d > remaining list length: return false
        subtract 1 from the next d values
        if any becomes negative: return false
```

The transformation preserves graphicality: a highest-degree vertex can be connected to vertices of the next-highest degrees without losing the existence of a realization. Straight sorting every round is polynomial (a simple implementation is $O(n^2 \log n)$). For a viva, distinguish the handshaking necessary test from this actual decision algorithm.

Excerpt source: 10_DISCRETE_MATHEMATICS_SLIDE_COMPLETE.md, original lines 2743–2775. Original section numbers are retained.

110. Walks, trails, paths, and cycles

Terminology varies, so state your convention:

- A **walk** is a vertex sequence where consecutive vertices are adjacent; vertices and edges may repeat.
- A **trail** is a walk with no repeated edge.
- A **path** is a walk with no repeated vertex.
- A **closed walk** starts and ends at the same vertex.
- A **cycle** is a closed path with no repeated vertex except first=last.

The deck informally calls any adjacent vertex sequence a “path” and then says **simple path** when vertices are distinct. If an examiner follows that convention, clarify rather than argue terminology.

A shortest unweighted $u-v$ walk is simple: if it repeats a vertex, the segment between repetitions is a cycle; removing it produces a shorter walk.

Excerpt source: 10_DISCRETE_MATHEMATICS_SLIDE_COMPLETE.md, original lines 2808–2821. Original section numbers are retained.

111. Connectedness and components

Vertices u, v are connected when a path joins them. An undirected graph is **connected** if every vertex pair is connected. A **connected component** is a maximal connected subgraph. A graph is connected iff it has exactly one component.

Breadth-first search marks exactly the vertices in the start vertex’s component:

```
BFS(G, s):
    mark s; Q.enqueue(s)
    while Q not empty:
        u=Q.dequeue()
        for each v in Adj[u]:
```

```

    if v unmarked:
        mark v
        parent[v]=u
        distance[v]=distance[u]+1
        Q.enqueue(v)

```

With adjacency lists, BFS is $\Theta(V+E)$ time and $\Theta(V)$ auxiliary space. It also finds shortest path lengths in an **unweighted** graph. Running BFS/DFS from every still-unmarked vertex counts and labels all components in the same $\Theta(V+E)$ total time.

Excerpt source: `10_DISCRETE_MATHEMATICS_SLIDE_COMPLETE.md`, original lines 2822–2842. Original section numbers are retained.

112. Trees and forests

A **forest** is an acyclic undirected graph. A **tree** is a connected acyclic undirected graph. A vertex of degree 1 is a **leaf**; a one-vertex tree is a special case whose only vertex has degree 0.

For a finite undirected graph T with n vertices, the following are equivalent:

1. T is connected and acyclic.
2. Every pair of vertices has exactly one simple path between them.
3. T is connected and has $n-1$ edges.
4. T is acyclic and has $n-1$ edges.
5. Removing any edge disconnects T —it is minimally connected.
6. Adding any missing edge creates exactly one cycle—it is maximally acyclic.

These are characterizations, not six unrelated facts.

Excerpt source: `10_DISCRETE_MATHEMATICS_SLIDE_COMPLETE.md`, original lines 2843–2857. Original section numbers are retained.

113. Core tree proofs from the slides

Unique path

Connectivity gives at least one $u-v$ path. If there were two distinct simple paths, follow them from u to their first divergence and then to their next common vertex; the two internally different subpaths form a cycle, contradicting acyclicity.

Adding or removing an edge

In a tree there is a unique path between u and v ; adding edge uv closes exactly that path into a cycle. Conversely, removing a tree edge uv destroys the unique $u-v$ path and disconnects the graph.

At least two leaves

In a tree with at least two vertices, take a longest simple path $v_1, \dots, v_k$. If v_1 had a neighbor other than v_2 , that neighbor either lies on the path—creating a cycle—or lies outside—extending the path. Both are impossible, so v_1 is a leaf; symmetrically v_k is another leaf.

A tree has $n-1$ edges

Induct on n . A one-vertex tree has 0 edges. For $n > 1$, remove a leaf and its incident edge. The remaining graph is still connected and acyclic, hence a tree with $n-1$ vertices and, by induction, $n-2$ edges. Restoring the leaf restores one edge, giving $n-1$.

Useful converses

If a connected graph on n vertices had fewer than $n-1$ edges, it could not connect all vertices; if it has exactly $n-1$, it cannot contain a cycle, because removing a cycle edge would leave a connected graph with too few edges. Dually, an acyclic graph with $n-1$ edges must be connected.

Any subgraph of an acyclic graph is acyclic, because a cycle in the subgraph would also be a cycle in the original graph. Consequently, every **connected** subgraph of a tree is itself a tree; connectedness cannot be omitted.

Excerpt source: 10_DISCRETE_MATHEMATICS_SLIDE_COMPLETE.md, original lines 2858–2881. Original section numbers are retained.

Seniors' workbook supplement — number theory for the board

Source: BRAC Ques Bank C5 (Diophantine equations) and Runtime Analysis [image4](#), which lists thirteen number-theory topics. This is a September 2026 workbook-driven supplement, not a claim that these were in all CSE 103 slides. Prioritize gcd, modular arithmetic, inverses, Diophantine equations and CRT; Hensel and reciprocity are optional deeper follow-ups explicitly listed there.

N1. Divisibility, Euclid, and extended Euclid

$a \mid b$ means $b = ak$ for some integer k . The gcd is the greatest positive common divisor (with $\gcd(0, b) = |b|$ for nonzero b).

Euclid uses $\gcd(a, b) = \gcd(b, a \bmod b)$, because the two pairs have exactly the same common divisors. Example:

```
30 = 1*18 + 12
18 = 1*12 + 6
12 = 2*6 + 0      -> gcd = 6
6 = 18 - 12 = 18 - (30 - 18) = -30 + 2*18
```

Extended Euclid also returns Bézout coefficients x, y with $ax + by = g$:

```
def egcd(a, b):                # nonnegative integers, not both zero
    if b == 0:
        return a, 1, 0
    g, x1, y1 = egcd(b, a % b)
    return g, y1, x1 - (a // b) * y1
```

The recurrence substitutes $a \% b = a - (a // b) * b$ into the returned identity. Euclid performs $O(\log \min(a, b))$ divisions for positive arguments; arbitrary-precision bit complexity also depends on the cost of each division. For nonzero integers, $\text{lcm}(a, b) = |ab| / \gcd(a, b)$; divide before multiplying to reduce overflow risk in fixed-width code.

N2. Modular arithmetic and inverse modulo

$a \equiv b \pmod{m}$ iff $m \mid (a - b)$, for $m > 0$. Addition, subtraction and multiplication preserve congruence. Division requires an inverse: a^{-1} exists modulo m exactly when $\gcd(a, m) = 1$.

```
def inverse_mod(a, m):
    if m <= 1:
        raise ValueError("modulus must exceed one")
    g, x, _ = egcd(a % m, m)
    if g != 1:
        raise ValueError("inverse does not exist")
    return x % m
```

For $3^{-1} \pmod{7}$, $3 \cdot 5 = 15 \equiv 1$, so the inverse is 5. For $6x \equiv 1 \pmod{9}$, no solution exists because gcd is 3. Never cancel a noninvertible factor blindly: $2x \equiv 2 \pmod{6}$ has both $x \equiv 1$ and $x \equiv 4$, not just $x \equiv 1 \pmod{6}$. See Conrad's modular arithmetic notes.

N3. Linear Diophantine equations — definition and real use

A Diophantine equation asks for **integer** solutions. For $ax + by = c$, with a, b not both zero, solutions exist iff $g = \gcd(a, b)$ divides c . Necessity: g divides both terms and hence their sum. Sufficiency: multiply a Bézout identity for g by c/g .

If (x_0, y_0) is one solution, all solutions are

$$x = x_0 + \frac{b}{g}t, \quad y = y_0 - \frac{a}{g}t, \quad t \in \mathbb{Z}.$$

Packaging use: can exactly 24 items be supplied using 6-item and 9-item boxes? Solve $6x + 9y = 24$ with $x, y \geq 0$. Gcd 3 divides 24. One solution is $(4, 0)$; all are $(4 + 3t, -2t)$. Nonnegativity requires $t = -1$ or 0, giving **1 six-box + 2 nine-boxes**, or **4 six-boxes**. Gcd divisibility guarantees integer solutions, **not** nonnegative ones; those inequalities are a separate step.

```
def diophantine(a, b, c):          # a,b positive in this classroom version
    if a <= 0 or b <= 0:
        raise ValueError("expected positive coefficients")
    g, x, y = egcd(a, b)
    if c % g:
        return None
    return x * (c // g), y * (c // g), b // g, -a // g
    # (x0, y0, x_step, y_step); add t times the two steps
```

Do not confuse with pigeonhole: Diophantine reasoning solves integer constraints; pigeonhole proves unavoidable sharing. For the latter say: "101 distinct keys go to 100 buckets, so a collision is guaranteed." Name the objects, boxes, mapping and forced conclusion, as in §§53–59.

N4. Bigmod / binary modular exponentiation

Repeated squaring computes $a^e \pmod{m}$ in $O(\log(e + 1))$ modular multiplications for nonnegative exponent, rather than e multiplications.

```
def bigmod(a, e, m):
    if e < 0 or m <= 0:
        raise ValueError("e >= 0 and m > 0 required")
    result = 1 % m
    a %= m
    while e:
        if e & 1:
            result = result * a % m
        a = a * a % m
        e //= 2
    return result
```

Invariant: `result * a**e` is congruent to the original requested power. For $3^{13} \bmod 7$, $13 = 8 + 4 + 1$, powers are $3^1 \equiv 3, 3^2 \equiv 2, 3^4 \equiv 4, 3^8 \equiv 2$; result $2 \cdot 4 \cdot 3 \equiv 3$. Fixed-width multiplication itself may overflow before `% m`; use a wider safe multiply/reduction technique for large moduli. This educational loop is not constant-time cryptographic code.

N5. Primes, sieve, and factorization

The fundamental theorem of arithmetic gives unique prime factorization of an integer greater than 1, up to ordering. Trial division needs only test through the square root: if $n = ab$ is composite, at least one factor is at most $\sqrt{n}$.

```
def factorize(n):
    if n < 1:
        raise ValueError("positive n required")
    factors, p = [], 2
    while p <= n // p:
        exponent = 0
        while n % p == 0:
            n //= p
            exponent += 1
        if exponent:
            factors.append((p, exponent))
        p += 1
    if n > 1:
        factors.append((n, 1))
    return factors

def sieve(n):
    if n < 2:
        return []
    prime = [True] * (n + 1)
    prime[0] = prime[1] = False
    p = 2
    while p * p <= n:
        if prime[p]:
            for multiple in range(p*p, n+1, p):
```

```

        prime[multiple] = False
    p += 1
    return [i for i in range(2, n+1) if prime[i]]

```

`factorize(360)` gives $2^3 3^2 5$. Trial division takes $O(\sqrt{n})$ divisions worst case, exponential in half the input's bit length. The sieve finds **all** primes through N in $O(N \log \log N)$ time and $O(N)$ space. Start crossing at p^2 because smaller multiples already have a smaller prime factor. Through 20: cross multiples of 2 from 4, then 3 from 9; survivors are **2, 3, 5, 7, 11, 13, 17, 19**. A smallest-prime-factor table also supports many later factorization queries efficiently.

N6. Fermat's little theorem and Euler's totient theorem

For prime p , $a^p \equiv a \pmod{p}$. If $p \nmid a$, divide by a to get $a^{p-1} \equiv 1 \pmod{p}$; hence $a^{-1} \equiv a^{p-2} \pmod{p}$. For composite m , replace $p-1$ by the totient **only when coprime**:

$$\gcd(a, m) = 1 \Rightarrow a^{\phi(m)} \equiv 1 \pmod{m}, \quad \phi(m) = m \prod_{p \mid m} \left(1 - \frac{1}{p}\right).$$

For $m = 12$, the coprime residues are **1, 5, 7, 11**, so $\phi(12) = 4$ and $5^4 \equiv 1 \pmod{12}$. Not every exponent may be reduced modulo $\phi(m)$ when the base is noncoprime: $2^4 \equiv 4 \pmod{8}$, but $2^0 \equiv 1$. Fermat congruences are not by themselves a proof of primality; pseudoprimes and Carmichael numbers exist. These theorems explain modular inverses and the RSA correctness intuition, not the full security of RSA.

N7. Chinese Remainder Theorem (CRT) — combining schedules

For pairwise coprime moduli $m_1, \dots, m_k$, every residue tuple has exactly one solution modulo $M = \prod m_i$:

$$x \equiv \sum_i a_i M_i y_i \pmod{M}, \quad M_i = M/m_i, \quad y_i = M_i^{-1} \pmod{m_i}.$$

Each term supplies the wanted residue in one modulus and zero in all others. Example: $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$. $M = 105$, with $M_i = (35, 21, 15)$ and inverses $(2, 1, 1)$:
 $x \equiv 140 + 63 + 30 = 233 \equiv 23 \pmod{105}$.

```

event A: remainder 2 on a 3-day cycle -> 2,5,8,11,...,23,...
event B: remainder 3 on a 5-day cycle -> 3,8,13,18,23,...
event C: remainder 2 on a 7-day cycle -> 2,9,16,23,...
all three agree first at 23; repeat every 105 days

```

```

def crt(residues, moduli):      # pairwise coprime moduli > 1
    if len(residues) != len(moduli):
        raise ValueError("length mismatch")
    M = 1
    for m in moduli:
        if m <= 1:
            raise ValueError("modulus must exceed one")
        M *= m
    answer = 0
    for a, m in zip(residues, moduli):

```

```

Mi = M // m
answer += a * Mi * inverse_mod(Mi, m)
return answer % M, M

```

For two noncoprime moduli, consistency instead requires $a \equiv b \pmod{\gcd(m, n)}$, with uniqueness modulo $\text{lcm}(m, n)$. For example $x \equiv 1 \pmod{4}$, $x \equiv 3 \pmod{6}$ has $x \equiv 9 \pmod{12}$; $x \equiv 1 \pmod{4}$, $x \equiv 2 \pmod{6}$ has no solution. See [Conrad's CRT notes](#).

N8. Wilson's theorem — exact but not an efficient large primality test

For integer $n > 1$, n is prime iff $(n - 1)! \equiv -1 \pmod{n}$. For a prime, pair every nonzero residue with its multiplicative inverse; only 1 and -1 are self-inverse, leaving product -1. Conversely, the congruence makes $(n - 1)!$ coprime to n , excluding any proper divisor between 2 and $n - 1$. Example: $4! = 24 \equiv -1 \pmod{5}$; $5! = 120 \equiv 0 \pmod{6}$. Computing the factorial directly costs $O(n)$ modular multiplications, so this beautiful characterization is not a practical large-input primality algorithm.

N9. Hensel's lemma — lift a simple root to a higher prime power

For integer polynomial f , prime p , and $f(a) \equiv 0 \pmod{p^k}$ with $f'(a) \not\equiv 0 \pmod{p}$, there is a unique lift $b = a + tp^k$ modulo p^{k+1} . Choose

$$t \equiv -\frac{f(a)}{p^k} [f'(a)]^{-1} \pmod{p}.$$

Example: $f(x) = x^2 - 2$, $p = 7$, $a = 3$, $k = 1$. Since $f(3) = 7$ and $f'(3) = 6$, $t \equiv -1 \cdot 6 \equiv 1 \pmod{7}$. Thus $b = 10$, and $10^2 - 2 = 98$ is divisible by 49. This builds solutions digit by digit in base p . If the derivative is zero modulo p , this simple uniqueness guarantee fails. The example is a root-lifting calculation, not an ordinary real-valued Newton iteration. [Conrad's Hensel notes](#).

N10. Quadratic reciprocity — when does a modular square root exist?

For an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, +1 if a is a nonzero square modulo p , and -1 otherwise. For distinct odd primes:

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{((p-1)/2)((q-1)/2)}.$$

The symbols agree unless both primes are 3 modulo 4, in which case their signs are opposite. Supplementary laws: $(-1/p) = (-1)^{(p-1)/2}$ and $(2/p) = (-1)^{(p^2-1)/8}$. Example: $(3/7) = -(7/3) = -(1/3) = -1$. Indeed the nonzero squares modulo 7 are **1, 2, 4**, so 3 has no square root modulo 7. Use it to decide solvability or simplify residue tests; it does not directly list the roots. [Conrad's quadratic reciprocity notes](#).

Excerpt source: 10_DISCRETE_MATHEMATICS_SLIDE_COMPLETE.md, original lines 3395–3658. Original section numbers are retained.

15.1 The pigeonhole principle – a solid answer

Basic principle: if more than k objects are placed into k boxes, some box contains at least two objects.

Generalized form: if N objects are placed into k boxes, some box contains at least

$$\left\lceil \frac{N}{k} \right\rceil$$

objects.

This is an existence theorem: it proves that a collision or concentration **must** occur, but does not identify which box.

Three defensible applications

1. **Circular queue state representation:** an n -cell array queue has $n + 1$ logical sizes but, for a fixed front, only n rear positions. Two sizes must share a representation unless we reserve a cell or store an extra count/full bit. This directly justifies the lecture design.
2. **Hash collision:** an m -bit hash has only 2^m possible outputs. Among $2^m + 1$ distinct inputs, at least two necessarily have the same output. This does not mean cryptographic collision search is easy; it proves collisions exist because the input domain is larger.
3. **Capacity planning:** assigning N jobs to k identical servers forces at least one server to receive at least $\lceil N/k \rceil$ jobs. This gives an unavoidable lower bound on maximum load, regardless of scheduling cleverness.

Board answer: "Objects are pigeons and categories are holes. With 13 people and 12 birth months, two share a birth month. In DSA, a circular queue gives a stronger use: $n + 1$ possible occupancies cannot be encoded by only n rear positions for fixed front, so empty and full collide unless we add state."

Trap: the principle says "at least one," not "exactly one," and it requires mapping every object to a defined box.

Excerpt source: 01_DSA_I_SLIDE_COMPLETE.md , original lines 2658–2679. Original section numbers are retained.

4. Data structures, searching, sorting and dynamic programming

Workbook coverage: DSA B7–B38/B58; Board B35/B37; Runtime Analysis sorting screenshot; BRAC array/list, circular array, DP, stock-profit and graph questions.

1. Algorithms, ADTs, correctness, and analysis

1.1 Problem, algorithm, program, data structure, and ADT

- A **problem** specifies a required input-output relationship, for example: given an array and a key, return an index containing the key or report absence.
- An **algorithm** is a finite, unambiguous, effective sequence of steps that solves every valid instance of a problem.
- A **program** is an implementation of one or more algorithms in a programming language, together with representation choices, input handling, and system details.
- A **data structure** is a way to organize data and its relationships so selected operations can be performed efficiently.
- An **abstract data type (ADT)** specifies observable values and operations, not their storage. A stack ADT promises **push**, **pop**, and **top**; it may be implemented by an array or a linked list.

Logical versus physical form. A list is logically an ordered sequence. Physically it may occupy consecutive array cells or scattered linked nodes. Clients should depend on the logical contract, not representation details.

Board answer – why study DSA? The same data can be represented in several ways, and no representation is best for every workload. Arrays provide constant-time indexing but costly middle insertion; linked lists give constant-time insertion once the predecessor is known but no constant-time random access. DSA is the study of choosing and proving the right trade-off.

Choosing a structure

Ask:

1. What operations dominate: access, search, insert, delete, minimum, FIFO, graph-neighbor iteration?
2. What must be guaranteed: order, uniqueness, stability, worst-case latency?
3. Is the input static or dynamic? How large can it grow?
4. What memory overhead and locality are acceptable?
5. Is average/expected performance enough, or is a worst-case bound required?

An ADT hides **how**; an algorithm chooses **steps**; a data structure supplies an efficient **representation**.

1.2 Correctness

An algorithm is correct when it terminates and returns the specified output for every valid input.

- **Partial correctness:** if it terminates, its result is correct.
- **Termination:** a well-founded measure decreases and cannot decrease forever.
- **Total correctness:** partial correctness plus termination.

Loop-invariant proof template

For a loop, state a property (P) and prove:

1. **Initialization:** (P) holds before the first iteration.
2. **Maintenance:** if (P) holds before an iteration, the body makes it hold before the next.
3. **Termination:** when the loop ends, (P) plus the exit condition implies the desired result.

Example for insertion sort: before iteration i , $a[0..i-1]$ is a sorted permutation of its original elements. Inserting $a[i]$ preserves that property. At $i=n$, the whole array is sorted.

Recursive proof template

Prove a base case, assume recursive calls correctly solve smaller instances, then show the combine step solves the current instance. Also identify a size measure that strictly decreases.

1.3 Why asymptotic analysis instead of seconds?

Experimental timing depends on processor, compiler, language, cache, system load, implementation skill, and chosen test data. It is useful after implementation but cannot cleanly compare algorithms on all input sizes. Asymptotic analysis counts a **basic operation** as a function of input size and studies growth as input becomes large.

Let $T(n)$ be the number of elementary steps for input size (n) . Constants and lower-order terms matter in engineering, but growth rate predicts scaling.

Cases and styles of analysis

- **Worst case:** maximum cost over all inputs of size (n) ; gives a guarantee.
- **Best case:** minimum cost; often not representative.
- **Average case:** expectation under an explicitly stated input distribution.
- **Expected/probabilistic:** expectation over random choices made by the algorithm and/or a stated input model.
- **Amortized:** average cost per operation over every possible sequence, without assuming a probability distribution.

Trap: amortized does not mean average-case. Dynamic-array append is amortized $O(1)$ even for an adversarial operation sequence, while a single resize still costs $\Theta(n)$.

1.4 Formal asymptotic notation

For eventually nonnegative functions:

- $f(n) \in O(g(n))$ if there exist $c > 0, n_0$ such that $0 \leq f(n) \leq cg(n)$ for all $n \geq n_0$. This is an asymptotic upper bound.
- $f(n) \in \Omega(g(n))$ if there exist $c > 0, n_0$ such that $0 \leq cg(n) \leq f(n)$ for all $n \geq n_0$. This is a lower bound.
- $f(n) \in \Theta(g(n))$ iff it is in both $O(g(n))$ and $\Omega(g(n))$; a tight bound.

Example: $3n^2 + 5n + 7 \in \Theta(n^2)$. For $n \geq 1$, it is at least $3n^2$, and for sufficiently large n , it is at most $15n^2$.

Language trap: "the complexity is at least $O(n)$ " mixes directions. Say "upper-bounded by $O(n)$ " or "lower-bounded by $\Omega(n)$." Big-O itself is not "the exact worst case"; it is a set of upper bounds.

Common growth order

$$1 < \log n < \sqrt{n} < n < n \log n < n^2 < n^3 < 2^n < n!$$

Changing the logarithm base only multiplies by a constant: $\log_a n = \log_b n / \log_b a$, so all fixed bases are $\Theta(\log n)$.

Counting patterns

```
// Theta(n): exactly n iterations
for (int i = 0; i < n; ++i) work();

// Theta(n^2): sum_{i=0}^{n-1} i = n(n-1)/2
for (int i = 0; i < n; ++i)
    for (int j = 0; j < i; ++j) work();

// Theta(log n): i takes 1,2,4,...
for (int i = 1; i < n; i *= 2) work();

// Theta(n log n): n choices of i, log n choices of j
for (int i = 0; i < n; ++i)
    for (int j = 1; j < n; j *= 2) work();
```

Nested loops are **not automatically** $O(n^2)$; count the actual ranges. Consecutive blocks add, nested independent blocks multiply, and dependent ranges often form a sum.

Input parameters matter

Graph time is naturally expressed using both $(|V|)$ and $(|E|)$. Matrix multiplication may use dimensions (m,n,p) . Do not silently compress independent parameters into one (n) .

1.5 Tractability and examples from the analysis notes

Polynomial-time problems are normally called **tractable** in this course context; exponential and factorial algorithms quickly become impractical. This is a growth-rate convention, not a claim that every high-degree polynomial is practical.

- Finding the maximum of (n) values needs $(n-1)$ comparisons: $\Theta(n)$.
- Binary search repeatedly halves the candidate range: $\Theta(\log n)$.
- Merging two sorted arrays of total length (n) is $\Theta(n)$.
- Brute-force closest pair checks $\binom{n}{2}$ pairs: $\Theta(n^2)$.
- Enumerating every subset takes $\Theta(2^n)$ subsets before the work inside each subset.

Time-space trade-off

Extra memory can avoid repeated work: memoized Fibonacci stores answers to obtain $O(n)$ time; a lookup table can accelerate queries; a hash table spends capacity for expected constant-time lookup. Conversely, recomputation may reduce storage.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 47–170. Original section numbers are retained.

2.1 Recursion mental model

A recursive function calls itself on a smaller instance. Every call creates an **activation record** containing parameters, local variables, return address, and bookkeeping. These frames form the runtime call stack.

A valid recursive solution needs:

1. one or more base cases;

2. progress toward a base case;
3. a correct combination of subproblem answers.

```
long long sumTo(long long n) {  
    if (n <= 0) return 0;    // base case  
    return n + sumTo(n - 1); // smaller instance  
}
```

Trace for `sumTo(4)` :

```
sumTo(4)  
= 4 + sumTo(3)  
= 4 + 3 + sumTo(2)  
= 4 + 3 + 2 + sumTo(1)  
= 4 + 3 + 2 + 1 + sumTo(0)  
= 10
```

Recurrence: $T(n)=T(n-1)+(1)=(n)$. Stack depth and auxiliary space are $\Theta(n)$.

Iterative version

```
long long sumToIter(long long n) {  
    long long ans = 0;  
    for (long long x = 1; x <= n; ++x) ans += x;  
    return ans;  
}
```

It has $\Theta(n)$ time and $\Theta(1)$ auxiliary space. The formula $n(n+1)/2$ is $\Theta(1)$ under a fixed-width arithmetic model, though arbitrary-precision bit complexity is different.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 173–214. Original section numbers are retained.

2.2 Recursion versus iteration

Neither is universally “better.”

Criterion	Recursion	Iteration
Natural fit	Trees, DFS, divide-and-conquer, backtracking	Sequential repetition, simple state machines
Extra state	Implicit call stack	Explicit variables/structure
Space	Usually proportional to depth	Often $O(1)$, unless an explicit stack is needed
Overhead	Function calls and possible stack overflow	Usually lower call overhead
Clarity	Can mirror recursive definitions	Can make control and limits explicit

Every recursion can be simulated with an explicit stack because that stack stores the same frames. Tail-recursive calls can sometimes be optimized into jumps, but C++ and Java do not guarantee tail-call optimization.

Board answer: compare implementations, not slogans. For factorial, iteration is simpler and constant-space. For tree traversal, recursion directly follows the structure; an iterative version needs an explicit stack and remains $O(h)$ space.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 215–230. Original section numbers are retained.

2.5 Solving recurrences

Expansion/iteration

For merge sort:

$$T(n) = 2T(n/2) + cn.$$

At recursion level (i), there are (2^i) subproblems of size $(n/2^i)$, so the nonrecursive work is $(2^i c(n/2^i) = cn)$. There are $(\log_2 n)$ levels, hence $\Theta(n \log n)$.

For $T(n) = T(n-1) + cn$:

$$T(n) = c(n + (n-1) + \dots + 1) + T(0) = \Theta(n^2).$$

Master theorem

For

$$T(n) = aT(n/b) + f(n), \quad a \geq 1, b > 1,$$

compare $(f(n))$ with $(n^{\log_b a})$:

1. If $(f(n) = O(n^{\log_b a - \epsilon}))$, then $(T(n) = \Theta(n^{\log_b a}))$.
2. If $(f(n) = \Theta(n^{\log_b a} \log^k n))$, then $(T(n) = \Theta(n^{\log_b a} \log^{k+1} n))$ for (k) .
3. If $(f(n) = \Omega(n^{\log_b a + \epsilon}))$ and the regularity condition $(af(n/b) \leq cf(n))$ for some $(c < 1)$ holds, then $(T(n) = \Theta(f(n)))$.

Examples:

- $(2T(n/2) + n = \Theta(n \log n))$.
- $(T(n/2) + 1 = \Theta(n))$.
- $(4T(n/2) + n = \Theta(n^2))$.
- $(2T(n/2) + n^2 = \Theta(n^2))$.
- $(3T(n/2) + n = \Theta(n^{\log_2 3}))$.

Trap: the standard Master theorem does not directly solve $(T(n) = T(n-1) + n)$, unequal subproblem sizes such as $(T(n/3) + T(2n/3) + n)$, or every recurrence containing a logarithm.

Generalized logarithmic form from the notes

For $T(n) = aT(n/b) + \Theta(n^c \log^p n)$, let $d = \log_b a$:

- if $d > c$, recursion leaves dominate: $T(n) = \Theta(n^d)$;
- if $d < c$ (with the usual regularity condition), combine work dominates: $T(n) = \Theta(n^c \log^p n)$;
- if $d = c$ and $p > -1$, $T(n) = \Theta(n^c \log^{p+1} n)$;
- if $d = c$ and $p = -1$, $T(n) = \Theta(n^c \log \log n)$;
- if $d = c$ and $p < -1$, $T(n) = \Theta(n^c)$.

This extension explains borderline logarithmic cases that the three-case classroom version does not state. Always verify that the recurrence actually has equal-sized subproblems before using it.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 269–324. Original section numbers are retained.

3.2 Array-based list

Representation:

```
template<class E>
struct AList {
    vector<E> a;           // or fixed E a[maxSize]
    int listSize = 0;
    int curr = 0;         // gap/index of current element
};
```

Slide-aligned fixed-capacity operations:

```
void insert(const E& x) {           // before a[curr]
    if (listSize == maxSize) throw overflow_error("full");
    for (int i = listSize; i > curr; --i)
        a[i] = a[i - 1];
    a[curr] = x;
    ++listSize;
}

void append(const E& x) {
    if (listSize == maxSize) throw overflow_error("full");
    a[listSize++] = x;
}

E remove() {
    if (curr < 0 || curr >= listSize) throw out_of_range("no current item");
    E removed = a[curr];
    for (int i = curr; i + 1 < listSize; ++i)
        a[i] = a[i + 1];
    --listSize;
    if (curr > listSize) curr = listSize;
    return removed;
}
```

Costs for length (n): random access $\Theta(1)$; append $\Theta(1)$ if capacity exists; insertion/removal at position (k), $\Theta(n - k)$ shifts; linear search $\Theta(n)$. A dynamic array occasionally reallocates and copies, so append is $\mathcal{O}(1)$ amortized, not worst-case.

Invariant for insertion: before copying index $i-1$ to i , cells $i..listSize$ already contain the original elements from $i-1..listSize-1$. Backward copying prevents overwriting unread data.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 354–397. Original section numbers are retained.

3.3 Singly linked list

Each node stores an element and the address of the next node; physical memory need not be contiguous.

```
template<class E>
struct Node {
    E value;
    Node* next;
    Node(const E& v, Node* n = nullptr) : value(v), next(n) {}
};
```

Basic operations and edge cases

```
void pushFront(Node*& head, int x) {
    head = new Node<int>(x, head);
}

void pushBack(Node*& head, Node*& tail, int x) {
    Node* p = new Node<int>(x);
    if (!head) head = tail = p;
    else { tail->next = p; tail = p; }
}

bool insertAfter(Node* p, int x) {
    if (!p) return false;
    p->next = new Node<int>(x, p->next);
    return true;
}

bool popFront(Node*& head, Node*& tail, int& out) {
    if (!head) return false;
    Node* dead = head;
    out = dead->value;
    head = head->next;
    if (!head) tail = nullptr;
    delete dead;
    return true;
}

bool eraseValue(Node*& head, Node*& tail, int x) {
    Node* prev = nullptr;
    Node* cur = head;
    while (cur && cur->value != x) {
        prev = cur;
        cur = cur->next;
    }
```

```

    if (!cur) return false;
    if (prev) prev->next = cur->next;
    else head = cur->next;
    if (tail == cur) tail = prev;
    delete cur;
    return true;
}

```

Search and access to position (k) are $O(n)$. Insertion/deletion is $O(1)$ **only when the necessary node/predecessor pointer is already known**. Finding that position may take $O(n)$. With a tail pointer, append is $O(1)$; without it, append requires traversal.

Why the lecture cursor points before the current item

If `curr` points to the predecessor of the logical current item, inserting and removing at the current position only rewires `curr->next`. A dummy header makes "before the first item" an ordinary node and avoids special cases.

```

// header carries no list element; tail == header in an empty list
void insertCurrent(const E& x) {
    curr->next = new Node<E>(x, curr->next);
    if (tail == curr) tail = curr->next;
    ++cnt;
}

E removeCurrent() {
    if (!curr->next) throw out_of_range("no current item");
    Node<E>* dead = curr->next;
    E x = dead->value;
    curr->next = dead->next;
    if (tail == dead) tail = curr;
    delete dead;
    --cnt;
    return x;
}

```

Representation invariant:

- `head` is a permanent dummy node;
- the data sequence starts at `head->next`;
- `tail` is the last node, or `head` if empty;
- `curr` is a reachable node immediately before the logical current element;
- exactly `cnt` data nodes are reachable.

Worked pointer trace

For `head -> 4 -> 7 -> 9 -> null`, delete 7:

1. `prev` points to 4, `cur` points to 7.
2. Save `cur` before changing links.

3. Set `prev->next = cur->next`, so 4 points to 9.
4. Delete `cur`. Never dereference it afterward.

Traps: losing the rest of the list by overwriting `next` before saving it; forgetting the empty/singleton case; leaving `tail` dangling; using a freed node; claiming $O(1)$ deletion when only a value—not its predecessor—is given.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 398–500. Original section numbers are retained.

3.6 Array versus linked representation

Question	Array list	Linked list
Index access	$\Theta(1)$	$\Theta(n)$
Insert/delete at known index	shifts, $\Theta(n)$ worst case	must locate predecessor, then $\Theta(1)$ rewiring
Append	$\Theta(1)$ with spare capacity; amortized for dynamic array	$\Theta(1)$ with tail
Memory	reserved capacity may be unused	pointer overhead per node
Locality	excellent	usually poor
Growth	reallocate/copy or fixed limit	node-by-node

The slide space comparison uses element size E , pointer size P , array capacity D , and current list length n : array storage is about DE ; singly linked storage is about $n(E + P)$. The linked form uses less space when $n(E + P) < DE$. This is only a storage model; allocator metadata and alignment can change real values.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 550–564. Original section numbers are retained.

4.1 Stack ADT and representations

A stack is **last in, first out (LIFO)**. Only the top is accessible.

```
push(x): add x at top
pop(): remove and return top
top()/peek(): return top without removing
empty(), size(), clear()
```

Calling `pop` or `top` on an empty stack is an underflow error; pushing into a fixed full array is overflow.

Array stack

Let `top` mean the first free index, so the current top element is `a[top-1]`.

```
template<class E>
class ArrayStack {
    vector<E> a;
    int topIndex = 0;
public:
    explicit ArrayStack(int cap) : a(cap) {}
```

```

bool empty() const { return topIndex == 0; }
int size() const { return topIndex; }

void push(const E& x) {
    if (topIndex == (int)a.size()) throw overflow_error("stack full");
    a[topIndex++] = x;
}

E pop() {
    if (empty()) throw underflow_error("empty stack");
    return a[--topIndex];
}

const E& top() const {
    if (empty()) throw underflow_error("empty stack");
    return a[topIndex - 1];
}
};

```

Put the top at the array tail: push and pop then take $\Theta(1)$. Putting it at index zero shifts all elements and costs $\Theta(n)$.

Linked stack

The head is the top, so no dummy header is necessary.

```

template<class E>
class LinkedStack {
    Node<E>* head = nullptr;
    int cnt = 0;
public:
    void push(const E& x) { head = new Node<E>(x, head); ++cnt; }
    E pop() {
        if (!head) throw underflow_error("empty stack");
        Node<E>* dead = head;
        E x = dead->value;
        head = head->next;
        delete dead;
        --cnt;
        return x;
    }
    const E& top() const {
        if (!head) throw underflow_error("empty stack");
        return head->value;
    }
};

```


Both implementations have $\Theta(1)$ core operations. The array has locality and no link per element but may waste unused capacity; the linked form grows one node at a time but pays pointer and allocator overhead.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 567–642. Original section numbers are retained.

5.1 Queue ADT

A queue is **first in, first out (FIFO)**.

```
enqueue(x): add at rear
dequeue(): remove and return front
front(): inspect front
empty(), size(), clear()
```

Real motivations include printer jobs, network packets awaiting service, BFS frontiers, and tasks waiting for a CPU. Queue order models arrival order, not importance.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 764–776. Original section numbers are retained.

5.2 Why a naive array queue fails

If every dequeue shifts all remaining elements left, dequeue costs $\Theta(n)$. If **front** simply drifts right and freed cells are never reused, the queue reports overflow despite free cells at the beginning. A circular array reuses them.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 777–780. Original section numbers are retained.

5.3 Circular queue: one-slot-empty design

Allocate `capacity + 1` cells. Let **front** index the first element and **rear** index the next free cell.

```
template<class E>
class CircularQueue {
    vector<E> a;                // requested capacity + 1
    int frontIndex = 0;
    int rearIndex = 0;
public:
    explicit CircularQueue(int capacity) : a(capacity + 1) {}

    bool empty() const { return frontIndex == rearIndex; }
    bool full() const { return (rearIndex + 1) % a.size() == frontIndex; }
    int size() const {
        return (rearIndex - frontIndex + a.size()) % a.size();
    }
    void enqueue(const E& x) {
        if (full()) throw overflow_error("queue full");
        a[rearIndex] = x;
        rearIndex = (rearIndex + 1) % a.size();
    }
    E dequeue() {
        if (empty()) throw underflow_error("queue empty");
```

```

        E x = a[frontIndex];
        frontIndex = (frontIndex + 1) % a.size();
        return x;
    }
    const E& front() const {
        if (empty()) throw underflow_error("queue empty");
        return a[frontIndex];
    }
};

```

Example with physical size 5 : enqueue at indices 3,4,0 ; logical order crosses the array boundary. Modulo arithmetic maps the logical next position back to zero.

Why sacrifice one slot? A pigeonhole argument

With an array of (n) cells and only **front** and **rear**, there are (n+1) possible queue sizes (0,1,,n). If **front** is fixed, **rear** has only (n) possible positions. By the pigeonhole principle, at least two sizes must share the same (**front**,**rear**) state—specifically, the ordinary modulo representation makes empty and full both look like **front==rear**. Therefore the representation needs extra information:

- reserve one cell, so maximum data size is (n-1);
- store a **count** from 0..n ;
- or keep a separate **full** flag.

This is also a concrete DSA use of the pigeonhole principle: it proves a representation with too few states cannot distinguish all abstract queue states.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 781–828. Original section numbers are retained.

5.4 Circular queue with a count

All array cells can be used if **count** disambiguates full and empty:

```

int frontIndex = 0, rearIndex = 0, count = 0;

void enqueue(int x) {
    if (count == capacity) throw overflow_error("full");
    a[rearIndex] = x;
    rearIndex = (rearIndex + 1) % capacity;
    ++count;
}

int dequeue() {
    if (count == 0) throw underflow_error("empty");
    int x = a[frontIndex];
    frontIndex = (frontIndex + 1) % capacity;
    --count;
    return x;
}

```

Choose one convention and derive all conditions from it. Many circular-queue bugs come from mixing "rear points to last item" with "rear points to first free cell."

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 829–853. Original section numbers are retained.

6. Binary trees and traversals

6.1 Definitions and vocabulary

A binary tree is either empty or consists of a root and two disjoint binary trees called its left and right subtrees. This recursive definition explains why recursive algorithms fit naturally.

- **Node, edge, root, parent, child, sibling.**
- **Leaf/external node:** no children. **Internal node:** at least one child.
- **Ancestor/descendant:** nodes along an upward/downward path.
- **Depth of a node:** number of edges from root to it; root depth is 0.
- **Height of a node:** longest downward edge path to a leaf. Under the edge convention, a leaf has height 0 and an empty tree has height (-1).
- **Height of tree:** height of its root / maximum depth.
- **Subtree:** a node with all descendants.

State the height convention because some texts count nodes rather than edges.

Full, perfect, complete, and balanced

- **Full/proper/strict:** every internal node has exactly two children.
- **Perfect:** all internal nodes have two children and all leaves have one depth.
- **Complete:** every level is full except possibly the last, which is filled left to right.
- **Balanced:** height is $O(\log n)$, or a more specific local balance rule in a named balanced-tree family.

These terms are not interchangeable. A heap is complete but need not be perfect; a full tree need not be complete.

6.2 Counting facts and proofs

Full binary tree: leaves = internal nodes + 1

Let (I) be internal nodes and (L) leaves. Each internal node contributes exactly two child edges, so a nonempty full tree has (2I) edges. Any tree with (I+L) nodes has (I+L-1) edges. Thus

$$2I = I + L - 1 \Rightarrow L = I + 1.$$

This can also be proved by structural induction.

Null pointers in a binary linked tree

With (n) nodes there are (2n) child-pointer fields. A tree has (n-1) real edges, hence

$$2n - (n - 1) = n + 1$$

null child pointers.

Perfect tree counts

At depth (d) there are (2^d) nodes. A perfect tree of height (h) has

$$1 + 2 + \dots + 2^h = 2^{h+1} - 1$$

nodes and (2^h) leaves.

6.3 Linked and array representations

```
template<class E>
struct BNode {
    E value;
    BNode *left = nullptr, *right = nullptr;
};
```

Linked representation fits arbitrary shapes but stores two pointers per node, many of which are null. A complete tree is compact in an array. Under 1-based indexing:

```
parent(i) = floor(i/2)
left(i)   = 2i
right(i)  = 2i+1
```

Under 0-based indexing:

```
parent(i) = floor((i-1)/2)
left(i)   = 2i+1
right(i)  = 2i+2
```

Do not mix conventions.

6.4 Traversals

```
void preorder(BNode<int>* r) {
    if (!r) return;
    visit(r); preorder(r->left); preorder(r->right);
}

void inorder(BNode<int>* r) {
    if (!r) return;
    inorder(r->left); visit(r); inorder(r->right);
}

void postorder(BNode<int>* r) {
    if (!r) return;
    postorder(r->left); postorder(r->right); visit(r);
}

void levelOrder(BNode<int>* root) {
    if (!root) return;
    queue<BNode<int>*> q;
    q.push(root);
    while (!q.empty()) {
        auto* u = q.front(); q.pop();
        visit(u);
        if (u->left) q.push(u->left);
    }
}
```

```

        if (u->right) q.push(u->right);
    }
}

```

- Preorder: root before subtrees; useful for copying/serialization with null markers.
- Inorder: left, root, right; yields sorted keys for a BST.
- Postorder: children before parent; useful for deleting a tree or computing subtree results.
- Level order: breadth first; uses a queue.

Example:



Preorder **A B D E C F**; inorder **D B E A C F**; postorder **D E B F C A**; level order **A B C D E F**.

Each visits each node exactly once: $\Theta(n)$ time. Recursive auxiliary space is $\Theta(h)$; worst case $\Theta(n)$, balanced case $\Theta(\log n)$. Level order can hold $\Theta(w)$ nodes where (w) is maximum width.

6.5 Recursive tree computations

```

int nodeCount(BNode<int>* r) {
    if (!r) return 0;
    return 1 + nodeCount(r->left) + nodeCount(r->right);
}

int height(BNode<int>* r) {           // edge-height convention
    if (!r) return -1;
    return 1 + max(height(r->left), height(r->right));
}

void destroy(BNode<int>* r) {
    if (!r) return;
    destroy(r->left);
    destroy(r->right);
    delete r;                          // postorder: children first
}

```

Invariant/proof: recursive calls return correct results for the two smaller subtrees; the root formula combines them. The null guard is essential.

Balanced-tree check

Computing height separately at every node can revisit subtrees and become $O(n^2)$ on a chain. Return height and failure together in one postorder traversal:

```

int heightOrFail(BNode<int>* r) {
    if (!r) return 0;
    int L = heightOrFail(r->left); if (L == -1) return -1;
    int R = heightOrFail(r->right); if (R == -1) return -1;
    if (abs(L - R) > 1) return -1;
    return 1 + max(L, R);
}
bool isHeightBalanced(BNode<int>* r) { return heightOrFail(r) != -1; }

```

This is $O(n)$: each node is processed once.

Excerpt source: [01_DSA_I_SLIDE_COMPLETE.md](#), original lines 918–1094. Original section numbers are retained.

7. Binary search trees

7.1 Property, duplicate convention, and motivation

For every node with key k , the lecture BST convention is:

- every key in the left subtree is $< k$;
- every key in the right subtree is $\geq k$.

Thus duplicates go right. Other implementations may reject duplicates or store a count, but the convention must be consistent in insertion, search, deletion, and validation.

A BST combines linked-tree flexibility with ordering. Operations follow one root-to-leaf path and therefore cost $\Theta(h)$, where h is tree height—not automatically $\Theta(\log n)$.

7.2 Search, minimum, maximum, and insertion

```

struct BSTNode {
    int key;
    BSTNode *left = nullptr, *right = nullptr;
};

BSTNode* search(BSTNode* r, int x) {
    while (r && r->key != x)
        r = (x < r->key) ? r->left : r->right;
    return r;
}

BSTNode* minimum(BSTNode* r) {
    if (!r) return nullptr;
    while (r->left) r = r->left;
    return r;
}

BSTNode* maximum(BSTNode* r) {

```

```

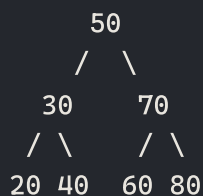
    if (!r) return nullptr;
    while (r->right) r = r->right;
    return r;
}

BSTNode* insert(BSTNode* r, int x) {
    if (!r) return new BSTNode{x};
    if (x < r->key) r->left = insert(r->left, x);
    else          r->right = insert(r->right, x); // duplicates right
    return r;
}

```

Search invariant: if key x occurs in the current subtree, the comparison identifies the only child subtree that can contain it. Discarding the other subtree cannot discard x .

Example insertion order 50, 30, 70, 20, 40, 60, 80:



Search 60 : compare with 50 (go right), 70 (go left), then find 60. Inorder traversal is 20, 30, 40, 50, 60, 70, 80.

Why inorder is sorted

By induction, inorder of the left subtree is sorted and all its keys are smaller than the root. The root comes next. Inorder of the right subtree is sorted and all its keys are at least the root. Concatenating those three sequences is nondecreasing.

7.3 Predecessor and successor

The **inorder predecessor** of a node is the greatest key strictly before it in inorder order. The **successor** is the smallest key strictly after it. With duplicates, clarify whether you mean a neighboring node in inorder sequence or the next distinct key.

If parent pointers exist:

- If x has a left subtree, predecessor is `maximum(x->left)`.
- Otherwise move upward while x is a left child; the first ancestor for which x lies in its right subtree is the predecessor.
- If x has a right subtree, successor is `minimum(x->right)`.
- Otherwise move upward while x is a right child; the first ancestor for which x lies in its left subtree is the successor.

Using the tree above:

- predecessor of 50 is 40 ; successor is 60 ;
- predecessor of 60 is 50 ; successor is 70 ;
- predecessor of minimum 20 does not exist; successor of maximum 80 does not exist.

Without parent pointers, search from root while remembering a candidate:

```
BSTNode* successor(BSTNode* root, int key) {
    BSTNode* candidate = nullptr;
    while (root) {
        if (key < root->key) {
            candidate = root;
            root = root->left;
        } else root = root->right;
    }
    return candidate;
}

BSTNode* predecessor(BSTNode* root, int key) {
    BSTNode* candidate = nullptr;
    while (root) {
        if (root->key < key) {
            candidate = root;
            root = root->right;
        } else root = root->left;
    }
    return candidate;
}
```

These functions return the next smaller/larger key relative to a query, not necessarily the neighboring duplicate node.

7.4 Deletion

There are three structural cases:

1. **Leaf:** replace it by null.
2. **One child:** replace it by its only child.
3. **Two children:** copy the minimum key from the right subtree (inorder successor) into the node, then delete that minimum from the right subtree.

```
BSTNode* erase(BSTNode* r, int x) {
    if (!r) return nullptr;
    if (x < r->key) {
        r->left = erase(r->left, x);
    } else if (x > r->key) {
        r->right = erase(r->right, x);
    } else {
        if (!r->left) {
            BSTNode* replacement = r->right;
            delete r;
            return replacement;
        }
        if (!r->right) {
            BSTNode* replacement = r->left;
            delete r;
            return replacement;
        }
        BSTNode* replacement = successor(r->right, r->key);
        replacement->left = r->left;
        r->right = erase(r->right, r->key);
        return replacement;
    }
}
```



```

        BSTNode* replacement = r->left;
        delete r;
        return replacement;
    }
    BSTNode* s = minimum(r->right);
    r->key = s->key;
    r->right = erase(r->right, s->key);
}
return r;
}

```

Slide-style `deleteMin`:

```

BSTNode* deleteMin(BSTNode* r) {
    if (!r->left) {
        BSTNode* replacement = r->right;
        delete r;
        return replacement;
    }
    r->left = deleteMin(r->left);
    return r;
}

```

Why can the promoted minimum of the right subtree have no left child? If it had a left child, that child would contain a still smaller key, contradicting minimality. It may have a right child, which must be reattached.

Worked deletion from the example tree:

- Delete leaf `20`: set `30.left=null`.
- Delete `30` after that: it has only child `40`, so `50.left=40`.
- Delete `50`: successor is `60`; copy `60` to the root, then remove the original `60` from under `70`. The ordering property remains true.

Traps: copying the successor key but forgetting to delete the original successor; failing to reconnect a successor's right child; and updating only a key but not its associated value in a key-value BST.

7.5 Height and degeneration

All operations above take $\Theta(h)$:

- in a height-balanced tree, $h = \Theta(\log n)$;
- inserting sorted keys into a plain BST can form a chain with $h = n - 1$, giving $\Theta(n)$ search/insert/delete.

Random insertion often gives logarithmic expected height, but a plain BST has no logarithmic worst-case guarantee. The DSA-I slides motivate maintaining balance but do not teach AVL or red-black invariants/rotations. Do not pretend those are slide-covered here.

7.6 BST validation

Checking only each node against its immediate children is insufficient: a deep descendant can violate an ancestor bound. Under this chapter's "duplicates go right" rule, carry the half-open interval `[lowInclusive, highExclusive)`:

```

bool validRange(BSTNode* r,
                optional<long long> lowInclusive,
                optional<long long> highExclusive) {
    if (!r) return true;
    long long k = r->key;
    if (lowInclusive && k < *lowInclusive) return false;
    if (highExclusive && k >= *highExclusive) return false;
    return validRange(r->left, lowInclusive, k) &&
           validRange(r->right, k, highExclusive);
}

bool validBST(BSTNode* root) {
    return validRange(root, nullopt, nullopt);
}

```

The left call makes the current key an exclusive upper bound; the right call makes it an inclusive lower bound. This also handles a duplicate that is in an ancestor's right subtree but later lies to the left of a larger descendant. Include `<optional>` and use `std::optional` / `std::nullopt` in non-pseudocode C++. Avoid `key+/-1`, which can overflow and mishandle noninteger keys. Another sound convention is to store `(key, insertion_id)` as a strict comparison key.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 1095–1296. Original section numbers are retained.

8. Binary heaps, priority queues, build-heap, and heapsort

8.1 Heap definition

A binary heap has two independent properties:

1. **Shape:** it is a complete binary tree.
2. **Order:** in a max-heap every parent key is at least its children; in a min-heap every parent is at most its children.

Consequences:

- the root is a global maximum/minimum;
- there is no ordering rule between siblings or across unrelated subtrees;
- searching for an arbitrary key can still be $\Theta(n)$;
- completeness gives height $\Theta(\log n)$ and compact array storage.

Heap versus BST: a max-heap quickly returns the maximum but cannot produce a search path for arbitrary keys. A BST orders the entire left/right relation and supports ordered search, predecessor, and successor when its height is controlled.

8.2 Height proof

For a complete tree of height h , levels `0..h-1` are full and the last level is nonempty:

$$2^h \leq n \leq 2^{h+1} - 1.$$

Taking logarithms gives $h = \lceil \log_2 n \rceil = \Theta(\log n)$.

8.3 Sift up / insertion

Append at the next free array position to preserve completeness, then swap upward while heap order is violated.

```
void pushMaxHeap(vector<int>& a, int x) { // 0-based
    a.push_back(x);
    int i = (int)a.size() - 1;
    while (i > 0) {
        int p = (i - 1) / 2;
        if (a[p] >= a[i]) break;
        swap(a[p], a[i]);
        i = p;
    }
}
```

Invariant: before each iteration, the only possible violation is between **i** and its parent; both child subheaps remain heaps. Each swap moves the new key one level, so time is $O(\log n)$.

Trace: push **50** into **[40,30,35,10,20]** :

```
[40,30,35,10,20,50]
swap with 35 -> [40,30,50,10,20,35]
swap with 40 -> [50,30,40,10,20,35]
```

8.4 Heapify / sift down

maxHeapify(i) assumes the child subtrees are already max-heaps, but **a[i]** may violate heap order.

```
void maxHeapify(vector<int>& a, int heapSize, int i) {
    while (true) {
        int largest = i;
        int left = 2*i + 1, right = 2*i + 2;
        if (left < heapSize && a[left] > a[largest]) largest = left;
        if (right < heapSize && a[right] > a[largest]) largest = right;
        if (largest == i) break;
        swap(a[i], a[largest]);
        i = largest;
    }
}
```

Swapping with the larger child fixes the root relative to both children. Only that child's subtree can now violate the property. The path has at most heap height, so time is $O(\log n)$. The slides also bound the recursive child subtree by at most $2n/3$:

$$T(n) \leq T(2n/3) + \Theta(1) = \Theta(\log n).$$

Source correction: swap/recurse only when **largest != i**; otherwise an unconditional recursive call may not terminate.

8.5 Extract maximum and increase key

```
int extractMax(vector<int>& a) {
    if (a.empty()) throw underflow_error("empty heap");
    int ans = a[0];
    a[0] = a.back();
    a.pop_back();
    if (!a.empty()) maxHeapify(a, a.size(), 0);
    return ans;
}

void increaseKey(vector<int>& a, int i, int larger) {
    if (larger < a[i]) throw invalid_argument("key decreased");
    a[i] = larger;
    while (i > 0 && a[(i-1)/2] < a[i]) {
        swap(a[i], a[(i-1)/2]);
        i = (i-1)/2;
    }
}
```

Replacing the root with the last item preserves completeness; sift-down restores order. `maximum` is $O(1)$; insert, extract, and increase-key are $O(\log n)$.

8.6 Build a heap in $O(n)$

Every index from `floor(n/2)` onward in a 1-based array is a leaf and already a one-element heap. Process internal nodes bottom-up:

```
void buildMaxHeap(vector<int>& a) {           // 0-based
    for (int i = (int)a.size()/2 - 1; i >= 0; --i)
        maxHeapify(a, a.size(), i);
}
```

Correctness invariant

Before processing index `i`, every subtree rooted at an index greater than `i` is a heap. Thus the child subtrees of `i` are heaps, exactly the precondition for `maxHeapify`. After it runs, the subtree at `i` is also a heap. When `i=0` finishes, the whole array is a heap.

Tight running-time derivation

There are $O(n)$ calls and each is at most $O(\log n)$, so $O(n \log n)$ is a valid but loose upper bound. Most calls are near leaves and move zero or one level.

At most $\lceil n/2^{h+1} \rceil$ nodes have height h . Therefore

$$T(n) \leq \sum_{h=0}^{\lfloor \log n \rfloor} \left\lceil \frac{n}{2^{h+1}} \right\rceil O(h) = O\left(n \sum_{h \geq 0} \frac{h}{2^{h+1}}\right) = O(n),$$

because the infinite series converges to a constant. Reading the input is $\Omega(n)$, so bottom-up build-heap is $\Theta(n)$.

Starting empty and inserting all n elements is correct but costs $O(n \log n)$ worst case.

8.7 Heapsort

```
void heapSort(vector<int>& a) {
    buildMaxHeap(a);                // Theta(n)
    for (int end = (int)a.size() - 1; end > 0; --end) {
        swap(a[0], a[end]);         // maximum reaches final place
        maxHeapify(a, end, 0);      // live heap is [0,end)
    }
}
```

Loop invariant: $a[0..end]$ is a max-heap and $a[end+1..n-1]$ is sorted in final positions, with every suffix element at least every live-heap element. Each iteration extends the suffix by one maximum.

Total time is $\Theta(n) + (n-1)O(\log n) = O(n \log n)$, with a matching comparison-sort lower bound. It is in-place with iterative heapify and is not stable. The slides note that quicksort often wins in practice because of locality and constants.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 1297–1446. Original section numbers are retained.

9. Graph language and representations

9.1 Definitions

A graph is $G = (V, E)$, a set of vertices and edges.

- **Undirected edge** $\{u, v\}$: symmetric connection.
- **Directed edge** (u, v) : goes from u to v .
- **Weighted graph**: edges carry costs, distances, capacities, etc.
- **Adjacent vertices / incident edge**.
- **Degree** in an undirected graph: incident-edge count. In a digraph distinguish indegree and outdegree.
- **Walk**: vertices/edges may repeat. **Trail**: no repeated edge. **Path**: normally no repeated vertex. **Cycle**: closed nontrivial path.
- **Connected**: every pair has a path in an undirected graph. A disconnected traversal forms a forest.
- **Subgraph**: selected vertices and edges from another graph.
- **Tree**: connected acyclic undirected graph; equivalently, a connected n -vertex graph with $n - 1$ edges, or one with a unique simple path between every pair.
- **Simple graph**: no self-loops or parallel edges.

Handshake facts

For an undirected graph,

$$\sum_{v \in V} \deg(v) = 2|E|,$$

because every edge contributes to two endpoint degrees. Therefore the number of odd-degree vertices is even.

For a directed graph,

$$\sum_v \text{indeg}(v) = \sum_v \text{outdeg}(v) = |E|.$$

9.2 Representations

Adjacency matrix

An $|V| \times |V|$ matrix stores whether or what edge connects each ordered pair.

- space $\Theta(V^2)$;
- edge-existence test $\Theta(1)$;
- enumerate neighbors of one vertex $\Theta(V)$;
- good for dense graphs or matrix-based algorithms;
- symmetric for a simple undirected graph.

Adjacency list

For each vertex store its outgoing neighbors.

- space $\Theta(V + E)$ for directed graphs and $\Theta(V + 2E) = \Theta(V + E)$ for undirected;
- enumerate neighbors in $\Theta(\deg(v))$;
- edge test $O(\deg(v))$ with a simple list, faster with a hash/set variant;
- good for sparse graphs and BFS/DFS.

Edge list

Store `(u, v[, weight])` records: $\Theta(E)$ space, excellent for algorithms that scan/sort all edges, poor for repeated neighbor queries.

Trap: saying BFS is $O(V + E)$ assumes adjacency lists. With an adjacency matrix, scanning a row for every visited vertex is $O(V^2)$.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 1447–1502. Original section numbers are retained.

10. BFS, DFS, bipartite testing, cycles, and topological ordering

10.1 Breadth-first search

BFS expands the frontier by distance from a source. Colors encode state:

- white: undiscovered;
- gray: discovered/enqueued but not fully scanned;
- black: fully scanned.

```
struct BFSResult {
    vector<int> distance;    // -1 means unreachable
    vector<int> parent;
};

BFSResult bfs(const vector<vector<int>>& adj, int s) {
    int n = adj.size();
    vector<int> color(n, 0), d(n, -1), parent(n, -1);
    queue<int> q;
    color[s] = 1; d[s] = 0; q.push(s);
```

```

while (!q.empty()) {
    int u = q.front(); q.pop();
    for (int v : adj[u]) {
        if (color[v] == 0) {
            color[v] = 1;           // mark when enqueued
            d[v] = d[u] + 1;
            parent[v] = u;
            q.push(v);
        }
    }
    color[u] = 2;
}
return {d, parent};
}

```

Mark when enqueueing, not when dequeuing; otherwise multiple neighbors may enqueue the same vertex.

Worked trace

Edges: A-B, A-C, B-D, C-D, C-E, source A.

```

start: Q=[A], d(A)=0
remove A: discover B,C -> Q=[B,C], d=1
remove B: discover D -> Q=[C,D], d(D)=2
remove C: D known; discover E -> Q=[D,E], d(E)=2
remove D, E: done

```

One valid BFS tree has parents $B \leftarrow A$, $C \leftarrow A$, $D \leftarrow B$, $E \leftarrow C$. Parent choice can depend on adjacency-list order, but distances do not.

Shortest-path property and queue invariant

In an unweighted graph, $d[v]$ equals the minimum number of edges from s to v . The lecture proves a useful queue invariant: queued distances are nondecreasing, and the last queued distance is at most the first queued distance plus 1. Newly discovered neighbors receive $d[u]+1$, so layers are processed in order. If there were a shorter path to v , its previous vertex would have been processed early enough to discover v at that shorter distance—a contradiction.

Reconstruct a path by following parents from target to source and reversing. If distance is -1 , no source-to-target path exists.

Complexity

Initialization is $O(V)$. Each vertex is enqueued/dequeued once; every adjacency-list entry is scanned once (directed) or twice (undirected). Thus time is $O(V + E)$ and auxiliary space is $O(V)$.

To traverse a disconnected graph, start BFS from every still-white vertex; the result is a BFS forest.

10.2 Depth-first search

DFS follows one path as deeply as possible, then backtracks.

```

int timer = 0;
vector<int> color, parent, discover, finish;

void dfsVisit(int u, const vector<vector<int>>& adj) {
    color[u] = 1;
    discover[u] = ++timer;
    for (int v : adj[u]) {
        if (color[v] == 0) {
            parent[v] = u;
            dfsVisit(v, adj);
        }
    }
    color[u] = 2;
    finish[u] = ++timer;
}

void dfs(const vector<vector<int>>& adj) {
    int n = adj.size();
    color.assign(n, 0); parent.assign(n, -1);
    discover.resize(n); finish.resize(n); timer = 0;
    for (int u = 0; u < n; ++u)
        if (color[u] == 0) dfsVisit(u, adj);
}

```

Time is $O(V + E)$ with adjacency lists. Recursive space is $O(V)$ in the worst case; an explicit stack avoids call-stack overflow but not the need to remember the frontier.

Timestamp / parenthesis theorem

Each vertex has interval $[discover[u], finish[u]]$. For any two vertices, their intervals are either disjoint or one is nested inside the other. Vertex v is a descendant of u exactly when

$$d[u] < d[v] < f[v] < f[u].$$

This follows because a recursive call for u cannot finish until all recursive descendants finish.

White-path theorem

At discovery time of u , vertex v becomes a descendant of u iff a path from u to v consists entirely of white vertices at that moment (allowing u). This characterizes the DFS tree independent of syntax.

10.3 DFS edge classification

In a directed graph:

- **tree edge:** discovers a white vertex;
- **back edge:** goes to a gray ancestor;
- **forward edge:** goes to a black descendant but is not a tree edge;
- **cross edge:** connects other completed branches/subtrees.

A directed graph is acyclic iff DFS finds no back edge.

In an undirected DFS, every edge is a tree edge or a back edge when each physical edge is interpreted once. The apparent edge from a child back to its parent is the same undirected tree edge, so skip the parent when

detecting a cycle.

10.4 Cycle detection

Directed:

```
bool cycleDirected(int u, const vector<vector<int>>& adj, vector<int>& color) {
    color[u] = 1;
    for (int v : adj[u]) {
        if (color[v] == 1) return true;
        if (color[v] == 0 && cycleDirected(v, adj, color)) return true;
    }
    color[u] = 2;
    return false;
}
```

Undirected:

```
bool cycleUndirected(int u, int p, const vector<vector<int>>& adj,
                    vector<bool>& seen) {
    seen[u] = true;
    for (int v : adj[u]) {
        if (!seen[v]) {
            if (cycleUndirected(v, u, adj, seen)) return true;
        } else if (v != p) return true;
    }
    return false;
}
```

Run from every unvisited vertex. A plain “visited neighbor means cycle” test is wrong for directed graphs and mistakes the parent edge in undirected graphs.

10.5 Bipartite graph testing

A graph is bipartite iff its vertices can be colored with two colors so every edge has different endpoint colors. Equivalently, an undirected graph is bipartite iff it has no odd cycle.

```
bool isBipartite(const vector<vector<int>>& adj) {
    int n = adj.size();
    vector<int> side(n, -1);
    for (int s = 0; s < n; ++s) {
        if (side[s] != -1) continue;
        queue<int> q;
        side[s] = 0; q.push(s);
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (int v : adj[u]) {
                if (side[v] == -1) {
                    side[v] = side[u] ^ 1;
                }
            }
        }
    }
    return true;
}
```

```

        q.push(v);
        } else if (side[v] == side[u]) return false;
    }
}
}
return true;
}

```

BFS level parity supplies the colors. A same-color edge plus the two BFS-tree paths gives an odd cycle. Conversely, colors must alternate around every cycle, so an odd cycle cannot return consistently to its starting color.

Source correction: the one-slide code starts from one source. That is sufficient only for its connected component. The outer loop is required for a disconnected graph.

10.6 Topological ordering

A topological order is a linear order of a directed graph in which every edge $u \rightarrow v$ places u before v . It exists iff the graph is a DAG.

DFS method

Run cycle-detecting DFS and list vertices in decreasing finish time. For every DAG edge $u \rightarrow v$, DFS ensures $\text{finish}[u] > \text{finish}[v]$.

Kahn's algorithm

```

vector<int> topo(const vector<vector<int>>& adj) {
    int n = adj.size();
    vector<int> indeg(n, 0), order;
    for (int u = 0; u < n; ++u)
        for (int v : adj[u]) ++indeg[v];

    queue<int> q;
    for (int u = 0; u < n; ++u) if (indeg[u] == 0) q.push(u);
    while (!q.empty()) {
        int u = q.front(); q.pop();
        order.push_back(u);
        for (int v : adj[u]) if (--indeg[v] == 0) q.push(v);
    }
    if ((int)order.size() != n) throw logic_error("directed cycle");
    return order;
}

```

Invariant: the queue contains unprocessed vertices with no incoming edge from the remaining graph. Removing one is safe; decrementing neighbors simulates deleting its outgoing edges. Complexity is $O(V + E)$.

For the lexicographically smallest valid order, replace the FIFO queue with a min-heap. A normal DFS order does not generally guarantee lexicographic minimality.

10.7 BFS versus DFS

Question	BFS	DFS
Frontier structure	queue	call stack / explicit stack
Unweighted shortest paths	yes	no
Memory tendency	may store a wide level	may store a deep path
Natural applications	levels, minimum-edge path, bipartite	timestamps, cycles, topological structure
Complexity with lists	$O(V + E)$	$O(V + E)$

Neither is inherently “faster”; both scan the reachable representation once. Choose by the required property.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 1503–1735. Original section numbers are retained.

11. Searching and sorting

11.1 Linear and binary search

Linear search

```
int linearSearch(const vector<int>& a, int key) {
    for (int i = 0; i < (int)a.size(); ++i)
        if (a[i] == key) return i;
    return -1;
}
```

It works on unsorted sequential data. Best case is $\Theta(1)$; worst case is $\Theta(n)$; extra space is $\Theta(1)$. On a singly linked list this is the natural search because reaching the middle is not constant-time.

Binary search

Contract: `a` is sorted in nondecreasing order. Maintain a half-open candidate interval `[lo, hi)`.

```
int binarySearch(const vector<int>& a, int key) {
    int lo = 0, hi = a.size();
    while (lo < hi) {
        int mid = lo + (hi - lo) / 2;
        if (a[mid] < key) lo = mid + 1;
        else if (key < a[mid]) hi = mid;
        else return mid;
    }
    return -1;
}
```

Invariant: if `key` exists, at least one occurrence lies in `[lo, hi)`. Each comparison discards a region that cannot contain it. The interval length strictly decreases. Recurrence $T(n) = T(n/2) + \Theta(1) = \Theta(\log n)$.

Trace on `[2,5,8,12,16,23,38]` , key `16` :

```
[0,7), mid=3, a[3]=12 < 16 -> [4,7)
mid=5, a[5]=23 > 16      -> [4,5)
mid=4, a[4]=16           -> found
```

Use `lo + (hi-lo)/2` to avoid overflow. Binary search on a linked list is not useful in the usual form: locating each midpoint is linear, so the random-access benefit disappears.

Lower bound / insertion position

`lower_bound` returns the first index whose value is at least `key` :

```
int lowerBound(const vector<int>& a, int key) {
    int lo = 0, hi = a.size();
    while (lo < hi) {
        int mid = lo + (hi - lo) / 2;
        if (a[mid] < key) lo = mid + 1;
        else hi = mid;
    }
    return lo;
}
```

Invariant: all indices below `lo` are too small; all indices at or above `hi` are known candidates. `upper_bound` instead finds the first value strictly greater than the key.

11.2 Sorting vocabulary

- **Stable:** equal keys retain original relative order. Important when sorting records by multiple fields.
- **In-place:** uses $O(1)$ or very small auxiliary storage, depending on convention.
- **Adaptive:** runs faster when data is already/nearly sorted.
- **Comparison sort:** learns order only through comparisons.
- **Internal/external:** data fits in memory / requires external storage.

Comparison table

Algorithm	Best	Average	Worst	Extra space	Stable?	Main idea
Bubble	$O(n)$ with early stop	$O(n^2)$	$O(n^2)$	$O(1)$	yes	swap adjacent inversions
Selection	$\Theta(n^2)$	$\Theta(n^2)$	$\Theta(n^2)$	$O(1)$	normally no	select minimum for each slot
Insertion	$O(n)$	$O(n^2)$	$O(n^2)$	$O(1)$	yes	grow sorted prefix
Merge	$\Theta(n \log n)$	same	same	$\Theta(n)$	yes if ties take left	split, sort, merge

Algorithm	Best	Average	Worst	Extra space	Stable?	Main idea
Quicksort	$\Theta(n \log n)$	expected $\Theta(n \log n)$	$\Theta(n^2)$	expected $O(\log n)$ stack	normally no	partition about pivot
Heapsort	$\Theta(n \log n)$	same	same	$O(1)$	no	repeatedly extract heap root

The elementary three are included as recall prerequisites; the local algorithm PDF develops merge sort and quicksort in depth, while heap decks develop heapsort.

11.3 Insertion sort

```
void insertionSort(vector<int>& a) {
    for (int i = 1; i < (int)a.size(); ++i) {
        int key = a[i], j = i - 1;
        while (j >= 0 && a[j] > key) {
            a[j + 1] = a[j];
            --j;
        }
        a[j + 1] = key;
    }
}
```

Invariant: before iteration `i`, `a[0..i-1]` is a sorted permutation of its original elements. Shift larger elements right and insert `key` into its unique position. The number of shifts equals the number of inversions involving the inserted item; therefore it is adaptive and runs in $\Theta(n + I)$ where I is the inversion count.

11.4 Selection and bubble sort

```
void selectionSort(vector<int>& a) {
    for (int i = 0; i < (int)a.size(); ++i) {
        int m = i;
        for (int j = i + 1; j < (int)a.size(); ++j)
            if (a[j] < a[m]) m = j;
        swap(a[i], a[m]);
    }
}

void bubbleSort(vector<int>& a) {
    for (int end = a.size() - 1; end > 0; --end) {
        bool changed = false;
        for (int i = 0; i < end; ++i) {
            if (a[i] > a[i + 1]) {
                swap(a[i], a[i + 1]);
                changed = true;
            }
        }
    }
}
```

```

        if (!changed) break;
    }
}

```

Selection sort makes $\Theta(n^2)$ comparisons even when already sorted, but only $O(n)$ swaps. Bubble sort's pass moves the largest remaining value to **end**; with early stopping it detects sorted input in $O(n)$.

11.5 Merge sort

Divide into halves, recursively sort both, then merge two sorted ranges.

```

void mergeRanges(vector<int>& a, vector<int>& tmp, int l, int m, int r) {
    int i = l, j = m, k = l;
    while (i < m && j < r) {
        if (a[i] <= a[j]) tmp[k++] = a[i++]; // <= preserves stability
        else tmp[k++] = a[j++];
    }
    while (i < m) tmp[k++] = a[i++];
    while (j < r) tmp[k++] = a[j++];
    for (int p = l; p < r; ++p) a[p] = tmp[p];
}

void mergeSort(vector<int>& a, vector<int>& tmp, int l, int r) {
    if (r - l <= 1) return;
    int m = l + (r - l) / 2;
    mergeSort(a, tmp, l, m);
    mergeSort(a, tmp, m, r);
    mergeRanges(a, tmp, l, m, r);
}

```

Merge invariant: **tmp[l..k)** contains the smallest **k-l** elements of the two input prefixes in sorted order; **i** and **j** point to the smallest unmerged element of each half. Each merge costs $\Theta(r - l)$.

The handwritten trace begins with **[10,5,3,2,7,11,6,1]**: recursively obtain **[2,3,5,10]** and **[1,6,7,11]**, then merge to **[1,2,3,5,6,7,10,11]**.

Recurrence $T(n) = 2T(n/2) + \Theta(n) = \Theta(n \log n)$. Standard array merge sort uses $\Theta(n)$ auxiliary memory and is stable when equal values come from the left first.

11.6 Quicksort with Lomuto partition

```

int partition(vector<int>& a, int lo, int hi) { // inclusive hi
    int pivot = a[hi];
    int i = lo - 1;
    for (int j = lo; j < hi; ++j) {
        if (a[j] <= pivot) {
            ++i;
            swap(a[i], a[j]);
        }
    }
}

```

```

        swap(a[i + 1], a[hi]);
        return i + 1;
    }

    void quickSort(vector<int>& a, int lo, int hi) {
        if (lo >= hi) return;
        int q = partition(a, lo, hi);
        quickSort(a, lo, q - 1);
        quickSort(a, q + 1, hi);
    }

```

Partition invariant during the scan:

- $a[lo..i] \leq pivot$;
- $a[i+1..j-1] > pivot$;
- $a[j..hi-1]$ is unclassified;
- pivot remains at hi .

After the final swap, the pivot is in a valid final rank position. Balanced partitions give $T(n) = 2T(n/2) + \Theta(n) = \Theta(n \log n)$. Repeated extreme pivots give $T(n) = T(n-1) + \Theta(n) = \Theta(n^2)$. With distinct keys, randomly choosing the pivot gives expected $\Theta(n \log n)$ independent of original input order, but worst case remains quadratic. This two-way partition is still quadratic on all-equal input; see §17.1 for three-way partitioning.

Quicksort is normally in-place apart from its recursion stack and normally unstable. Tail-recursing on the smaller side and iterating over the larger limits stack depth to $O(\log n)$ even when partitions are poor.

11.7 Comparison-sort lower bound

A deterministic comparison sort corresponds to a decision tree whose leaves distinguish the $n!$ input permutations. A binary tree of height h has at most 2^h leaves, so

$$2^h \geq n! \Rightarrow h \geq \log_2(n!) = \Omega(n \log n).$$

Therefore no comparison sort can guarantee asymptotically better than $\Omega(n \log n)$ for arbitrary distinct keys. Counting/radix methods can beat it only by exploiting additional key structure, not comparisons alone.

Excerpt source: 01_DSA_I_SLIDE_COMPLETE.md, original lines 1736–1940. Original section numbers are retained.

12.1 Pattern and proof method

Divide-and-conquer algorithms:

1. **divide** an instance into smaller independent instances;
2. **conquer** them recursively;
3. **combine** their answers.

The common recurrence is $T(n) = aT(n/b) + f(n)$. Correctness normally follows by strong induction: recursive calls solve smaller inputs, and the combine step is proved to transform their answers into the full answer.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 1943–1952. Original section numbers are retained.

12.3 Maximum subarray by divide and conquer

Problem: find a contiguous nonempty subarray with maximum sum.

A maximum subarray lies entirely left, entirely right, or crosses the midpoint. A crossing optimum is the best suffix of the left half plus the best prefix of the right half.

```
struct Segment { long long sum; int l, r; }; // inclusive endpoints
```

```
Segment maxCrossing(const vector<int>& a, int l, int m, int r) {
    long long sum = 0, bestL = LLONG_MIN, bestR = LLONG_MIN;
    int leftIndex = m, rightIndex = m + 1;
    for (int i = m; i >= l; --i) {
        sum += a[i];
        if (sum > bestL) { bestL = sum; leftIndex = i; }
    }
    sum = 0;
    for (int j = m + 1; j <= r; ++j) {
        sum += a[j];
        if (sum > bestR) { bestR = sum; rightIndex = j; }
    }
    return {bestL + bestR, leftIndex, rightIndex};
}
```

```
Segment maxSubarrayDC(const vector<int>& a, int l, int r) {
    if (l == r) return {a[l], l, r};
    int m = l + (r-l)/2;
    Segment L = maxSubarrayDC(a, l, m);
    Segment R = maxSubarrayDC(a, m+1, r);
    Segment C = maxCrossing(a, l, m, r);
    return max({L,R,C}, [](const Segment& x, const Segment& y) {
        return x.sum < y.sum;
    });
}
```

Recurrence $T(n) = 2T(n/2) + \Theta(n) = \Theta(n \log n)$. The brute-force start/end enumeration is $O(n^2)$ if sums are extended incrementally, or $O(n^3)$ if every sum is recomputed. Kadane's DP later improves this to $O(n)$.

Edge case: do not initialize to zero if the subarray must be nonempty; an all-negative array should return its largest (least negative) element.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 1974–2013. Original section numbers are retained.

14.1 When DP applies

Dynamic programming solves problems with:

- **optimal substructure:** an optimal answer can be formed from optimal answers to subproblems;
- **overlapping subproblems:** the same state is requested repeatedly.

The design recipe:

1. define a state in a complete sentence;
2. identify the decision made at that state;
3. write a recurrence from smaller states;
4. specify base cases;
5. choose memoization or a valid table order;
6. recover choices if the actual solution—not only its value—is required;
7. count states times work per state and account for memory.

DP versus divide-and-conquer: D&C subproblems are usually independent; DP caches repeated states. **DP versus greedy:** DP considers competing choices and retains their best result; greedy commits to one choice and needs a special proof.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 2293–2311. Original section numbers are retained.

14.2 Fibonacci: the smallest overlap example

Naive recursion:

```
long long fib(int n) {
    if (n <= 1) return n;
    return fib(n-1) + fib(n-2);
}
```

It recomputes the same values and takes exponential time (more tightly $\Theta(\varphi^n)$), with $O(n)$ stack depth.

Memoization: top down

```
long long fibMemo(int n, vector<long long>& memo) {
    if (n <= 1) return n;
    if (memo[n] != -1) return memo[n];
    return memo[n] = fibMemo(n-1, memo) + fibMemo(n-2, memo);
}
```

Tabulation: bottom up

```
long long fibTab(int n) {
    if (n <= 1) return n;
    long long prev2=0, prev1=1;
    for (int i=2; i<=n; ++i) {
        long long cur=prev1+prev2;
        prev2=prev1; prev1=cur;
    }
    return prev1;
}
```

```

    }
    return prev1;
}

```

Both take $O(n)$ time. Memoization computes only reached states but pays recursion; tabulation controls order and can compress space to $O(1)$ because only two earlier states are needed.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 2312–2350. Original section numbers are retained.

14.3 Longest common subsequence (LCS)

A subsequence preserves order but need not be contiguous. Define

$$dp[i][j] = \text{length of an LCS of } X[0..i] \text{ and } Y[0..j].$$

Recurrence:

$$dp[i][j] = \begin{cases} 0, & i = 0 \text{ or } j = 0, \\ 1 + dp[i-1][j-1], & X[i-1] = Y[j-1], \\ \max(dp[i-1][j], dp[i][j-1]), & X[i-1] \neq Y[j-1]. \end{cases}$$

```

pair<int,string> lcs(const string& X, const string& Y) {
    int n=X.size(), m=Y.size();
    vector<vector<int>> dp(n+1, vector<int>(m+1));
    for (int i=1; i<=n; ++i) {
        for (int j=1; j<=m; ++j) {
            if (X[i-1]==Y[j-1]) dp[i][j]=1+dp[i-1][j-1];
            else dp[i][j]=max(dp[i-1][j],dp[i][j-1]);
        }
    }
    string answer;
    int i=n,j=m;
    while (i>0 && j>0) {
        if (X[i-1]==Y[j-1]) {
            answer.push_back(X[i-1]); --i; --j;
        } else if (dp[i-1][j] >= dp[i][j-1]) --i;
        else --j;
    }
    reverse(answer.begin(),answer.end());
    return {dp[n][m],answer};
}

```

For $X=ABCBDAB$, $Y=BDCABA$, the length is 4; valid LCSs include `BCBA` and `BDAB`. Multiple optimal answers are normal.

Correctness: if last characters match, an optimal solution can include that common character after an LCS of both prefixes. If not, any common subsequence omits at least one of those two last characters, so the maximum of the two smaller states is exhaustive.

Time and table space are $O(nm)$. If only length is needed, retain two rows for $O(\min(n, m))$ space; ordinary reconstruction needs the table or extra techniques.

Excerpt source: `01_DSA_I_SLIDE_COMPLETE.md`, original lines 2351–2396. Original section numbers are retained.

14.5 0/1 knapsack

Each item may be chosen at most once. Define

$dp[i][w]$ = maximum value using the first i items within capacity w .

For item $i-1$ of weight w_i and value v_i :

$$dp[i][w] = \begin{cases} dp[i-1][w], & w_i > w, \\ \max(dp[i-1][w], v_i + dp[i-1][w - w_i]), & w_i \leq w. \end{cases}$$

```
pair<int, vector<int>> knapsack01(const vector<int>& wt,
                                const vector<int>& val, int W) {
    int n=wt.size();
    vector<vector<int>> dp(n+1, vector<int>(W+1));
    for (int i=1; i<=n; ++i) {
        for (int w=0; w<=W; ++w) {
            dp[i][w]=dp[i-1][w];
            if (wt[i-1]<=w)
                dp[i][w]=max(dp[i][w], val[i-1]+dp[i-1][w-wt[i-1]]);
        }
    }

    vector<int> chosen;
    for (int i=n, w=W; i>0; --i) {
        if (dp[i][w] != dp[i-1][w]) {
            chosen.push_back(i-1);
            w -= wt[i-1];
        }
    }
    reverse(chosen.begin(), chosen.end());
    return {dp[n][W], chosen};
}
```

The image-only notes use capacity 5, weights `[2,3,4,5]`, values `[3,4,5,6]`. The optimum is 7 by taking weights 2 and 3. A brute-force subset search has 2^n choices; DP has $O(nW)$ time and $O(nW)$ table space.

One-dimensional compression and the crucial loop direction

```
vector<int> dp(W+1, 0);
for (int i=0; i<n; ++i)
    for (int w=W; w>=wt[i]; --w)
        dp[w]=max(dp[w], val[i]+dp[w-wt[i]]);
```

Capacity runs **downward** so an item is not reused in the same iteration. Running upward instead implements unbounded reuse. This is a classic viva trap.

Like coin change, $O(nW)$ is pseudo-polynomial.

Excerpt source: 01_DSA_I_SLIDE_COMPLETE.md , original lines 2426–2481. Original section numbers are retained.

14.12 Memoization versus tabulation

Point	Memoization	Tabulation
Direction	top down	bottom up
States	only reached states	usually all table states
Dependencies	implicit through recursion	must choose valid order
Overhead	function calls/stack	loops, often better locality
Reconstruction	store choices	store choices

Both implement the same recurrence. A memoized algorithm is not automatically polynomial: count the number of distinct states and work per state.

Excerpt source: 01_DSA_I_SLIDE_COMPLETE.md , original lines 2642–2655. Original section numbers are retained.

17. Seniors’ workbook supplement — search, sorting, and board simulations

Source: Varsity_Topics_List_In Memory of Masud.xlsx , DSA B7–B38, BRAC Ques Bank C21/C39, and Runtime Analysis image17 . Added September 2026. These are explicitly workbook-driven additions, not newly claimed slide pages.

17.1 A complete quicksort partition, including the duplicate-key trap

Use the Lomuto code in §11.6 on [4,2,7,3,1,5] , pivot 5 :

Scan	Decision	Array after decision	Last position <= pivot
j=0 , value 4	keep in left partition	[4,2,7,3,1,5]	0
j=1 , value 2	keep in left partition	[4,2,7,3,1,5]	1
j=2 , value 7	leave in right region	[4,2,7,3,1,5]	1
j=3 , value 3	swap positions 2 and 3	[4,2,3,7,1,5]	2
j=4 , value 1	swap positions 3 and 4	[4,2,3,1,7,5]	3
end	put pivot at position 4	[4,2,3,1,5,7]	pivot rank 4

Recursion now sorts [4,2,3,1] and [7] ; it never moves the rank-4 pivot again. On the left, pivot 1 gives [1,2,3,4] , then pivot 4 and pivot 3 finish it. The final array is [1,2,3,4,5,7] .

Randomization removes dependence on a predictable input order; it does not remove the quadratic worst case. The usual expected $\Theta(n\log n)$ proof assumes distinct keys or a partition scheme that handles equality appropriately. The two-way Lomuto code above is quadratic on **all equal keys even with random pivots**. Three-way partition fixes that particular problem:

```

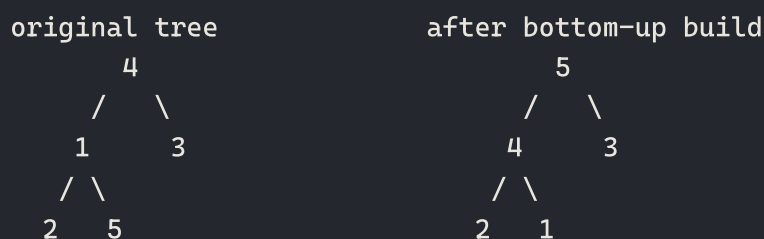
choose pivot value p
lt = lo; i = lo; gt = hi
while i <= gt:
    if a[i] < p: swap(a[lt], a[i]); lt++; i++
    else if a[i] > p: swap(a[i], a[gt]); gt--
    else: i++
recurse on [lo, lt-1] and [gt+1, hi]

```

Invariant: $< p \mid = p \mid \text{unknown} \mid > p$. Equal keys need no further sorting. All-equal input takes one linear scan. Ordinary two-way recursion can use $O(n)$ stack in its worst case; recurse on the smaller side and loop over the larger side to guarantee $O(\log n)$ stack.

17.2 Build-heap and heapsort are different operations: full trace

For `[4,1,3,2,5]`, zero-based child indices are `2i+1` and `2i+2`:



Operation	Heap prefix	Sorted suffix
sift down index 1	<code>[4,5,3,2,1]</code>	empty
sift down index 0	<code>[5,4,3,2,1]</code>	empty
swap root/end; sift size 4	<code>[4,2,3,1]</code>	<code>[5]</code>
swap root/end; sift size 3	<code>[3,2,1]</code>	<code>[4,5]</code>
swap root/end; sift size 2	<code>[2,1]</code>	<code>[3,4,5]</code>
swap root/end	<code>[1]</code>	<code>[2,3,4,5]</code>

`heapify(i)` assumes its child subtrees are already heaps and repairs **one** root, costing $O(h_i)$. `buildHeap` repairs roots in reverse level order, costing $\Theta(n)$ by the height sum in §8.6. Heapsort then performs repeated extractions, with $O(n \log n)$ worst-case time and $O(1)$ auxiliary space for iterative sift-down. Heap order alone is not sorted order.

17.3 Exponential search when the array length is unknown

Contract: random-access sorted data; `read(i)` returns an integer or `None` after the finite end. For a truly infinite sequence, termination needs a finite index whose value is at least the target. An infinite sequence of zeros cannot certify absence of 1 by this procedure. Never read outside a real C array.

```

def exponential_search(read, target):
    first = read(0)
    if first is None or first > target:
        return -1
    if first == target:

```

```

        return 0
    hi = 1
    while True:
        value = read(hi)
        if value is None or value >= target:
            break
        hi *= 2
    lo = hi // 2 + 1
    while lo <= hi:
        mid = lo + (hi - lo) // 2
        value = read(mid)
        if value is None or value > target:
            hi = mid - 1
        elif value < target:
            lo = mid + 1
        else:
            return mid
    return -1

```

For `[2,5,8,12,16,23,38,56,72]`, target 23, inspect indices `0,1,2,4,8`, then binary-search `[5,8]`: mid 6 gives 38; mid 5 gives 23. Doubling takes $O(\log(p+1))$ probes to bracket position p , then binary search takes the same order. Extra state is $O(1)$ machine words under bounded-index arithmetic; Python integers avoid index overflow but grow in bit length.

For one query in unsorted data use $O(n)$ linear search. For k queries compare $O(kn)$ with sorting plus search, $O(n\log n + k\log n)$, **or** an expected $O(n+k)$ hash-table strategy for equality only. Sorting is not automatically best for every repeated-query workload; ordered/range queries change the choice.

17.4 One buy, one later sell: explain the invariant, not “two pointers”

Given nonnegative prices, choose at most one purchase and one **later** sale. Brute force tries all ordered pairs in $O(n^2)$. For each sale day, the best purchase is simply the minimum price seen on an earlier day.

```

def max_profit(prices):
    if not prices:
        return 0
    lowest = prices[0]
    best = 0
    for day in range(1, len(prices)):
        price = prices[day]
        best = max(best, price - lowest)
        lowest = min(lowest, price)
    return best

```

Trace `[7,1,5,3,6,4]`:

Sale price	Earlier minimum	Candidate profit	Best
1	7	-6	0

Sale price	Earlier minimum	Candidate profit	Best
5	1	4	4
3	1	2	4
6	1	5	5
4	1	3	5

Invariant: after processing day j , **best** is the optimum over all sales up to j , and **lowest** is the minimum price through j . Every possible sale is considered with its best legal purchase, so the answer is globally optimal. $O(n)$ time, $O(1)$ space; a descending list returns 0. Fees, cooldowns, multiple transactions, or mandatory trading are different contracts, often needing DP.

17.5 Counting, radix, bucket, and Shell sort — missing screenshot topics

Stable counting sort

Keys must be integers in $[0, k)$. Count keys, compute cumulative end positions, then place records **right to left** to preserve equal-key order.

```
def counting_sort(a, k):
    count = [0] * k
    for x in a:
        if not 0 <= x < k:
            raise ValueError("key outside [0,k)")
        count[x] += 1
    for i in range(1, k):
        count[i] += count[i-1]
    out = [0] * len(a)
    for x in reversed(a):
        count[x] -= 1
        out[count[x]] = x
    return out
```

$[2, 1, 2, 0]$ gives counts $[1, 1, 2]$, cumulative counts $[1, 2, 4]$, output $[0, 1, 2, 2]$. If the 2s carry labels A then B, reverse placement keeps A before B. Time/space $\Theta(n + k)$. Huge key range can make it inappropriate.

LSD radix sort

For nonnegative integers, stably sort by units, tens, hundreds, and so on. After pass t , numbers are ordered by their lowest t digits.

```
def radix_sort(a, base=10):
    if base < 2 or any(x < 0 for x in a):
        raise ValueError("base >= 2; nonnegative integer keys")
    a = list(a)
    maximum = max(a, default=0)
    place = 1
    while maximum // place:
        buckets = [[] for _ in range(base)]
        for x in a:
```

```

        buckets[(x // place) % base].append(x)
    a = [x for bucket in buckets for x in bucket]
    place *= base
return a

```

```

[170, 45, 75, 90, 802, 24, 2, 66]
units:    [170, 90, 802, 2, 24, 45, 75, 66]
tens:     [802, 2, 24, 45, 66, 170, 75, 90]
hundreds: [2, 24, 45, 66, 75, 90, 170, 802]

```

With d digits and radix b , time $O(d(n + b))$, space $O(n + b)$. This code uses stable bucket appends; a stable counting pass is another implementation. The spreadsheet uses **k** ambiguously for both digits and range—define both instead.

Bucket sort

Assume independent uniform inputs in $[0, 1)$. Scatter into n ordered buckets, insertion-sort each, concatenate. Membership in earlier buckets implies a smaller value, so only within-bucket sorting is needed.

```

def bucket_sort(a):
    n = len(a)
    if n == 0:
        return []
    buckets = [[] for _ in range(n)]
    for x in a:
        if not 0 <= x < 1:
            raise ValueError("expected values in [0,1)")
        buckets[int(n*x)].append(x)
    out = []
    for bucket in buckets:
        for i in range(1, len(bucket)):
            x, j = bucket[i], i - 1
            while j >= 0 and bucket[j] > x:
                bucket[j+1] = bucket[j]
                j -= 1
            bucket[j+1] = x
        out.extend(bucket)
    return out

```

`[.42, .05, .91, .33]` enters buckets `[.05] | [.42, .33] | [] | [.91]`; sorting the second bucket yields `[.05, .33, .42, .91]`. Expected $\Theta(n)$ under the distribution assumption; worst $\Theta(n^2)$ when one bucket contains a badly ordered large cluster. Space $O(n)$. Stability depends on stable distribution and stable within-bucket sorting; it is not a universal property of the name.

Shell sort

Insertion-sort interleaved subsequences with decreasing gaps, ending in gap 1. Large gaps move distant out-of-order values earlier.


```
def shell_sort(a):
    gap = len(a) // 2
    while gap:
        for i in range(gap, len(a)):
            x, j = a[i], i
            while j >= gap and a[j-gap] > x:
                a[j] = a[j-gap]
                j -= gap
            a[j] = x
        gap //= 2
```

[8,5,3,7,6,2,1,4] becomes [6,2,1,4,8,5,3,7] after gap 4, [1,2,3,4,6,5,8,7] after gap 2, and sorted after gap 1. For this halving sequence: $O(n^2)$ worst case, $\Theta(n \log n)$ best case on already sorted input, $O(1)$ auxiliary space, generally unstable. Do not quote a universal average bound for Shell sort: the gap sequence matters.

Which sort for a sorted array or disk-resident data?

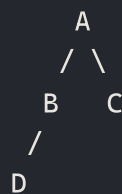
Already certified sorted? No sorting is needed. Need to verify it? One pass is $O(n)$. Nearly sorted? Insertion sort is adaptive. Data larger than RAM? External merge sort produces sorted runs and merges sequentially, giving good block-I/O locality. Ordinary array heapsort jumps around a heap and is **not** generally the preferred disk-sorting method merely because it is in-place.

17.6 Tree shapes and deletion complexity: fix the workbook shorthand

full, not complete



complete, not full



perfect



Full means every node has 0 or 2 children. Complete means every level is full except possibly the last, which is left packed. Perfect means all levels are full. A one-child chain is degenerate; consistently one-sided is skewed. Balance bounds height; it does **not** imply key ordering.

BST search/delete is $O(h)$, $O(n)$ for a chain and $O(\log n)$ when height is logarithmic. A simple splice can be $O(1)$ only after obtaining the node and the parent/link to update; finding a predecessor/successor can still cost $O(h)$. In an **unordered** binary tree, even balanced, locating a key can require inspecting all n nodes. There is no general $O(n \log n)$ balanced binary-tree deletion rule. Clarify what "delete" must preserve before choosing an implementation.

Underflow means removing/reading from an empty structure; capacity overflow means inserting into a full bounded structure. A dynamically allocated list can still fail allocation. Numeric underflow/overflow are different meanings.

Excerpt source: 01_DSA_I_SLIDE_COMPLETE.md, original lines 2873–3177. Original section numbers are retained.

5. Graph algorithms, hashing and amortized analysis

Workbook coverage: DSA B5/B44–B56; Board B9–B15; Runtime Analysis MST/shortest-path screenshots; hidden Niche Topics C4 (Johnson).

3.1 Disjoint-set union (union–find)

DSU maintains a partition under:

- make-set;
- find: representative of an element's set;
- union: merge two sets.

Use **union by rank/size** and **path compression** together:

```
struct DSU {
    vector<int> p, sz;
    explicit DSU(int n) : p(n), sz(n, 1) {
        iota(p.begin(), p.end(), 0);
    }
    int find(int x) {
        return p[x] == x ? x : p[x] = find(p[x]);
    }
    bool unite(int a, int b) {
        a = find(a); b = find(b);
        if (a == b) return false;
        if (sz[a] < sz[b]) swap(a, b);
        p[b] = a;
        sz[a] += sz[b];
        return true;
    }
};
```

A sequence of M operations on n elements takes $O(M\alpha(n))$, where inverse Ackermann α is below 5 for any practical input. This is amortized, not a claim that each individual operation is worst-case constant.

Undirected cycle detection with DSU — workbook DSA B5

Process each **undirected edge once**, not both copies in an adjacency list. Before inserting (u, v) , if $\text{find}(u) == \text{find}(v)$, an existing path already joins them, so the new edge closes a cycle. Otherwise merge their components.

```
bool hasUndirectedCycle(int n, const vector<pair<int, int>>& edges) {
    DSU d(n);
    for (auto [u, v] : edges)
        if (!d.unite(u, v)) return true;
    return false;
}
```

edge	components afterwards	cycle?
0--1	{0,1} {2} {3}	no
1--2	{0,1,2} {3}	no
2--0	0--1--2--0 closes a path	yes

Initialization $O(V)$, operations $O(E\alpha(V))$ amortized, storage $O(V)$. A self-loop is immediately detected; a second parallel edge closes a length-2 cycle under multigraph conventions. DSU cannot detect **directed** cycles: $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow C$ is a DAG although its underlying undirected graph is cyclic. Use gray-stack DFS or topological sorting for directed graphs.

The workbook's "constant versus near constant" needs two different functions: $\log^* n$ counts repeated base-2 logs until the result is at most 1 ($65536 \rightarrow 16 \rightarrow 4 \rightarrow 2 \rightarrow 1$, four logs). It is unbounded, unlike a constant. The tight standard DSU bound uses the still slower inverse Ackermann $\alpha(n)$, not literally $\log^* n$. Amortized bounds describe a sequence; one **find** need not take constant time.

Excerpt source: `02_DSA_II_SLIDE_COMPLETE.md`, original lines 345–411. Original section numbers are retained.

3.2 MST definitions and facts

For a connected, undirected, weighted graph, a **minimum spanning tree** (MST) is a spanning tree with minimum total edge weight.

- Negative edges are fine.
- If the graph is disconnected, the algorithms produce a minimum spanning forest.
- Distinct edge weights imply a unique MST; equal weights may still happen to have a unique MST.
- An MST minimizes total tree weight, not the path distance between every pair; shortest-path trees and MSTs solve different objectives.

A **cut** $(S, V - S)$ partitions vertices. An edge crosses it if its endpoints are on opposite sides.

Cut property. A lightest edge crossing any cut that respects a partial MST forest is safe to add. Exchange proof: take an MST T . If it lacks light edge e , adding e creates a cycle containing some other crossing edge f . Since $w(e) \leq w(f)$, replace f by e ; the result is no heavier and remains a tree.

Cycle property. A strictly heaviest edge on a cycle belongs to no MST. If an MST used it, replace it by a lighter edge from the same cycle.

These are the correctness engines behind Kruskal, Prim, and reverse-delete.

Excerpt source: `02_DSA_II_SLIDE_COMPLETE.md`, original lines 412–428. Original section numbers are retained.

3.3 Kruskal

Sort edges from light to heavy and add an edge iff it connects two different components.

```
struct WEdge { int u, v; long long w; };

pair<long long, vector<WEdge>>
kruskal(int n, vector<WEdge> edges) {
    sort(edges.begin(), edges.end(),
```

```

        [](const WEdge& a, const WEdge& b) { return a.w < b.w; });
DSU dsu(n);
long long total = 0;
vector<WEdge> chosen;
for (const auto& e : edges) {
    if (dsu.unite(e.u, e.v)) { // equivalent to find(u) != find(v)
        chosen.push_back(e);
        total += e.w;
        if ((int)chosen.size() == n - 1) break;
    }
}
if ((int)chosen.size() != n - 1)
    throw runtime_error("graph is disconnected");
return {total, chosen};
}

```

The set test must be unequal representatives. One compact slide version accidentally suggests the opposite; that would deliberately create cycles.

Invariant: chosen edges are a forest extendible to some MST. The next accepted edge is lightest across the cut defined by two current components, so the cut property makes it safe.

Complexity:

- sorting: $O(E \log E) = O(E \log V)$ for a simple graph;
- DSU: $O(E \alpha(V))$;
- total: $O(E \log E)$.

With small bounded integer weights, counting/radix sorting can reduce sorting, potentially to $O(E)$.

Tiny trace

Edges AB : 1, BC : 2, AC : 3, CD : 4, BD : 5:

1. accept AB ;
2. accept BC ;
3. reject AC because A, C are already connected;
4. accept CD .

The MST weight is $1 + 2 + 4 = 7$.

Excerpt source: `02_DSA_II_SLIDE_COMPLETE.md`, original lines 429–477. Original section numbers are retained.

3.4 Prim

Prim grows one tree. For each outside vertex v , its key is the lightest edge connecting it to the current tree.

```

pair<long long, vector<int>>
prim(const vector<vector<pair<int,int>>>& adj, int start = 0) {
    int n = adj.size();
    const long long INF = numeric_limits<long long>::max()/4;
    vector<long long> key(n, INF);

```

```

vector<int> parent(n, -1);
vector<char> used(n);
using State = pair<long long, int>;
priority_queue<State, vector<State>, greater<State>> pq;
key[start] = 0;
pq.push({0, start});
long long total = 0;
int taken = 0;

while (!pq.empty()) {
    auto [ku, u] = pq.top(); pq.pop();
    if (used[u] || ku != key[u]) continue; // stale lazy-heap entry
    used[u] = true;
    total += ku;
    ++taken;
    for (auto [v, w] : adj[u]) {
        if (!used[v] && w < key[v]) {
            key[v] = w;
            parent[v] = u;
            pq.push({key[v], v});
        }
    }
}
if (taken != n) throw runtime_error("graph is disconnected");
return {total, parent};
}

```

Invariant: vertices already extracted form a tree contained in an MST; every outside key is the cheapest crossing edge currently known. The extracted minimum is globally lightest across the tree/outside cut and is safe.

Complexity:

- adjacency matrix + array minimum: $O(V^2)$, often best for dense graphs;
- adjacency list + binary heap: $O((V + E)\log V)$, written $O(E\log V)$ for connected graphs;
- Fibonacci heap: $O(E + V\log V)$ amortized because decrease-key is $O(1)$ amortized.

Lazy Prim stores candidate edges and skips obsolete entries; **eager Prim** stores one best key per outside vertex. The code above is eager logically but uses duplicate heap entries as a practical substitute for decrease-key.

Excerpt source: `02_DSA_II_SLIDE_COMPLETE.md`, original lines 478–525. Original section numbers are retained.

3.6 MST comparison viva table

Question	Kruskal	Prim
Grows	forest of components	one tree
Main structure	sorted edge list + DSU	adjacency structure + min-PQ
Sparse graph	excellent	excellent

Question	Kruskal	Prim
Dense matrix graph	sorting all edges less attractive	$O(V^2)$ version excellent
Directed graph	not an ordinary MST problem	not an ordinary MST problem
Negative weights	allowed	allowed
Disconnected input	minimum spanning forest	restart per component for a forest

Excerpt source: `02_DSA_II_SLIDE_COMPLETE.md`, original lines 532–545. Original section numbers are retained.

4. Single-source shortest paths

4.1 Definitions and relaxation

For a path $p = \langle v_0, \dots, v_k \rangle$,

$$w(p) = \sum_{i=1}^k w(v_{i-1}, v_i).$$

The shortest-path distance $\delta(s, v)$ is the minimum path weight from s to v , or ∞ if unreachable. If a reachable negative-weight cycle can lead to v , there is no finite minimum: walks can loop around the cycle to decrease weight without bound.

Core facts:

- **Optimal substructure:** every subpath of a shortest path is itself shortest between its endpoints.
- **Triangle inequality:** $\delta(s, v) \leq \delta(s, u) + w(u, v)$.
- A shortest simple path has at most $V - 1$ edges when no reachable negative cycle is relevant.

Initialize

$$d[s] = 0, \quad d[v] = \infty \ (v \neq s), \quad \pi[v] = \text{NIL}.$$

Relaxation tests whether going through u improves v :

```
bool relax(int u, int v, long long w,
          vector<long long>& d, vector<int>& parent) {
    const long long INF = numeric_limits<long long>::max()/4;
    if (d[u] == INF || d[v] <= d[u] + w) return false;
    d[v] = d[u] + w;
    parent[v] = u;
    return true;
}
```

Relaxation never makes a tentative distance smaller than the true shortest distance if initialization was correct. Once a shortest path's edges have been relaxed in order, its destination has the correct distance.

4.2 Shortest paths in a DAG

Topologically sort the DAG, then relax every outgoing edge in that order. Every predecessor of v is handled before v , so when v is reached all possible last edges into it have been considered.

```

vector<long long> dagShortestPaths(
    const vector<vector<pair<int,int>>>& adj, int s,
    const vector<int>& topo) {
    const long long INF = numeric_limits<long long>::max()/4;
    vector<long long> d(adj.size(), INF);
    vector<int> parent(adj.size(), -1);
    d[s] = 0;
    for (int u : topo) {
        if (d[u] == INF) continue;
        for (auto [v, w] : adj[u])
            relax(u, v, w, d, parent);
    }
    return d;
}

```

Time $O(V + E)$. Negative edges are allowed because a DAG cannot contain a cycle.

4.3 Dijkstra

Assumption: every edge reachable from s has $w(e) \geq 0$. Zero is fine; one negative edge invalidates the greedy proof even if a particular run happens to return the right answer.

Algorithm: keep tentative distances in a min-priority queue. Extract the unsettled vertex of least distance and relax its outgoing edges.

```

vector<long long> dijkstra(
    const vector<vector<pair<int,int>>>& adj, int s,
    vector<int>& parent) {
    const long long INF = numeric_limits<long long>::max()/4;
    int n = adj.size();
    vector<long long> d(n, INF);
    parent.assign(n, -1);
    using State = pair<long long,int>;
    priority_queue<State, vector<State>, greater<State>> pq;
    d[s] = 0;
    pq.push({0, s});

    while (!pq.empty()) {
        auto [du, u] = pq.top(); pq.pop();
        if (du != d[u]) continue; // stale entry
        for (auto [v, w] : adj[u]) {
            if (w < 0) throw invalid_argument("Dijkstra needs nonnegative edges");
            if (d[v] > du + w) {
                d[v] = du + w;
                parent[v] = u;
                pq.push({d[v], v});
            }
        }
    }
}

```

```

    }
    return d;
}

```

Settling invariant. When u is extracted with minimum tentative distance, $d[u] = \delta(s, u)$.

Proof by contradiction: suppose the true shortest path to u first leaves settled set S on edge (x, y) . Then $x \in S$, so that edge has already been relaxed and $d[y] \leq \delta(s, y) \leq \delta(s, u)$, since all remaining edges are nonnegative. The queue chose u , so $d[u] \leq d[y]$. Together with $d[u] \geq \delta(s, u)$, equality follows.

The slide trace on the standard s, t, x, y, z example settles distances:

$$d(s) = 0, \quad d(y) = 5, \quad d(z) = 7, \quad d(t) = 8, \quad d(x) = 9.$$

Complexity:

- array / adjacency matrix: $O(V^2)$;
- adjacency list + binary heap: $O((V + E)\log V)$;
- Fibonacci heap: $O(E + V\log V)$ amortized.

Do not mark a vertex “final” when first inserted. A later relaxation may improve it; it becomes final only when the current minimum entry is extracted.

4.4 Bellman–Ford

Bellman–Ford permits negative edges and detects a **reachable** negative cycle.

```

struct BFEdge { int u, v; long long w; };

bool bellmanFord(int n, const vector<BFEdge>& edges, int s,
                 vector<long long>& d, vector<int>& parent) {
    const long long INF = numeric_limits<long long>::max()/4;
    d.assign(n, INF);
    parent.assign(n, -1);
    d[s] = 0; // Do this after initializing all vertices.

    for (int pass = 1; pass <= n - 1; ++pass) {
        bool changed = false;
        for (auto [u, v, w] : edges) {
            if (d[u] != INF && d[v] > d[u] + w) {
                d[v] = d[u] + w;
                parent[v] = u;
                changed = true;
            }
        }
        if (!changed) break;
    }
    for (auto [u, v, w] : edges)
        if (d[u] != INF && d[v] > d[u] + w)
            return false; // reachable negative cycle
    return true;
}

```


Invariant after pass i : every shortest path using at most i edges has been found. A finite shortest simple path uses at most $V - 1$ edges, hence $V - 1$ passes suffice. Any further improvement proves a reachable negative cycle.

Time $O(VE)$, space $O(V)$ beyond the edge list. Early termination improves easy instances but not the worst-case bound.

To recover an actual negative cycle, remember a vertex relaxed on the V -th pass, follow parent pointers V times to enter the cycle, then follow parents until it repeats.

4.5 Which SSSP algorithm?

Situation	Algorithm	Time
Unweighted / equal weights	BFS	$O(V + E)$
DAG, even negative edges	topological relaxation	$O(V + E)$
Nonnegative weights	Dijkstra	$O(E \log V)$ with binary heap
Negative edges, no reachable negative cycle	Bellman–Ford	$O(VE)$
Weights only 0 or 1	0–1 BFS with deque	$O(V + E)$

An MST does not answer SSSP: the globally cheapest tree may omit a direct edge needed for a shortest route from one source.

Excerpt source: `02_DSA_II_SLIDE_COMPLETE.md`, original lines 546–712. Original section numbers are retained.

5.2 Floyd–Warshall

Define $D^{(k)}[i, j]$ as the shortest path from i to j whose internal vertices are drawn only from $\{1, \dots, k\}$. Either the path avoids k , or it passes through k :

$$D^{(k)}[i, j] = \min \left(D^{(k-1)}[i, j], D^{(k-1)}[i, k] + D^{(k-1)}[k, j] \right).$$

```
vector<vector<long long>>
floydWarshall(vector<vector<long long>> d) {
    const long long INF = numeric_limits<long long>::max()/4;
    int n = d.size();
    for (int i = 0; i < n; ++i) d[i][i] = min(d[i][i], 0LL);
    for (int k = 0; k < n; ++k)
        for (int i = 0; i < n; ++i)
            for (int j = 0; j < n; ++j)
                if (d[i][k] != INF && d[k][j] != INF)
                    d[i][j] = min(d[i][j], d[i][k] + d[k][j]);
    return d;
}
```

The loop order must have k outermost. A compact slide uses inconsistent indices in the conditional; the correct update is $d[i][j]$ against $d[i][k] + d[k][j]$.

Time $O(V^3)$, space $O(V^2)$. Afterward, any $d[i][i] < 0$ signals a negative cycle reachable from and back to i .

Path reconstruction: maintain `next[i][j]=j` for an original edge. Whenever route through k improves (i,j) , set `next[i][j]=next[i][k]`; repeatedly follow next until j .

For transitive closure, replace min by OR and addition by AND:

$$reach[i][j] \leftarrow reach[i][j] \vee (reach[i][k] \wedge reach[k][j]).$$

Excerpt source: `02_DSA_II_SLIDE_COMPLETE.md`, original lines 733–769. Original section numbers are retained.

5.3 Johnson's algorithm

Johnson handles sparse directed graphs with negative edges but no negative cycle.

1. Add new source q and zero-weight edge $q \rightarrow v$ for every original vertex.
2. Run Bellman–Ford from q . If it detects a negative cycle, stop.
3. Let potential $h(v) = \delta(q, v)$.
4. Reweight every original edge:

$$w'(u, v) = w(u, v) + h(u) - h(v).$$

5. Run Dijkstra from every original source using w' .
6. Convert answers back:

$$\delta(u, v) = \delta'(u, v) - h(u) + h(v).$$

The signs matter. Several shorthand notes are easy to miscopy; these are the correct two formulas.

Why all reweighted edges are nonnegative:

$$h(v) \leq h(u) + w(u, v) \Rightarrow w(u, v) + h(u) - h(v) \geq 0.$$

Why shortest paths are preserved: for a path $p = u = v_0, \dots, v_k = v$, potentials telescope:

$$w'(p) = w(p) + h(u) - h(v).$$

Every u -to- v path gets the same endpoint-dependent offset, so their ordering is unchanged.

Complexity:

- Bellman–Ford: $O(VE)$;
- V binary-heap Dijkstra runs: $O(VE \log V)$;
- commonly stated as $O(VE + V^2 \log V)$ with Fibonacci heaps, or $O(VE \log V)$ with binary heaps on sparse graphs;
- space $O(V + E)$ plus the $O(V^2)$ output if all distances are stored.

Johnson worked simulation — hidden workbook sheet “Niche Topics”, C4

Consider edges `A→B:-2`, `B→C:3`, `A→C:4`, `C→A:1`. The cycle `A→B→C→A` weighs 2, not negative. A zero-edge super-source reaches every vertex. Bellman–Ford gives `h(A)=0`, `h(B)=-2`, `h(C)=0`.

Edge	Original weight	<code>h(u)-h(v)</code>	Rewighted
<code>A→B</code>	-2	2	0
<code>B→C</code>	3	-2	1
<code>A→C</code>	4	0	4

Edge	Original weight	$h(u)-h(v)$	Reweighted
C→A	1	0	1

Dijkstra from A returns reweighted distances $[0, 0, 1]$. Restore with $d(A, v) = d'(A, v) - h(A) + h(v)$ to obtain $[0, -2, 1]$. The route $A \rightarrow B \rightarrow C$ beats the direct $A \rightarrow C$ edge in both graphs.

```
Johnson(G):
    add q with zero edges to every v
    h = BellmanFord(G + q, q)
    if any negative cycle: report no finite all-pairs solution
    for every original edge (u,v):  $w_2(u,v) = w(u,v) + h(u) - h(v)$ 
    for each original source s:
        d2 = Dijkstra(G with  $w_2$ , s)
        for each v:  $answer[s,v] = d2[v] - h(s) + h(v)$ 
    return answer
```

For a general sparse graph including many isolated vertices, keep the vertex terms: binary-heap bound $O(V(V + E)\log V)$ plus the super-source Bellman–Ford work, not just $O(VE\log V)$ when E might be smaller than V . With Fibonacci heaps the conventional bound is $O(VE + V^2\log V)$. Explicitly storing all answers costs $\Theta(V^2)$ regardless of sparsity.

Excerpt source: [02_DSA_II_SLIDE_COMPLETE.md](#), original lines 770–848. Original section numbers are retained.

13.1 What it is—and is not

Amortized analysis bounds the total actual cost of **every** sequence of operations, then divides that total across the sequence. It is not:

- average-case analysis over an input distribution;
- expected analysis over random choices;
- permission to ignore a rare expensive operation.

An individual dynamic-array append may cost $\Theta(n)$, yet any n appends from empty cost $\Theta(n)$ total, hence $O(1)$ amortized per append.

Excerpt source: [02_DSA_II_SLIDE_COMPLETE.md](#), original lines 1940–1949. Original section numbers are retained.

13.2 The three methods

Aggregate method

Directly bound

$$\sum_{i=1}^m c_i \leq T(m).$$

Then average amortized cost is $T(m)/m$. Different operation types receive the same average charge.

Accounting method

Assign artificial charge $\hat{c}_i$. If $\hat{c}_i > c_i$, store the surplus as credit; later expensive operations spend it. Maintain nonnegative credit after every prefix:

$$\sum_{i=1}^k \hat{c}_i \geq \sum_{i=1}^k c_i \quad \text{for every } k.$$

Then total charged cost upper-bounds actual cost.

Potential method

Potential $\Phi(D_i)$ is stored energy in data-structure state after operation i :

$$\hat{c}_i = c_i + \Phi(D_i) - \Phi(D_{i-1}).$$

Telescoping gives

$$\sum_{i=1}^m \hat{c}_i = \sum_{i=1}^m c_i + \Phi(D_m) - \Phi(D_0).$$

If $\Phi(D_0) = 0$ and $\Phi(D_i) \geq 0$, total amortized cost upper-bounds actual cost.

Potential belongs to the **state**, not to the operation number. A negative potential can still be used if final potential is always at least initial potential, but nonnegative/zero-initial is the easiest safe design.

Excerpt source: 02_DSA_II_SLIDE_COMPLETE.md, original lines 1950–1991. Original section numbers are retained.

13.3 Binary counter

Increment a k -bit binary counter by flipping trailing 1s to 0 and the first 0 to 1.

Aggregate count: bit 0 flips n times, bit 1 at most $n/2$, bit 2 at most $n/4$, so

$$n + n/2 + n/4 + \dots < 2n.$$

Thus n increments cost $O(n)$, at most 2 amortized bit flips per increment.

Potential proof: Φ = number of 1 bits. If an increment clears t trailing ones and sets one zero:

$$c = t + 1, \quad \Delta\Phi = 1 - t, \quad \hat{c} = (t + 1) + (1 - t) = 2.$$

The expensive carry chain spends potential saved in those 1 bits.

Excerpt source: 02_DSA_II_SLIDE_COMPLETE.md, original lines 1992–2012. Original section numbers are retained.

13.4 Stack with multipop

Operations:

- PUSH costs 1;
- POP costs 1 when nonempty;
- MULTIPOP(k) pops $\min(k, |S|)$ elements.

No element can be popped more than once after being pushed. Across m operations, total pushes and pops are $O(m)$.

Accounting: charge PUSH 2—one pays now and one credit remains on the item to pay for its eventual pop. POP/MULTIPOP charge 0.

Potential: $\Phi(S) = |S|$. PUSH has amortized $1 + 1 = 2$; popping t items has $t - t = 0$.

Excerpt source: `02_DSA_II_SLIDE_COMPLETE.md`, original lines 2013–2026. Original section numbers are retained.

13.5 Dynamic array/table

Doubling

When an array of capacity m is full, allocate capacity $2m$, copy m items, then insert. During n appends, copying costs

$$1 + 2 + 4 + \dots < 2n.$$

Together with n writes, total is $O(n)$. A convenient potential in the at-least-half-full region is

$$\Phi = 2 \text{ num} - \text{size}.$$

Ordinary insertion builds potential; expansion consumes it.

Deletion and avoiding thrashing

If we double when full and halve immediately when half-full, alternating one insertion and one deletion around the threshold can resize every operation. Use hysteresis:

- grow when load reaches 1;
- shrink only when load reaches $1/4$, reducing capacity by half.

The post-shrink table is half full, leaving a large gap before the next resize. A standard nonnegative potential is

$$\Phi(T) = \begin{cases} 2 \text{ num} - \text{size}, & \text{num} \geq \text{size}/2, \\ \text{size}/2 - \text{num}, & \text{num} < \text{size}/2. \end{cases}$$

It proves $O(1)$ amortized insert and delete, while each resize remains $\Theta(n)$ actual.

Excerpt source: `02_DSA_II_SLIDE_COMPLETE.md`, original lines 2027–2063. Original section numbers are retained.

17. Hashing

17.1 The two-stage idea

A hash table maps a large key universe into array indices:

$$\text{key} \xrightarrow{\text{hash code}} \text{machine integer} \xrightarrow{\text{compression}} \{0, \dots, M-1\}.$$

The ideal practical hash is fast, deterministic for the lifetime of the table, uses the whole key, and spreads actual keys close to uniformly. It is impossible to avoid collisions when the key universe is larger than the table, so a collision-resolution policy is part of the data structure.

Load factor:

$$\alpha = \frac{N}{M},$$

where N is stored keys and M table slots/buckets.

17.2 Equality and hash-code contract

For Java-style keys:

$$x.\text{equals}(y) \Rightarrow x.\text{hashCode}() = y.\text{hashCode()}.$$

The converse is false: equal hash codes do not imply equal objects. Returning a constant hash is legal but destroys performance.

Rules for user-defined keys:

- include every field used by equality;
- use the same fields in equality and hashing;
- do not mutate a key in a way that changes its hash while it is in the table;
- recursively hash reference/array fields;
- handle null consistently.

A common combination is

$$h \leftarrow 31h + \text{fieldHash}.$$

For string $s_0s_1\dots s_{L-1}$, Horner's rule computes

$$h = s_0 31^{L-1} + s_1 31^{L-2} + \dots + s_{L-1}$$

with L multiply-add operations:

```
uint32_t stringHash(const string& s) {
    uint32_t h = 0;
    for (unsigned char c : s) h = 31u * h + c;
    return h; // unsigned overflow is defined modulo 2^32
}
```

Immutable strings can cache the result. A cached value of zero needs a separate “computed” bit if zero itself is a valid hash.

Java modular compression should not use plain absolute value: `abs(INT_MIN)` is still negative. A traditional nonnegative form is

```
(key.hashCode() & 0x7fffffff) % M
```

Modern code can use a floor-mod operation. For powers of two, libraries often mix high bits into low bits before masking.

17.3 Uniform hashing and unavoidable collisions

The **uniform hashing assumption** says each key is independently/equally likely to land in each bucket. It is an analysis model, not automatically guaranteed by choosing a prime.

Balls-and-bins intuition:

- birthday paradox: collisions appear after only $\Theta(\sqrt{M})$ random keys;
- coupon collector: filling every bucket takes about $M \ln M$ keys;
- after M random keys, maximum bucket load is about $\Theta(\log M / \log \log M)$.

A family of **universal hash functions** chooses a function randomly so that for distinct x, y ,

$$\Pr[h(x) = h(y)] \leq 1/M.$$

Randomizing the chosen function also prevents an adversary from knowing one permanent collision pattern.

17.4 Separate chaining

Each array cell points to a collection of all keys hashing there.

```
template<class K, class V, class Hash = std::hash<K>>
class ChainedHash {
    vector<list<pair<K,V>>> bucket;
    size_t count = 0;
    Hash hasher;

    size_t index(const K& key) const {
        return hasher(key) % bucket.size();
    }

    void rehashTo(size_t m) {
        vector<list<pair<K,V>>> old = move(bucket);
        bucket.assign(m, {});
        count = 0;
        for (auto& chain : old)
            for (auto& [k,v] : chain) put(k, v);
    }

public:
    explicit ChainedHash(size_t m = 8) : bucket(m) {}

    V* get(const K& key) {
        for (auto& [k,v] : bucket[index(key)])
            if (k == key) return &v;
        return nullptr;
    }

    void put(const K& key, const V& value) {
        auto& chain = bucket[index(key)];
        for (auto& [k,v] : chain)
            if (k == key) { v = value; return; }
        chain.push_front({key, value});
        ++count;
        if ((double)count / bucket.size() >= 8.0)
            rehashTo(2 * bucket.size());
    }

    bool erase(const K& key) {
        auto& chain = bucket[index(key)];
        for (auto it = chain.begin(); it != chain.end(); ++it)
            if (it->first == key) {
                chain.erase(it); --count; return true;
            }
    }
}
```

```

        return false;
    }
};

```

Under simple uniform hashing:

- expected chain length is $\alpha = N/M$;
- unsuccessful search is $\Theta(1 + \alpha)$;
- successful search is also $\Theta(1 + \alpha)$, roughly half a chain after the bucket access;
- worst case is $\Theta(N)$ when all keys collide.

The lecture resizing policy keeps average chain length in a constant band—for example double around $\alpha \geq 8$, halve around $\alpha \leq 2$. Rehash every key because the final index depends on new M .

17.5 Open addressing

All keys live in the table array. A probe function

$$h(k, i), \quad i = 0, \dots, M - 1$$

generates candidate slots. Necessarily $\alpha < 1$.

Search follows exactly the same probe sequence as insertion:

- stop with success on equal key;
- stop with failure at a never-used EMPTY cell;
- continue past a DELETED tombstone.

Linear probing

$$h(k, i) = (h(k) + i) \bmod M.$$

Excellent cache locality, but contiguous runs create **primary clustering**: longer clusters attract even more insertions.

Under the usual uniform-hashing approximation, average probes are:

$$\begin{aligned} \text{successful} &\approx \frac{1}{2} \left(1 + \frac{1}{1 - \alpha} \right), \\ \text{unsuccessful/insert} &\approx \frac{1}{2} \left(1 + \frac{1}{(1 - \alpha)^2} \right). \end{aligned}$$

At $\alpha = 1/2$, these are about 1.5 and 2.5. A compact summary source can be read as swapping these; the squared denominator belongs to unsuccessful search/insert.

Quadratic probing

$$h(k, i) = (h_1(k) + c_1 i + c_2 i^2) \bmod M.$$

It removes primary clustering but keys with the same initial hash still share a probe sequence (**secondary clustering**). Constants/table size must be chosen so enough slots are visited; a common guarantee uses prime M and keeps $\alpha \leq 1/2$.

Double hashing

$$h(k, i) = (h_1(k) + i h_2(k)) \bmod M.$$

Require $h_2(k) \neq 0$ and $\gcd(h_2(k), M) = 1$, so the sequence can visit every slot. For prime M , a common choice is

$$h_2(k) = 1 + (k \bmod (M - 1)).$$

Different keys usually get different step sizes, greatly reducing clustering.

Random probing

Use a key-determined pseudorandom permutation/offset sequence. Search must reproduce the same sequence; genuine fresh randomness on each search would make stored keys impossible to locate.

17.6 Linear-probing implementation and deletion

```
template<class K, class V, class Hash = std::hash<K>>
class LinearHash {
    enum State : unsigned char { EMPTY, OCCUPIED, DELETED };
    struct Slot { K key{}; V value{}; State state = EMPTY; };
    vector<Slot> a;
    size_t used = 0;           // occupied
    size_t filled = 0;         // occupied + tombstones
    Hash hash;

    size_t locate(const K& key) const {
        size_t m = a.size(), i = hash(key) % m;
        while (a[i].state != EMPTY) {
            if (a[i].state == OCCUPIED && a[i].key == key) return i;
            i = (i + 1) % m;
        }
        return m;
    }

    void rebuild(size_t m) {
        vector<Slot> old = move(a);
        a.assign(m, {});
        used = filled = 0;
        for (auto& s : old) if (s.state == OCCUPIED)
            put(move(s.key), move(s.value));
    }

public:
    explicit LinearHash(size_t m = 8) : a(m) {}

    V* get(const K& key) {
        size_t i = locate(key);
        return i == a.size() ? nullptr : &a[i].value;
    }

    void put(K key, V value) {
        if ((filled + 1.0) / a.size() > 0.5) rebuild(2 * a.size());
        size_t m = a.size(), i = hash(key) % m, firstDeleted = m;
        while (a[i].state != EMPTY) {
            if (a[i].state == OCCUPIED && a[i].key == key) {
                a[i].value = move(value); return;
            }
            if (a[i].state == DELETED) firstDeleted = i;
            i = (i + 1) % m;
        }
        a[i].key = move(key); a[i].value = move(value); a[i].state = OCCUPIED;
        used++; if (firstDeleted < filled) filled--;
    }
};
```

```

        if (a[i].state == DELETED && firstDeleted == m) firstDeleted = i;
        i = (i + 1) % m;
    }
    if (firstDeleted != m) i = firstDeleted;
    else ++filled;
    a[i] = {move(key), move(value), OCCUPIED};
    ++used;
}

bool erase(const K& key) {
    size_t i = locate(key);
    if (i == a.size()) return false;
    a[i].state = DELETED;
    --used;
    if (a.size() > 8 && used * 8 < a.size()) rebuild(a.size()/2);
    else if (filled > 2*used + 8) rebuild(a.size()); // clear tombstones
    return true;
}
};

```

Why simply clearing a deleted slot fails: a later key displaced past that slot would become unreachable because search stops at the first never-used empty slot. Alternatives:

- leave tombstones and periodically rebuild;
- for linear probing, remove and reinsert the following cluster until an empty slot.

Keep a low load, often at most $1/2$. The slides' grow-at- $1/2$, shrink-near- $1/8$ thresholds supply hysteresis.

Hashing corrections and tombstone simulation — workbook DSA B46–B56

For table size 7, $h(k)=k \bmod 7$, linear probing inserts keys 10, 17, 24:

index	0	1	2	3	4	5	6
after insert	.	.	.	10	17	24	.
delete 17	.	.	.	10	DEL	24	.
search 24:	check 3 -> skip DEL at 4 -> found at 5						

Changing slot 4 to "never used" would incorrectly terminate the search. Insertion may reuse **DEL**, but must still check the remainder of the probe sequence for an already-present equal key before inserting a duplicate map key.

The load factor is n/m . Chaining permits $n/m > 1$; conventional open addressing requires $n/m < 1$. Thresholds such as .9 or .5 are implementation policies, **not mathematical laws**. Tombstones affect effective probe occupancy even after live load falls. Growing/rebuilding costs $O(n)$ at that moment; controlled geometric resizing gives amortized bounds under the hashing assumptions.

"Uniform hashing" is a probability model, not a separate everyday algorithm: simple uniform hashing models uniform independent bucket assignments; uniform open-address probing is the stronger model of equally likely probe permutations. A practical hash need not satisfy these idealizations. Primary clustering appears in linear probing; quadratic probing can share probe sequences for equal initial hashes (secondary clustering). For double

hashing, $h_2(k)$ must be nonzero and coprime to m : with $m=7$, $step=3$, probes from 1 are 1,4,0,3,6,2,5; with $m=8$, $step=2$, only 1,3,5,7 are visited.

Perfect hashing is collision-free on the **chosen fixed key set**, not on every possible key. An absent key may map to an occupied slot, so verify stored-key equality. Compared with a direct-index array, hashing supports arbitrary sparse keys but adds hashing, metadata and collision costs; arrays have excellent locality and deterministic $O(1)$ indexed access. Neither is always more memory-efficient, and dynamic resizing is an implementation feature, not part of the abstract definition of a hash table.

17.7 Variants

- **Two-choice chaining:** hash to two buckets and insert in the shorter; maximum chain length drops dramatically, to about $O(\log\log N)$ under the model.
- **Cuckoo hashing:** each key has two candidate positions. Insertion may evict a resident to its alternative; lookup checks constant positions. A cycle triggers rehash. Expected constant operations, worst-case rebuild.
- **Perfect hashing:** for a fixed static set, two-level randomized hashing can give worst-case $O(1)$ lookup with linear expected space.
- **Robin Hood hashing:** when inserting, a key with larger probe distance steals a slot from one with smaller distance; it reduces variance.
- **Bloom filter:** not a symbol table, but a compact probabilistic membership filter. It has false positives but no false negatives if items are only inserted.

17.8 Security and cryptographic hashes

Known deterministic functions can enable **algorithmic complexity attacks**. Java's base-31 string scheme has families such as "Aa" and "BB" with equal hash codes; concatenating such blocks yields exponentially many colliding strings. If an attacker controls keys, a nominal $O(1)$ web-server table can degrade to $O(N)$ per operation.

Mitigations include secret per-process seeding/universal hashing, collision-bucket treeification, and resource limits.

A cryptographic hash aims for preimage/collision resistance and is intentionally more expensive than an ordinary table hash. MD5 and SHA-1 are obsolete for collision security. Passwords should not be stored as a fast unsalted hash at all; use a salted, slow password KDF such as Argon2, scrypt, bcrypt, or PBKDF2.

17.9 Hash table versus balanced tree

Need	Prefer
exact lookup, no adversarial concern	hash table, expected $O(1)$
guaranteed worst-case	AVL/RB tree, $O(\log n)$
sorted iteration, predecessor/successor, ranges	balanced BST
excellent array cache locality	open addressing
graceful high load/easy deletion	chaining
attacker-controlled keys	randomized hash or worst-case tree fallback

Hashing is good for equality lookup, not ordering. A prime table size can help some compression patterns but cannot rescue a poor hash that ignores important key structure.

Excerpt source: 02_DSA_II_SLIDE_COMPLETEE.md , original lines 2423–2778. Original section numbers are retained.

21. Coping with hardness: special cases, DP, approximation, and exact exponential algorithms

NP-hard does not mean “never solvable.” The slides emphasize:

- exploit a special graph class or small parameter;
- use dynamic programming/pseudo-polynomial time;
- use an approximation algorithm with proof;
- use a heuristic with no formal guarantee;
- use backtracking/branch-and-bound;
- use a faster exact exponential algorithm.

Excerpt source: 02_DSA_II_SLIDE_COMPLETEE.md , original lines 3372–3382. Original section numbers are retained.

21.5 Approximation versus heuristic

For minimization, an r -approximation satisfies

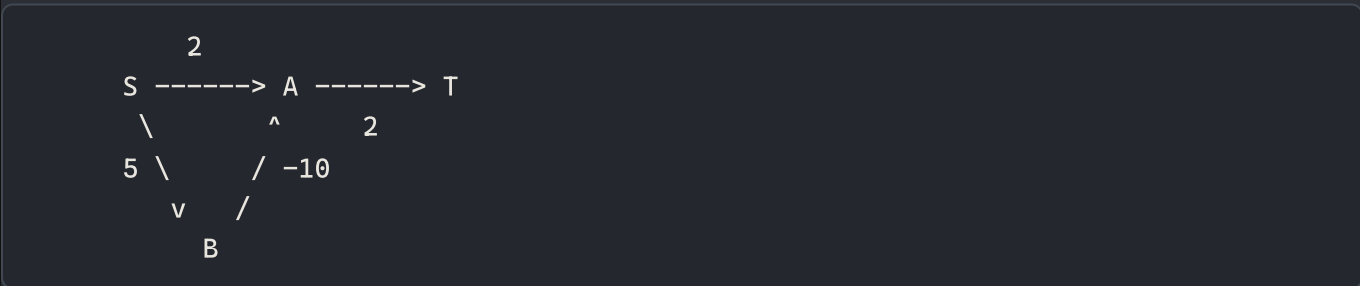
$$ALG(I) \leq r \cdot OPT(I).$$

For maximization, common conventions are $ALG \geq OPT/r$ or a $(1 - \epsilon)$ guarantee. A heuristic seeks a good answer quickly but has no proved worst-case quality. Both can be useful; only the approximation algorithm carries a theorem.

Excerpt source: 02_DSA_II_SLIDE_COMPLETEE.md , original lines 3452–3461. Original section numbers are retained.

24. Workbook board drill — negative edges, negative cycles, and runtime

Source: “Things to Explain on Board” B9–B15; BRAC Ques Bank C25; Runtime Analysis images 15/16. Rehearse the example before quoting a table.



There is no cycle. A settled-once Dijkstra run can finalize A at 2 and T at 4 before B reveals $A=-5$ and $T=-3$. The greedy assumption fails on negative edges; the existence of a cycle is not the deciding issue. The guarded Dijkstra code in §4.3 rejects negative edges instead of silently returning such an answer.

Bellman–Ford with edge order $A \rightarrow T$, $B \rightarrow A$, $S \rightarrow B$, $S \rightarrow A$:

Pass	$d(S)$	$d(A)$	$d(B)$	$d(T)$
initialize	0	∞	∞	∞
1	0	2	5	∞

Pass	$d(S)$	$d(A)$	$d(B)$	$d(T)$
2	0	-5	5	4
3	0	-5	5	-3
extra check	0	-5	5	-3

Add $A \rightarrow B:1$: the cycle $A \rightarrow B \rightarrow A$ weighs -9. Repeating it decreases walk cost without bound; affected reachable distances have no finite minimum. An extra Bellman–Ford relaxation after $V - 1$ passes detects this. To detect a cycle **anywhere**, not just from one source, add a super-source or initialize all distance labels to zero. To identify affected destinations, propagate forward from vertices improvable on the extra pass and mark them – **infinity**.

Floyd–Warshall also detects negative cycles through negative diagonal entries. An answer (i, j) is unbounded iff there is a vertex k with $d[k][k] < 0$, i reaching k , and k reaching j . Other pairs may still have valid finite answers. An undirected negative edge yields an arbitrarily repeatable negative walk by traversing it back and forth; “no negative simple cycle” is insufficient for the usual shortest-**walk** problem.

Floyd numerical mini-trace for $A \rightarrow B:2$, $B \rightarrow C:-1$, $A \rightarrow C:5$:

initial				after allowing B internally			
	A	B	C		A	B	C
A	0	2	5	A	0	2	1
B	∞	0	-1	B	∞	0	-1
C	∞	∞	0	C	∞	∞	0

$d[A,C] = \min(5, 2 + (-1)) = 1$. The outer k loop enforces the allowed-intermediate invariant; moving it innermost generally breaks the dynamic program.

Algorithm	Assumptions/implementation	Time	Auxiliary memory
BFS/DFS traversal	finite graph, adjacency list, visited set	$O(V + E)$	$O(V)$
BFS/DFS traversal	adjacency matrix	$O(V^2)$	$O(V)$ beyond matrix
Bellman–Ford	edge list, negative edges allowed	$O(VE)$	$O(V)$ beyond edges
Dijkstra	nonnegative, binary heap	$O((V + E)\log V)$	$O(V)$ indexed heap; lazy heap can use $O(E)$
Dijkstra	dense matrix, array min selection	$O(V^2)$	$O(V)$ beyond matrix
Prim	binary heap / Fibonacci heap	$O(E\log V)$ / $O(E + V\log V)$ on connected input	implementation-dependent heap storage
Kruskal	sort edges plus DSU	$O(E\log E)$	$O(V)$ DSU plus sorting storage
Floyd–Warshall	matrix	$O(V^3)$	$O(V^2)$

The screenshot’s **Prim requires connected input** is only for a single spanning tree: restart from each unvisited component to obtain a minimum spanning forest. The implementations in §3 deliberately throw for

disconnected input when their contract asks for **one** spanning tree. Do not compare one algorithm's auxiliary space with another's total graph storage.

Excerpt source: 02_DSA_II_SLIDE_COMPLETE.md , original lines 4334–4406. Original section numbers are retained.

6. Database topics from the interviews

Workbook coverage: DB tab is work-in-progress; BRAC C19 (ACID), C41 (NoSQL to SQL); Board B41 (SQL injection).

1.6 Relational versus non-relational

[SLIDE]

- A **relational database** represents data as relations/tables with rows and named columns; keys identify rows and constraints connect tables.
- Non-relational systems may use key–value, document (JSON/XML), wide-column, or graph models.

Do not answer “NoSQL has no schema.” Better: many NoSQL systems permit a flexible or application-enforced schema; they still have data shape and constraints, but may not enforce them relationally.

Excerpt source: 05_DBMS_SLIDE_COMPLETE.md , original lines 83–93. Original section numbers are retained.

2.1 Relation vocabulary

For schema Product(PName, Price, Category, Manufacturer):

- **Relation schema:** name plus attributes.
- **Relation instance:** current set of tuples.
- **Tuple:** one row/record.
- **Attribute:** named column.
- **Domain:** permitted atomic values for an attribute.
- **Degree/arity:** number of attributes.
- **Cardinality:** number of tuples.

In the mathematical relational model:

- a relation is a **set**, so duplicate tuples do not exist;
- tuples are unordered;
- attribute order is irrelevant under the named perspective;
- values are atomic in 1NF.

In SQL, query results are normally **bags/multisets**, so duplicates can occur unless DISTINCT or a duplicate-eliminating set operation is used.

Excerpt source: 05_DBMS_SLIDE_COMPLETE.md , original lines 96–116. Original section numbers are retained.

2.2 Keys—say these precisely

- **Superkey:** any attribute set that uniquely identifies every possible tuple.
- **Candidate key:** a **minimal** superkey; remove any attribute and uniqueness is lost.
- **Primary key:** candidate key chosen as the main identifier.
- **Alternate key:** candidate key not chosen as primary.
- **Composite key:** key containing multiple attributes.

- **Foreign key:** attributes in one relation whose non-NULL values must match a candidate/primary key in a referenced relation.
- **Surrogate key:** artificial identifier such as an auto-generated integer.
- **Prime attribute:** belongs to at least one candidate key.

“Minimal” means minimal by **set inclusion**, not necessarily fewest bytes.

Example:

Movie(title, year, length, genre)

Two films can share a title and many films share a year, but (title, year) may uniquely identify a film. Then:

- (title, year) is a candidate key;
- (title, year, genre) is a superkey but not a candidate key;
- title alone is not a key.

Excerpt source: *05_DBMS_SLIDE_COMPLETE.md*, original lines 117–139. Original section numbers are retained.

18.1 Transaction and ACID

[SLIDE] A transaction is a logical unit of program execution that reads and possibly updates database items.

Fund transfer of 50 from A to B:

```
read(A)
A = A - 50
write(A)
read(B)
B = B + 50
write(B)
commit
```

- **Atomicity:** all effects happen or none do. A crash after debit must not lose money.
- **Consistency:** if the transaction starts in a valid state and its logic is correct, it preserves declared/business invariants, e.g. $A+B$ unchanged. Consistency is a joint responsibility of constraints, correct transaction logic, and isolation—not magic performed solely by the DBMS.
- **Isolation:** concurrent outcome appears as allowed by the chosen isolation model, ideally equivalent to some serial execution.
- **Durability:** after commit is acknowledged, effects survive failures covered by the recovery model.

Atomicity concerns failure within a transaction; isolation concerns interference among transactions. Durability begins at commit; consistency is the application/schema property the other mechanisms help protect.

Excerpt source: *05_DBMS_SLIDE_COMPLETE.md*, original lines 2894–2916. Original section numbers are retained.

18.2 Transaction states

[SLIDE]


```

active -> partially committed -> committed -> terminated
|           |
+-----> failed -> aborted -> terminated
                \\-> restarted (new execution)

```

- **Active:** executing.
- **Partially committed:** last statement ran, but commit is not yet durable.
- **Committed:** commit record/effects satisfy durability protocol.
- **Failed:** cannot continue normally.
- **Aborted:** effects have been rolled back.

After abort, a transaction may be retried if failure was transient, or terminated if its input/logic is invalid. A compensating transaction semantically reverses an already committed business action; it is not identical to low-level rollback.

Excerpt source: 05_DBMS_SLIDE_COMPLETE.md, original lines 2917–2935. Original section numbers are retained.

26. Workbook interview case — migrate a document database to SQL

Source: BRAC Ques Bank C41. The DB sheet itself says only “Work in progress”; it is not evidence that DBMS is unimportant. The full SQL, indexing, ACID, normalization, concurrency and recovery material above remains essential.

Suppose a document stores:

```

{"order_id": 9, "customer": {"id": 2, "name": "Rina"},
 "items": [{"sku": "P1", "qty": 2, "unit_price": 120},
            {"sku": "P2", "qty": 1, "unit_price": 50}]}

```

Do not simply replace JSON syntax with a **CREATE TABLE**. Identify entities, relationships, keys, constraints and which values are historical snapshots.

```

Customer 1 ---- many OrderHeader 1 ---- many OrderLine many ---- 1 Product
customer_id      order_id              (order_id, line_no)      sku

```

```

CREATE TABLE Customer (
    customer_id INTEGER PRIMARY KEY,
    name VARCHAR(100) NOT NULL
);
CREATE TABLE Product (
    sku VARCHAR(40) PRIMARY KEY
);
CREATE TABLE OrderHeader (
    order_id INTEGER PRIMARY KEY,
    customer_id INTEGER NOT NULL REFERENCES Customer(customer_id)
);
CREATE TABLE OrderLine (

```

```

    order_id INTEGER REFERENCES OrderHeader(order_id),
    line_no INTEGER,
    sku VARCHAR(40) NOT NULL REFERENCES Product(sku),
    qty INTEGER NOT NULL CHECK (qty > 0),
    unit_price DECIMAL(12,2) NOT NULL CHECK (unit_price >= 0),
    PRIMARY KEY (order_id, line_no)
);

```

Keep **unit_price** on the line if it records the price **at purchase**; joining to the product's current price would rewrite history. **line_no** preserves array order and permits repeated SKUs. Use stable source IDs or an explicit mapping table, and decide how missing fields, explicit nulls, type inconsistencies, duplicate customers and unknown referenced products will be handled.

The example order total is $2 \cdot 120 + 1 \cdot 50 = 290$:

```

SELECT order_id, SUM(qty * unit_price) AS total
FROM OrderLine
GROUP BY order_id;

```

Migration sequence: profile data → agree on schema/mapping → stage and validate → load in dependency order → reconcile → catch up concurrent changes → cut over with a rollback plan. Reconcile counts, distinct keys, foreign-key orphans, totals and sampled records; order count alone does not prove line-item fidelity. Use idempotent batch keys so retries do not duplicate orders. For a live system, design a write pause or coordinated change-data capture; uncontrolled dual writes can diverge. SQL versus NoSQL does not by itself decide whether transactions, joins or consistency are supported; inspect the actual product/model.

26.1 Demo and follow-up questions

For a 4–5 minute demo, draw the single document, extract the four entities, show the 1:N relationships, and calculate the total. Save full production migration detail for Q&A. Expected follow-ups: why a composite key? Why keep price on the line? How to preserve nested-array order? What validates migration? How can a retry be safe? Which operation must be atomic?

Excerpt source: 05_DBMS_SLIDE_COMPLETE.md, original lines 3927–4001. Original section numbers are retained.

11.1 SQL injection

Cause: untrusted input changes SQL syntax.

Wrong:

```
String sql = "SELECT * FROM users WHERE email='" + email + "'";
```

Correct pattern:

```

PreparedStatement ps = connection.prepareStatement(
    "SELECT id, role FROM users WHERE email = ?"
);

```

```
ps.setString(1, email);
ResultSet rs = ps.executeQuery();
```

Parameterized queries separate code from data. Also use least-privilege DB accounts and allow-list identifiers when a value cannot be parameterized, such as a selected sort column.

Excerpt source: [13_SECURITY_CORE_COMPLETE.md](#), original lines 379–400. Original section numbers are retained.

19.4 SQL injection, prepared statements, and identifiers

Parameter binding keeps values out of SQL grammar:

```
row = db.execute(
    "SELECT id, role FROM users WHERE email = ?",
    (email,)
).fetchone()
```

Parameters cannot usually stand for table/column/order keywords. Map an external choice to a fixed allow-list:

```
columns = {"newest": "created_at DESC", "price": "price ASC"}
order_sql = columns.get(user_choice)
if order_sql is None:
    raise ValueError("invalid sort")
query = "SELECT id, price FROM products ORDER BY " + order_sql
```

Stored procedures are safe only if they avoid unsafe dynamic SQL. Input escaping is database/encoding-specific and inferior to separating code/data.

Excerpt source: [13_SECURITY_CORE_COMPLETE.md](#), original lines 923–945. Original section numbers are retained.

7. Software engineering

Workbook coverage: BRAC C5 (Agile versus waterfall); BRAC C11 professional ethics.

2. What software engineering is

Software engineering is the systematic, disciplined, and measurable application of engineering principles to the development, operation, maintenance, and evolution of software.

Programming produces code; software engineering also manages:

- stakeholder needs and changing requirements;
- architecture, interfaces, data, and quality attributes;
- teamwork, process, estimates, schedule, cost, and risk;
- verification, validation, deployment, operation, maintenance;
- security, privacy, safety, ethics, and retirement.

Software does not physically wear out like hardware, but it degrades structurally when repeated changes accumulate coupling, duplication, obsolete assumptions, and technical debt.

2.1 The framework activities from the slides

```
communication -> planning -> modeling -> construction -> deployment
                        ^                               |
                        +----- feedback -----+
```

- **Communication:** understand stakeholders and scope.
- **Planning:** estimate, schedule, assign resources, track risk.
- **Modeling:** analyze requirements and design solution.
- **Construction:** code and test.
- **Deployment:** deliver, support, collect feedback, evolve.

Umbrella activities such as quality assurance, configuration management, measurement, review, documentation, and risk management span the lifecycle.

Excerpt source: 08_SOFTWARE_ENGINEERING_SLIDE_COMPLETE.md , original lines 24–55. Original section numbers are retained.

3. SDLC and process models

3.1 Seven-phase SDLC sequence in the Elin slides

1. **Planning:** define problem, feasibility, scope, resources, milestones.
2. **Analysis:** elicit, understand, prioritize, and document requirements.
3. **Design:** architecture, data, interfaces, components, processes.
4. **Development:** build technical environment, database, and program.
5. **Coding/testing:** implement and verify units/integration/system/acceptance.
6. **Implementation:** deploy, document, train, and migrate users/data.

7. **Maintenance:** correct, adapt, perfect, and prevent future problems.

These are activities, not always rigid one-time boxes. Iterative methods revisit them.

3.2 Scope creep and feature creep

- **Scope creep:** uncontrolled expansion of agreed project scope.
- **Feature creep/gold plating:** extra features are added without validated need or planned tradeoff.

Control with a baseline, traceable change request, impact analysis, prioritization, stakeholder decision, and updated schedule/budget/test plan. "Never change requirements" is not realistic; unmanaged change is the problem.

3.3 Implementation/conversion strategies

Strategy	Mechanism	Advantage	Risk/cost
Direct/plunge	stop old and start new	fast, low parallel cost	highest failure/recovery risk
Parallel	old and new run together	fallback and result comparison	duplicate cost/work, inconsistency
Pilot	deploy to limited group/site	limits blast radius	pilot may not represent all users
Phased	roll out modules/groups gradually	controlled learning	integration/mixed-version complexity

3.4 Waterfall

Sequential phase gates work best when requirements are stable, technology is understood, and strong documentation/regulation matters. Benefits: clear milestones and artifacts. Weaknesses: late feedback/testing, expensive requirement change, and an apparently complete document can hide misunderstanding.

3.5 Incremental and iterative

- **Incremental:** deliver usable slices over time.
- **Iterative:** revisit/refine the solution repeatedly.

Many processes are both. Early value/feedback and reduced one-shot risk are benefits; architecture, integration, and cross-cutting quality need deliberate coordination.

3.6 Prototyping

Useful when requirements or interaction are unclear. A throwaway prototype learns and is discarded; an evolutionary prototype becomes the product and therefore must be engineered, not casually promoted from demo code. Users may mistake UI progress for a production-ready backend/security system.

3.7 Spiral

Risk-driven cycles:

```
set objectives/alternatives
-> identify and resolve risks
-> develop/verify next level
-> plan next iteration
```

Strong for large high-risk projects; costly/complex and requires risk expertise.

3.8 V-model

Pairs development artifacts with corresponding test levels: requirements ↔ acceptance, system design ↔ system test, architecture ↔ integration, module design/code ↔ unit test. Its value is early test planning and traceability; it remains less flexible when interpreted rigidly.

Excerpt source: 08_SOFTWARE_ENGINEERING_SLIDE_COMPLETE.md , original lines 56–119. Original section numbers are retained.

4. Agile, XP, and Scrum

4.1 Agile is not “no plan or documentation”

Agile values individuals/interactions, working software, customer collaboration, and responding to change, while still valuing the items on the other side. It uses short feedback loops and adaptive plans. The correct amount of documentation is the amount needed for value, safety, compliance, onboarding, maintenance, and coordination.

The slide sequence gives all twelve principles. Recall them by purpose rather than as disconnected slogans:

1. satisfy the customer through early and continuous delivery of valuable software;
2. welcome changing requirements, even late, when change gives the customer an advantage;
3. deliver working software frequently, preferring shorter feedback intervals;
4. business/client representatives and developers collaborate regularly throughout the project;
5. build around motivated people, give them support and environment, and trust them;
6. prefer the richest effective direct communication—the original principle names face-to-face conversation;
7. use working software as the primary progress measure;
8. maintain a sustainable pace rather than repeated burnout;
9. continuously attend to technical excellence and good design;
10. practice simplicity: maximize work deliberately not done;
11. let strong architecture, requirements, and design emerge from self-organizing teams;
12. reflect regularly, then tune behavior and process.

These principles do not ban plans, documentation, architecture, or remote tools. They subordinate them to feedback, working outcomes, sustainable delivery, and the needs of the system and organization.

4.2 Extreme Programming (XP)

Practices include user stories/planning game, small releases, simple design, test-first/TDD, refactoring, pair programming, continuous integration, collective ownership, coding standards, and sustainable pace.

Pair programming has driver and navigator roles and is a continuous review/knowledge-sharing practice. It does not mean two people independently type the same code.

4.3 Scrum accountabilities

- **Product Owner:** maximizes value and orders product backlog.
- **Scrum Master:** helps the team understand/improve Scrum and remove impediments; not merely a command-and-control manager.
- **Developers:** create the usable increment and manage sprint work.

Artifacts/commitments:

- product backlog / product goal;

- sprint backlog / sprint goal;
- increment / definition of done.

Events:

Sprint Planning -> Sprint with Daily Scrums
-> Sprint Review -> Retrospective -> next Sprint

- Review inspects product/outcome with stakeholders.
- Retrospective improves team/process.
- Daily Scrum is for developers to inspect progress/adapt plan, not a status interrogation.

4.4 User story and acceptance criteria

As a <role>, I want <capability>, so that <benefit>.

Acceptance criteria make behavior testable:

Given an authenticated borrower with no overdue fine,
When they borrow an available book,
Then one active loan is created and inventory becomes unavailable.

INVEST heuristic: independent, negotiable, valuable, estimable, small, testable.

4.5 Kanban

Kanban literally evokes a visible signboard; in software work it is a continuous-flow method that makes work and bottlenecks visible. A card represents a task/subtask and its relevant requirement or user story. Cards are **pulled** to the next state when downstream capacity exists rather than pushed into an already overloaded stage.

Backlog -> Ready [WIP <= 3] -> Build [WIP <= 2]
-> Review/Test [WIP <= 2] -> Done

Core practices:

1. visualize the workflow and current cards;
2. limit work in progress (WIP);
3. manage flow and blocked work;
4. make entry/exit policies explicit;
5. use feedback loops such as stand-ups and retrospectives;
6. improve collaboratively and experimentally.

A WIP limit exposes congestion: when Review is full, developers should help unblock/review/test rather than start unlimited new work. Important flow measures are:

- **lead time:** request/commitment to delivery;
- **cycle time:** active start to completion;
- **throughput:** items completed per unit time;
- **WIP:** items started but not completed.

Under a stable system with consistent units, Little's Law gives $L = \lambda W$: average WIP equals average throughput times average flow time. It is a relationship for observing a stable flow, not permission to manipulate estimates mechanically.

Scrum	Kanban
timeboxed sprints and a sprint goal	continuous flow; no required sprint boundary
defined accountabilities/events	roles/events may be retained or adapted
sprint backlog is normally stabilized for the sprint	priorities/cards can change as capacity frees
controls commitment through sprint planning	controls overload primarily through pull and WIP limits

Both can use visual boards, daily coordination, retrospectives, user stories, and incremental delivery. Kanban's strengths are transparency, flexibility, and bottleneck control; without explicit policies, WIP limits, service expectations, and active flow management, a board can degrade into a passive to-do list with unpredictable completion.

Excerpt source: 08_SOFTWARE_ENGINEERING_SLIDE_COMPLETE.md , original lines 120–225. Original section numbers are retained.

17.4 Software engineering ethics principles

The slides organize professional duties around:

1. public interest;
2. client and employer, consistent with public interest;
3. product quality/professional standards;
4. independent professional judgment;
5. ethical management;
6. integrity of profession;
7. fairness/support for colleagues;
8. lifelong learning and ethical practice.

Ethical judgment is not a mechanical algorithm. For a conflict: identify stakeholders/harms, duties/law/policy, evidence/uncertainty, alternatives, proportional action, escalation, documentation, and protection of the public. Confidentiality does not justify concealing imminent serious harm or illegal behavior; follow lawful professional escalation.

Excerpt source: 08_SOFTWARE_ENGINEERING_SLIDE_COMPLETE.md , original lines 976–992. Original section numbers are retained.

8. Operating systems

Workbook coverage: OS B3–B15; BRAC C11/C13/C31/C37/C49: scheduling, deadlock, semaphores, page tables, replacement and context switching.

1. The operating-system mental model

P0 — What is an operating system?

An operating system is privileged software that sits between applications and hardware. It has two complementary roles:

1. **Resource manager:** allocates CPU time, memory, storage, and I/O devices while enforcing protection and fairness.
2. **Abstraction provider:** turns awkward hardware into useful abstractions—processes, virtual address spaces, files, sockets, and virtual machines.

Virtualization does not mean everything is a hardware VM. The OS virtualizes the CPU by rapidly scheduling processes, virtualizes memory through per-process address spaces, and virtualizes persistent storage through files and directories.

P0 — Kernel mode and user mode

The processor provides at least two privilege levels. Application code normally executes in **user mode**, where privileged instructions and arbitrary physical-memory access are forbidden. The kernel executes in **kernel/supervisor mode**.

A transition to the kernel can occur through:

- a **system call** deliberately requested by a program;
- an **exception/trap**, such as divide-by-zero or a page fault, caused synchronously by the current instruction;
- an **interrupt**, such as timer or I/O completion, caused asynchronously by hardware.

The CPU saves enough state to resume later, switches to a protected kernel entry point, and changes privilege. A system call is therefore not an ordinary function call even if a library exposes it as one.

Trap: do not say that every mode switch is a process context switch. A system call may enter and leave the kernel while returning to the same process. A context switch changes the running execution context and is usually more expensive.

P0 — Why is the timer interrupt essential?

Without a timer interrupt, a CPU-bound or malicious process could keep the CPU indefinitely in a preemptive system. The OS programs a timer; when it fires, control returns to the kernel, which may continue the process or schedule another one.

P1 — Common kernel structures

- **Monolithic kernel:** most services run in kernel space. Calls between subsystems are fast, but a bug has a large failure domain. Linux is monolithic but modular.
- **Microkernel:** keeps a small mechanism-focused kernel and moves services such as drivers/filesystems to user processes. Isolation and extensibility improve; IPC/context-transition overhead and design complexity

can increase.

- **Layered/modular/hybrid:** practical systems combine ideas rather than matching one pure category.

Mechanism answers **how** something can be done; policy answers **which choice** should be made. For example, a context-switch mechanism lets the OS change processes, while the scheduling policy chooses the next process.

Excerpt source: 06_OPERATING_SYSTEMS_SLIDE_COMPLETE.md, original lines 20–56. Original section numbers are retained.

2. Programs, processes, and threads

P0 — Program vs process

A program is passive code and static data stored in a file. A process is a running instance with execution state and resources: program counter, registers, address space, stack, heap, open files, credentials, and scheduling state. Multiple processes may run the same program but have separate state.

Typical process states are:

```
new -> ready -> running -> terminated
      ^       |
      |       +--> blocked/waiting --event completes--> ready
+-----+
|         |
+-----+ preemptio/time slice -----+
```

- **Ready:** can run but is waiting for CPU.
- **Blocked:** cannot make progress until an event, usually I/O or synchronization.

P0 — PCB and context switch

The **process control block** stores kernel metadata such as PID, state, saved registers/program counter, scheduling information, memory-map references, credentials, and open-resource references.

During a context switch, the kernel saves the outgoing context, chooses another runnable task, restores its context, and may switch address spaces. Costs include kernel work, pipeline disruption, cache/TLB disturbance, and lost locality. The exact hardware state is architecture-dependent.

P0 — Process vs thread

Threads within a process share code, heap, address space, and open resources, but each thread has its own program counter, registers, stack, and scheduling state.

Separate processes	Threads in one process
Stronger isolation	Cheap communication through shared memory
IPC is explicit	Races are easier to create
Failure often contained	One corrupting thread can damage the process
Separate address spaces	Shared address space

Use processes for isolation/security boundaries; use threads when concurrent activities need efficient shared state—while controlling synchronization.

P1 — User-level vs kernel-level threads

User-level threading can schedule without entering the kernel and can support custom runtimes. If the kernel sees only one schedulable entity, however, one blocking call may block the entire process and parallel execution on several cores may be unavailable. Kernel threads are visible to the OS and can run in parallel, but creation and switching involve kernel management. Modern systems/runtimes use various one-to-one or multiplexed mappings.

P1 — `fork`, `exec`, and `copy-on-write`

On Unix-like systems, `fork` creates a child process with a logically copied address space; implementations normally use **copy-on-write**, sharing physical pages read-only until one side writes. `exec` replaces the current process image with a new program while retaining the process identity and selected resources. A shell often performs `fork`, redirection setup, then `exec`.

Excerpt source: 06_OPERATING_SYSTEMS_SLIDE_COMPLETE.md, original lines 57–101. Original section numbers are retained.

3. CPU scheduling

P0 — Metrics

For process `i`:

```
turnaround_i = completion_i - arrival_i
waiting_i     = turnaround_i - total_CPU_burst_i
response_i    = first_run_i - arrival_i
```

Other goals include throughput, CPU utilization, deadline satisfaction, fairness, and low variance. Interactive systems value response time; batch systems may value throughput/turnaround; real-time systems value predictable deadline behavior.

P0 — Preemptive vs non-preemptive

- **Non-preemptive:** a running task keeps the CPU until it blocks, exits, or voluntarily yields.
- **Preemptive:** the OS may stop it, usually on a timer or higher-priority arrival.

Preemption improves responsiveness but adds synchronization concerns and switching overhead.

P0 — Scheduling algorithms

FCFS/FIFO

Runs in arrival order. It is simple and starvation-free, but a long CPU-bound job can delay short jobs—the **convoy effect**.

SJF and SRTF

Shortest Job First minimizes average waiting time when all jobs are available together and burst lengths are known. Shortest Remaining Time First is its preemptive form. Real systems estimate future bursts, often using exponential averaging:

```
tau_next = alpha * actual_burst + (1-alpha) * tau_previous
```

Long jobs may starve without aging.

Round Robin

Each ready task receives at most quantum q in cyclic order. A very large q approaches FCFS; a very small q improves responsiveness but raises context-switch overhead. It does not automatically minimize average waiting time.

Priority scheduling

Runs the highest-priority task. It may be preemptive or non-preemptive. Low-priority starvation can be reduced by **aging**, which gradually increases the priority of waiting tasks.

P0 — Multilevel Feedback Queue (MLFQ)

MLFQ tries to approximate SJF/SRTF without knowing burst lengths and preserve interactive response:

1. multiple queues have decreasing priority and usually increasing time quanta;
2. the scheduler chooses from the highest nonempty queue;
3. a job that uses its CPU allotment is demoted, suggesting CPU-bound behavior;
4. a job that blocks/yields after a short burst stays high under a carefully designed accounting rule;
5. periodic priority boosts prevent starvation and adapt to phase changes.

Gaming problem: if a task can yield just before its quantum expires and retain priority, it can game the policy. Track total CPU allotment at a level, not only uninterrupted bursts.

Board answer: draw three queues Q_0 , Q_1 , Q_2 with quanta 4, 8, 16; simulate an interactive job that repeatedly blocks and a CPU-bound job that gets demoted. Then mention periodic boost.

P1 — Multiprocessor scheduling

With several cores, the OS must balance load while preserving **processor affinity** because a task may have warm caches on its previous core. Per-core run queues reduce contention but require load balancing. NUMA systems also care which memory node holds a task's pages.

P1 — Real-time scheduling

- **Rate Monotonic:** fixed priority; shorter period means higher priority, under standard periodic-task assumptions.
- **Earliest Deadline First:** dynamic priority; earliest absolute deadline first. On an ideal preemptive uniprocessor under its assumptions, EDF can use the processor up to full utilization.

Real-time correctness concerns deadlines and bounded latency, not merely high average speed.

Excerpt source: 06_OPERATING_SYSTEMS_SLIDE_COMPLETE.md, original lines 102–171. Original section numbers are retained.

4. Concurrency and synchronization

P0 — Race condition and critical section

A race condition occurs when the result depends on an uncontrolled interleaving of concurrent operations. `counter++` is not necessarily atomic; it is conceptually read, modify, write. Two threads can lose an update.

A correct critical-section solution aims for:

- **mutual exclusion:** at most one participant inside;
- **progress:** if nobody is inside, eligible contenders can decide without irrelevant indefinite delay;
- **bounded waiting:** a requester is not postponed forever.

P0 — Atomic operations and locks

Hardware instructions such as test-and-set or compare-and-swap allow an atomic state transition. A **spinlock** repeatedly checks the lock and wastes CPU while waiting, but can be reasonable for extremely short kernel critical sections or when sleeping is impossible. A blocking mutex lets the waiter sleep, which is better for longer waits but adds scheduler overhead.

Locks protect **invariants**, not merely lines of code. Define which shared data and relationship the lock protects. Keep a consistent lock order to reduce deadlock risk.

P1 — Peterson's two-process mutual exclusion

Peterson's algorithm is a software-only teaching solution for **exactly two participants**. Each participant announces interest through **flag**, then gives the other participant priority through **turn**:

```
/* Shared: atomic reads/writes, sequentially consistent memory model */
bool flag[2] = {false, false};
int turn;

void enter(int i) {
    int j = 1 - i;
    flag[i] = true;           /* I want to enter */
    turn = j;                 /* let the other go first if both want it */
    while (flag[j] && turn == j) {
        /* busy wait */
    }
}

void leave(int i) {
    flag[i] = false;
}
```

If only one process wants the critical section, the other flag is false and it enters. If both want it, the final value of **turn** makes at most one wait, providing mutual exclusion; the leaving process clears its flag, providing progress and bounded waiting under the model.

Do not present Peterson as a production lock. It assumes two participants, atomic shared-variable access, and a sufficiently strong memory-order model. Optimizing compilers and weakly ordered multiprocessors may reorder/cache ordinary accesses; real code uses language atomics with specified ordering or tested mutex primitives. It also busy-waits.

P0 — Semaphore

A semaphore is an integer synchronization object changed only through atomic operations:

- **wait/P/down** : decrement if possible; otherwise block;
- **signal/V/up** : increment and wake an eligible waiter.

A binary semaphore can provide mutual exclusion, but ownership semantics may differ from a mutex. A counting semaphore represents multiple identical resources or available items.

Producer-consumer with buffer capacity **N**:

```
semaphore empty = N, full = 0, mutex = 1
```

```
producer: wait(empty); wait(mutex); insert(); signal(mutex); signal(full)
consumer: wait(full); wait(mutex); remove(); signal(mutex); signal(empty)
```

The order matters. Holding the mutex while blocking on `empty / full` can deadlock.

P0 — Monitor and condition variable

A monitor packages shared state and procedures with implicit mutual exclusion. A condition variable lets a thread sleep until a predicate may have become true:

```
lock(m)
while (!condition())
    wait(cv, m)    // atomically releases m and sleeps; reacquires before return
use_or_modify_state()
unlock(m)
```

Use `while`, not usually `if`, because wakeups may be spurious or another thread may consume the condition before this one reacquires the lock. `signal` means “the condition may now be true,” not “transfer ownership of the lock immediately” in common Mesa-style semantics.

P1 — Classic synchronization problems

- **Readers-writers:** allow concurrent readers but exclusive writers; policies trade reader/writer starvation and fairness.
- **Dining philosophers:** naïve acquisition of left then right forks can deadlock. Impose resource ordering, allow at most `N-1` contenders, or use a waiter.
- **Sleeping barber:** coordinates bounded waiting-room capacity, customer availability, and barber sleep/wakeup.

P0 — IPC mechanisms

- **Pipe/FIFO:** byte stream, simple; ordinary pipe is often related-process and one-direction oriented.
- **Message queue/mailbox:** preserves message boundaries and decouples sender/receiver.
- **Shared memory:** fastest data path after setup because processes access common pages, but requires synchronization.
- **Socket:** local or network communication with a standard endpoint abstraction.
- **Signals/events:** lightweight notification, poor for transferring complex data.
- **RPC:** presents remote communication like a procedure call, but must not hide partial failure, latency, serialization, retries, or duplicate execution.

Excerpt source: `06_OPERATING_SYSTEMS_SLIDE_COMPLETE.md`, original lines 172–265. Original section numbers are retained.

5. Deadlock, starvation, and livelock

P0 — Necessary conditions for deadlock

All four Coffman conditions must hold:

1. **mutual exclusion** for some resource;

2. **hold and wait**;
3. **no preemption** of held resources;
4. **circular wait**.

Breaking any one prevents deadlock. A cycle in a resource-allocation graph is necessary; with one instance per resource type it is also sufficient. With multiple instances, a cycle alone may not prove deadlock.

P0 — Four handling strategies

1. **Ignore**: acceptable if rare and recovery/restart is cheap.
2. **Prevention**: structurally break a Coffman condition, e.g. total resource ordering breaks circular wait.
3. **Avoidance**: grant a request only if the resulting state remains safe; Banker's algorithm needs declared maximum demand.
4. **Detection and recovery**: allow deadlock, detect it, then preempt/rollback/terminate selected work.

A **safe state** has some completion ordering for all processes. Unsafe does not mean already deadlocked; it means future requests may force deadlock.

P0 — Banker's algorithm idea

Given **Available**, **Allocation**, and **Max**, compute **Need** = **Max** - **Allocation**. Repeatedly find an unfinished process whose **Need** ≤ **Work**; pretend it completes and returns its allocation. If all can finish, a safe sequence exists.

P0 — Distinctions

- **Deadlock**: a set waits in a cycle and none can progress.
- **Starvation**: a task is continually denied service while others progress.
- **Livelock**: participants keep reacting/changing state but do no useful work.
- **Priority inversion**: a high-priority task waits for a lock held by a low-priority task while medium-priority tasks run. Priority inheritance can temporarily boost the lock holder.

Excerpt source: 06_OPERATING_SYSTEMS_SLIDE_COMPLETE.md, original lines 266–298. Original section numbers are retained.

6. Address spaces and virtual memory

P0 — Why virtual memory?

Each process sees a private, contiguous-looking virtual address space even though pages may be scattered in RAM, shared, protected, or temporarily absent. Benefits include isolation, relocation, controlled sharing, sparse address spaces, and the ability to run working sets larger than available physical memory—with performance limits.

An address generated by the CPU is virtual; the MMU translates it. Protection bits enforce read/write/execute and user/kernel access.

P0 — Base-and-bound and segmentation

With dynamic relocation, **physical** = **base** + **virtual** if **virtual** < **bound**. It is simple but a single contiguous region limits growth and causes external fragmentation.

Segmentation represents logical regions such as code, heap, and stack. A virtual address includes segment number and offset; each segment has base, limit, and protection. Segments support logical sharing/protection but variable sizes cause external fragmentation.

Internal fragmentation wastes space inside an allocated unit; **external fragmentation** leaves enough total free space but split into unsuitable holes.

P0 — Paging and address translation

Paging divides virtual memory into fixed-size pages and physical memory into equal-size frames. For page size 2^p , split a virtual address into virtual page number and p -bit offset. The page-table entry maps VPN to PFN plus valid, protection, referenced/accessed, dirty, and other bits.

Example: a 32-bit virtual address with 4 KiB (2^{12}) pages has a 12-bit offset and 20-bit VPN. A flat table has 2^{20} entries; at 4 bytes each, it needs 4 MiB per address space, motivating multilevel/sparse page tables.

P0 — TLB

The Translation Lookaside Buffer caches recent virtual-to-physical translations. On a TLB hit, translation is fast; on a miss, hardware or software walks the page table and may insert the result. A TLB miss is **not necessarily a page fault**: the page may be in RAM but its translation is not cached.

Context switches may flush non-global entries or use address-space identifiers to distinguish processes. TLB reach is approximately $\text{number_of_entries} \times \text{page_size}$.

P1 — Page-table organizations

- **Multilevel**: allocates lower-level tables only for used virtual regions.
- **Hashed/inverted**: reduce per-process table size in very large address spaces, with different lookup complexity.
- **Huge pages**: increase TLB reach and reduce table overhead, but raise internal fragmentation and allocation/migration cost.

P0 — Demand paging and page-fault path

On access to a nonresident valid page:

1. hardware traps to the kernel;
2. kernel validates the address and permissions;
3. selects a free frame or victim;
4. writes a dirty victim if necessary;
5. reads/constructs the needed page;
6. updates page table/TLB state;
7. restarts the faulting instruction.

Page faults are extremely expensive relative to memory accesses, so locality and working-set control matter.

P0 — Page replacement

- **Optimal/MIN**: evict the page used farthest in the future; theoretical benchmark, not implementable online.
- **FIFO**: simple; can show **Belady's anomaly**, where more frames cause more faults.
- **LRU**: uses past recency as a locality predictor; exact LRU is costly, so systems approximate it.
- **Clock/second chance**: circularly scans reference bits, clearing recent pages before evicting an unreferenced one.

Stack algorithms such as true LRU and OPT do not exhibit Belady's anomaly because the pages resident with n frames are a subset of those with $n+1$ under the same reference sequence.

P0 — Thrashing

Thrashing occurs when active working sets exceed physical memory, so the system spends most time paging rather than executing. Symptoms include high fault rate and low useful CPU progress. Adding more processes can worsen it. Responses include reducing multiprogramming, allocating more frames, working-set/page-fault-frequency control, or adding RAM.

P1 — Copy-on-write and memory mapping

Copy-on-write shares pages until a write fault creates a private copy. Memory-mapped files map file contents into a process address space; ordinary loads/stores then access cached file pages. `mmap` can simplify random access and sharing, but durability still requires appropriate synchronization/flush semantics.

Excerpt source: 06_OPERATING_SYSTEMS_SLIDE_COMPLETE.md, original lines 299–363. Original section numbers are retained.

Workbook supplement — single-core threads, process states, and RR cost

Source: Operating System B3–B15 and BRAC Ques Bank C19/C47/C49.

Multiprogramming keeps multiple jobs available so one can use the CPU while another waits. Time-sharing adds frequent preemption for interactive response. Concurrency means overlapping progress; parallelism means simultaneous execution. Multiple runnable threads do not execute simultaneously on one core.

```
one core:  | UI thread 0..2 | worker 2..4 | UI 4..5 | worker 5..7 |
I/O:      |                | worker may block independently
```

Threads can still improve responsiveness, structure and I/O overlap. They do not intrinsically speed up pure CPU computation on one core; synchronization and switching can add overhead. A many-to-one user-thread runtime whose only kernel thread blocks in a blocking call can block all its user threads.

Process state transitions, with unambiguous arrow labels:

```
new --admit--> ready --dispatch--> running --exit--> terminated
                ^                               |
                |<----preempt-----|
                |                       |
                |                       | wait for event
                |                       v
                +--event done----- blocked
```

I/O completion normally moves blocked→ready, not directly to running. A syscall can switch user→kernel→user for the **same** process without a scheduling context switch. The PCB keeps PID/state, saved registers/PC, scheduling metadata, memory-map references, credentials and resource references; a thread also needs its own execution state/stack even when sharing process resources.

For quantum q and switch cost c , a rough saturated RR useful-CPU fraction is $q/(q + c)$ when a switch occurs after each full quantum. If $q = 2$ ms and $c = .1$ ms, it is about 95.2%; if $q = .1$ ms, it is 50%. Real I/O blocking and scheduling behavior change the estimate. Very large q approaches FCFS; very small q may improve response but can waste CPU and disrupt caches/TLBs. Temporal locality is reuse of the same location; spatial locality is reuse of nearby locations. These help explain why switching overhead is more than just the instructions that copy registers.

14. Core-subject self-test

You are ready only if you can do these without notes:

- ☐ Draw user-to-kernel transitions and distinguish syscall, trap, interrupt, and context switch.
- ☐ Draw process states and explain PCB, `fork`, `exec`, and copy-on-write.
- ☐ Solve FCFS, SJF/SRTF, RR, and priority scheduling tables accurately.
- ☐ Teach MLFQ, including gaming, allotment, boost, and starvation.
- ☐ Explain races and implement bounded producer-consumer with semaphores/conditions.
- ☐ Compare locks, semaphores, monitors, and IPC mechanisms.
- ☐ State all Coffman conditions and work a Banker safe-sequence example.
- ☐ Translate a virtual address through paging and explain TLB/page-fault behavior.
- ☐ Compare FIFO, LRU, Clock, and OPT and explain Belady's anomaly/thrashing.
- ☐ Explain interrupts, DMA, HDD/SSD trade-offs, and RAID 0/1/5/6/10.
- ☐ Explain inode/directory structure, hard/symbolic links, journaling, and LFS.
- ☐ Give a careful one-minute answer on multiprocessor scheduling or distributed partial failure.

Excerpt source: `06_OPERATING_SYSTEMS_SLIDE_COMPLETE.md`, original lines 544–608. Original section numbers are retained.

15.1 Scheduling metrics and one complete trace

For process `i`:

$$T_i = C_i - A_i, \quad W_i = T_i - B_i, \quad R_i = F_i - A_i,$$

where `A` is arrival, `B` total CPU burst, `F` first scheduled time, and `C` completion. The waiting formula assumes the simple model in which the listed burst is all CPU service; with repeated I/O bursts, waiting must be accumulated from time actually spent ready.

Example:

Process	Arrival	Burst
P1	0	5
P2	1	3
P3	2	1

FCFS:

0	5	8	9
P1	P2	P3	

- completion: `C1=5, C2=8, C3=9` ;
- turnaround: `5, 7, 7` ;
- waiting: `0, 4, 6` ;
- response equals waiting here because each process runs only once.

SRTF:

```

0  1  2  3    5      9
|P1|P2|P3| P2 |   P1  |

```

At each arrival/completion choose the smallest remaining time. Recalculate carefully: a preempted process’s response is still its **first** run delay, while waiting accumulates across ready intervals.

Round Robin with quantum **q** maintains a FIFO ready queue. New arrivals join according to the stated event convention; ambiguity at an exact quantum boundary can change the trace, so state the convention. Small **q** improves response but raises context-switch overhead; as **q**→∞, RR approaches FCFS.

For the same processes, Round Robin with **q=2** gives the following trace. Assume arrivals at an exact quantum boundary join before the expired process is requeued:

```

0      2      4  5      7  8  9
|  P1  |  P2  |P3|  P1  |P2 |P1 |

```

Process	First run	Completion	Turnaround C-A	Waiting T-B	Response first-A
P1	0	9	9	4	0
P2	2	8	7	4	1
P3	4	5	3	2	2

Trace the ready queue to justify it: after P1’s first slice the queue is **P2,P3,P1** ; after P2 it is **P3,P1,P2** ; after P3 it is **P1,P2** . Round Robin is fair by time slices, but the quantum and arrival-boundary convention affect the exact schedule.

Excerpt source: `06_OPERATING_SYSTEMS_SLIDE_COMPLETE.md` , original lines 611–670. Original section numbers are retained.

15.2 MLFQ as an explicit algorithm

An implementable policy follows five rules:

1. if priorities differ, run the higher-priority ready job;
2. among equal priorities, use round robin;
3. a new job starts at the highest queue;
4. after consuming its CPU allotment at a level—even across voluntary yields—it is demoted;
5. periodically boost all jobs to the top to prevent starvation and forget stale behavior.

Why it works: interactive/I/O-bound jobs often relinquish CPU early and remain responsive; CPU-bound jobs consume allotments and migrate down. Why naive MLFQ fails: a job may game the scheduler by yielding just before its quantum, or a changed workload may remain permanently low. Cumulative allotment and periodic priority boost address these failures.

Excerpt source: `06_OPERATING_SYSTEMS_SLIDE_COMPLETE.md` , original lines 671–682. Original section numbers are retained.

15.5 Race condition trace

`count++` is conceptually read–modify–write:

```
T1: read count=5
T2: read count=5
T1: write 6
T2: write 6
```

Two increments produced one. A critical-section solution aims for mutual exclusion, progress, and bounded waiting under its model. Disabling interrupts is not a general user-space/multiprocessor lock: another core still runs, and arbitrary user code must not control interrupts.

Excerpt source: 06_OPERATING_SYSTEMS_SLIDE_COMPLETE.md , original lines 711–723. Original section numbers are retained.

15.6 Semaphore bounded buffer

For `N` slots:

```
semaphore empty = N
semaphore full  = 0
semaphore mutex = 1

producer(item):
    wait(empty)
    wait(mutex)
    put(item)
    signal(mutex)
    signal(full)

consumer():
    wait(full)
    wait(mutex)
    item = get()
    signal(mutex)
    signal(empty)
    return item
```

Acquiring `mutex` before `empty/full` can deadlock: a producer may hold the mutex while waiting for space, preventing the consumer from acquiring mutex to create space. Counting semaphores represent resource counts; the binary `mutex` protects buffer state.

Excerpt source: 06_OPERATING_SYSTEMS_SLIDE_COMPLETE.md , original lines 724–750. Original section numbers are retained.

15.8 Deadlock detection and Banker safety

Coffman conditions: mutual exclusion, hold-and-wait, no preemption, circular wait. All are necessary for the classic model; breaking one prevents that form of deadlock.

Banker safety algorithm:

```

Work = Available
Finish[i] = false for every process
repeat:
    find unfinished i with Need[i] <= Work
    if none exists: stop
    Work += Allocation[i]
    Finish[i] = true
safe iff every Finish[i] is true

```

Need=Max-Allocation. A **safe state** has at least one completion order under declared maximum claims. An unsafe state is not necessarily already deadlocked; it lacks a guaranteed safe sequence. A resource request is tentatively allocated only if it does not exceed Need/Available and the resulting state remains safe.

Excerpt source: `06_OPERATING_SYSTEMS_SLIDE_COMPLETE.md`, original lines 764–782. Original section numbers are retained.

15.10 Paging numeric example

With 4 KiB pages, offset has $\log_2(4096)=12$ bits. For a 32-bit virtual address:

```

VPN = VA >> 12          (20 bits)
offset = VA & 0xFFF      (12 bits)
PTE = page_table[VPN]
if invalid: page fault
PA = (PFN << 12) | offset

```

Example `VA=0x12345`:

```

VPN = 0x12
offset = 0x345
if PTE maps VPN 0x12 to PFN 0xA7,
PA = 0xA7345

```

The TLB caches the mapping/permissions. On a TLB miss, hardware/software walks the page table; a valid PTE produces a TLB fill. A **page fault** means the translation requires OS handling (not-present, protection, copy-on-write, etc.), which is distinct from a mere TLB miss.

Two-level page-table bit walk

With a 32-bit virtual address, 4 KiB pages, and 4-byte entries in 4 KiB page-table pages, each table holds $4096/4 = 1024 = 2^{10}$ entries. A common two-level teaching split is therefore:

31	22	21	12	11	0
+-----+-----+-----+					
directory index: 10 b		table index: 10 b		offset: 12 b	
+-----+-----+-----+					

For `VA = 0x12345345`:

```

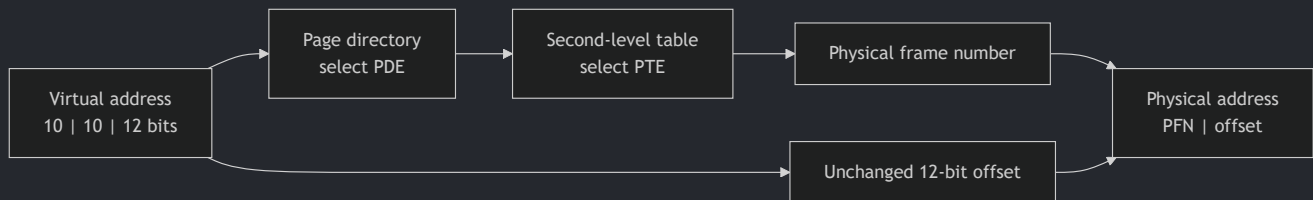
directory index = VA[31:22] = 0x048 = 72
table index     = VA[21:12] = 0x345 = 837
offset          = VA[11:0]  = 0x345

```

```

PDE = page_directory[72]      -> address of a second-level table
PTE = second_level_table[837] -> PFN plus valid/protection bits
PA  = (PFN << 12) | 0x345

```



The gain is sparsity: a process needs second-level tables only for virtual regions it actually uses. The cost of a TLB miss is a longer walk; more levels trade smaller sparse tables for more dependent memory references. If the PDE/PTE is invalid, distinguish an unmapped/protection fault from a valid but nonresident page that can be brought into memory.

Excerpt source: `06_OPERATING_SYSTEMS_SLIDE_COMPLETE.md`, original lines 794–851. Original section numbers are retained.

15.11 TLB effective access time

For a simplified single-level table where TLB lookup overlaps/negligibly costs and memory access is M , hit ratio h :

$$EAT = hM + (1 - h)(2M) = (2 - h)M.$$

If TLB lookup cost T is separate:

$$EAT = h(T + M) + (1 - h)(T + 2M),$$

excluding page faults. State the assumed page-table depth and lookup overlap; blindly memorizing $hM + (1 - h)2M$ is unsafe for multilevel walks.

Excerpt source: `06_OPERATING_SYSTEMS_SLIDE_COMPLETE.md`, original lines 852–867. Original section numbers are retained.

15.12 Page-fault service sequence

1. hardware detects invalid/not-present/protection condition and traps;
2. kernel validates the address/access;
3. illegal access terminates/signals the process;
4. otherwise locate the backing data or create zero/COW page;
5. select a free frame or victim;
6. write back a dirty victim if required;
7. read/prepare the page and block the faulting task during I/O;
8. update PTE/TLB state;
9. restart the faulting instruction.

This cost is orders of magnitude above a normal memory access, motivating locality and low page-fault rate.

Excerpt source: `06_OPERATING_SYSTEMS_SLIDE_COMPLETE.md`, original lines 868–881. Original section numbers are retained.

15.13 Replacement algorithms and Belady anomaly

- **OPT/MIN:** evict page used farthest in future; unattainable online, useful lower-bound benchmark.
- **FIFO:** evict oldest arrival; simple, may show Belady’s anomaly.
- **LRU:** evict least recently used; stack property avoids Belady anomaly, exact implementation expensive.
- **Clock/second chance:** circular hand checks reference bit; referenced pages get a second chance by clearing bit.

For a reference string, show the frame contents after **every** reference and mark hits/faults. Do not conflate number of distinct pages with number of frames. Working-set/clock-like policies approximate recency to avoid thrashing.

Worked comparison with three initially empty frames:

```
Reference: 7 0 1 2 0 3 0 4 2 3 0 3 2
FIFO:      F F F F H F F F F F F H H   -> 10 faults
LRU:      F F F F H F H F F F F H H   -> 9 faults
OPT:      F F F F H F H F H H F H H   -> 7 faults
```

The full frame trace below shows physical frame slots; slot order is not FIFO/LRU priority order:

Step/reference	FIFO	LRU	OPT
1 / 7	[7,-,-] F	[7,-,-] F	[7,-,-] F
2 / 0	[7,0,-] F	[7,0,-] F	[7,0,-] F
3 / 1	[7,0,1] F	[7,0,1] F	[7,0,1] F
4 / 2	[2,0,1] F	[2,0,1] F	[2,0,1] F
5 / 0	[2,0,1] H	[2,0,1] H	[2,0,1] H
6 / 3	[2,3,1] F	[2,0,3] F	[2,0,3] F
7 / 0	[2,3,0] F	[2,0,3] H	[2,0,3] H
8 / 4	[4,3,0] F	[4,0,3] F	[2,4,3] F
9 / 2	[4,2,0] F	[4,0,2] F	[2,4,3] H
10 / 3	[4,2,3] F	[4,3,2] F	[2,4,3] H
11 / 0	[0,2,3] F	[0,3,2] F	[2,0,3] F
12 / 3	[0,2,3] H	[0,3,2] H	[2,0,3] H
13 / 2	[0,2,3] H	[0,3,2] H	[2,0,3] H

For FIFO, after loading `7,0,1`, reference `2` evicts the oldest page `7`; a hit does not change FIFO arrival order. LRU updates recency on every hit, so the hit on `0` changes its next victim. OPT looks ahead and evicts the resident page whose next use is farthest away or nonexistent; it is the unattainable lower-bound benchmark. The

comparison is for this reference string only—LRU does not have a universal fixed fault advantage over FIFO, though unlike FIFO it is a stack algorithm and cannot show Belady's anomaly.

Excerpt source: 06_OPERATING_SYSTEMS_SLIDE_COMPLETE.md , original lines 882–919. Original section numbers are retained.

9. Artificial intelligence

Workbook coverage: AI B4–B34; DSA BFS/DFS completeness claims; BRAC rationality, search, minimax and alpha-beta; ML horizon effect.

1. Four views of AI

View	Goal	Typical basis	Limitation
thinking humanly	model how people actually think	cognitive science	thought is hard to observe; humans are not always rational
acting humanly	behavior indistinguishable from humans	Turing test	imitation is not understanding or optimality
thinking rationally	derive correct conclusions	logic/laws of thought	uncertainty and intractability remain
acting rationally	choose actions with best expected performance	rational-agent approach	needs performance measure/model/resource assumptions

The course chiefly uses the rational-agent view. Rational does not mean omniscient or always successful: it means maximizing expected performance from the percept history, prior knowledge, available actions, and computational resources. Bad luck does not prove irrationality.

To act humanly in a text Turing test, a system needs NLP, knowledge representation, reasoning, and learning. A total Turing test adds vision/perception and robotics. Passing establishes behavioral indistinguishability under a test, not consciousness.

Excerpt source: 09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md, original lines 25–37. Original section numbers are retained.

3. Agent, percept, action, rationality

An agent perceives through sensors and acts through actuators. A percept is current input; a percept sequence is the full history. An abstract agent function is

$$f: P^* \rightarrow A.$$

An agent program implements that function on an architecture. Rationality depends on performance measure, percept history, prior knowledge/model, and available actions. Autonomy means behavior increasingly reflects experience, not that an agent begins with zero built-in knowledge.

Excerpt source: 09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md, original lines 55–62. Original section numbers are retained.

4. PEAS

PEAS = **P**erformance measure, **E**nvironment, **A**ctuators, **S**ensors.

Autonomous taxi:

- P: safety, legality, time, comfort, fuel/cost;

- E: road, traffic, pedestrians, weather, passengers;
- A: steering, accelerator, brake, indicators, display/speech;
- S: camera/lidar/radar, GPS, speed/engine sensors, microphone.

Medical diagnosis:

- P: accuracy/outcome, low risk/cost, timely decision;
- E: patient, diseases, clinic/lab;
- A: questions, tests, diagnosis/treatment recommendation;
- S: symptoms, history, physical findings, results.

Do not list a sensor as a performance measure or confuse the environment with observations of it.

Excerpt source: 09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md , original lines 63–82. Original section numbers are retained.

5. Environment dimensions

Dimension	Contrast	Question
observability	fully vs partially observable	does the percept expose all relevant state?
determinism	deterministic vs stochastic	do state+action uniquely fix next state?
episode relation	episodic vs sequential	does today's action affect later decisions?
change	static vs dynamic/semidynamic	can the world/performance change while deliberating?
representation	discrete vs continuous	are state/action/time/percepts discrete?
participants	single vs multi-agent	do strategic others affect performance?
model	known vs unknown	are action outcomes initially known?

Crossword solving is mostly fully observable, deterministic, sequential, static, discrete, single-agent. Driving is partially observable, stochastic, sequential, dynamic, mostly continuous, multi-agent. A chess clock makes performance semidynamic even while the board waits.

Excerpt source: 09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md , original lines 83–96. Original section numbers are retained.

6. Agent architectures

1. table-driven: conceptual percept-history lookup, not scalable;
2. simple reflex: condition–action rules from current percept;
3. model-based reflex: internal state plus transition/sensor model;
4. goal-based: consider future states to reach a goal;
5. utility-based: rank trade-offs/uncertainty by expected utility;

6. learning agent: performance element acts, learning element improves, critic supplies feedback, problem generator encourages exploration.

A goal says acceptable/unacceptable; utility says how desirable outcomes are, enabling trade-offs.

Excerpt source: `09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md`, original lines 97–109. Original section numbers are retained.

7. Search-problem formulation

A classical problem needs initial state s_0 , actions $A(s)$, transition $Result(s, a)$, goal test, and step cost $c(s, a, s')$ whose sum is path cost $g(n)$. A **state** is a world configuration; a **node** is a record containing state, parent, action, depth, and cost. Different nodes may contain the same state.

Slide examples:

- Romania: state=current city, action=connected road, goal=Bucharest, cost=distance;
- 8-puzzle: tile arrangement and legal blank moves, usually unit cost;
- 8-queens: incremental placements or complete-state local search;
- water jugs: (x, y) amounts with fill/empty/pour actions;
- TSP: current city+visited set, goal=tour, cost=distance;
- assembly/VLSI: configurations and geometry/quality cost;
- classifier learning as optimization over parameter settings.

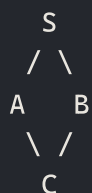
Abstraction discards irrelevant details while retaining solution validity. Too much detail is infeasible; too little yields plans invalid in reality.

Excerpt source: `09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md`, original lines 112–127. Original section numbers are retained.

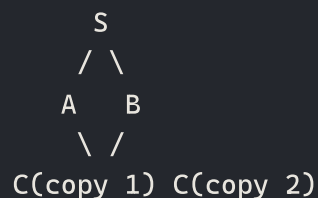
8. Tree search versus graph search

Tree search can regenerate states and loop. Graph search uses explored/frontier records, but duplicates are cost-sensitive: BFS with unit costs can mark on first discovery; UCS/A* must replace a state when a cheaper g is found; A* with an inconsistent heuristic may need to reopen an expanded state. Failure to detect repeats can turn a small state space into an exponential/infinite tree.

State graph:



Tree-search nodes:



The world has one state C. Tree search may create two node records for it. Graph search keeps a state-keyed best record, but may replace/reopen it when a cheaper path is discovered.

Excerpt source: `09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md`, original lines 128–145. Original section numbers are retained.

9. Generic best-first graph search

Invariant: `best_g[s]` is the cheapest discovered cost to `s`; stale queue entries are ignored and improvements reopen states.

```
from dataclasses import dataclass
from heapq import heappush, heappop
from itertools import count
from math import inf

@dataclass
class Node:
    state: object
    parent: "Node | None"
    action: object
    g: float
    depth: int

def best_first(start, is_goal, successors, priority):
    ticket = count()
    root = Node(start, None, None, 0, 0)
    frontier = [(priority(root), next(ticket), root)]
    best_g = {start: 0}
    while frontier:
        _, _, node = heappop(frontier)
        if node.g != best_g.get(node.state, inf):
            continue
        if is_goal(node.state):          # pop, not merely generate, for UCS/A*
            return node
        for action, nxt, cost in successors(node.state):
            new_g = node.g + cost
            if new_g < best_g.get(nxt, inf):
                child = Node(nxt, node, action, new_g, node.depth + 1)
                best_g[nxt] = new_g
                heappush(frontier, (priority(child), next(ticket), child))
    return None
```

Use priority g for UCS, h for greedy, and $g + h$ for A*. Goal testing on generation only proves some path reached the goal; popping the valid minimum establishes the cost-ordering claim.

Excerpt source: `09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md`, original lines 146–187. Original section numbers are retained.

10. Criteria and notation

b =branching factor, d =shallowest-goal depth, m =maximum depth, ℓ =depth limit, C^* =optimal cost, and $\epsilon > 0$ =minimum positive step cost. Judge completeness, optimality, time, and space; complexity can differ slightly by goal-test convention.

Excerpt source: `09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md`, original lines 190–193. Original section numbers are retained.

11. Breadth-first search

BFS expands the shallowest node with FIFO.

```
from collections import deque

def bfs(start, is_goal, neighbors):
    q, parent = deque([start]), {start: None}
    while q:
        u = q.popleft()
        if is_goal(u): return parent, u
        for v in neighbors(u):
            if v not in parent:
                parent[v] = u
                q.append(v)
    return parent, None
```

For finite b : complete; optimal only for equal/unit costs (or the corresponding monotone-by-depth condition); time and space $O(b^{d+1})$ in the slide convention. On $A: \{B, C\}$, $B: \{D, E\}$, $C: \{F, G\}$, the frontier is $[A]$, $[B, C]$, $[C, D, E]$, $[D, E, F, G]$.

Excerpt source: `09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md`, original lines 194–214. Original section numbers are retained.

12. Uniform-cost search

UCS expands minimum $g(n)$. It is complete if each step costs at least $\epsilon > 0$ and relevant branching is finite; optimal with nonnegative costs and correct duplicate/decrease handling; common bound $O(b^{1+\lceil C^*/\epsilon \rceil})$ time/space. It equals BFS when all costs are equal.

Counterexample to generation-goal testing: $S-G=10$, $S-A=1$, $A-G=2$. UCS pops A, improves G to 3, then pops G. Negative edges invalidate the settled-minimum argument; infinite zero-cost regions break the usual ϵ completeness proof.

Excerpt source: `09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md`, original lines 215–220. Original section numbers are retained.

13. Depth-first search

DFS uses recursion/LIFO and expands deepest first. Tree DFS is incomplete in infinite-depth/cyclic spaces, not optimal, takes $O(b^m)$ time and $O(bm)$ depth-first space. Its benefit is memory and sometimes fast discovery when solutions are dense.

Excerpt source: `09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md`, original lines 221–224. Original section numbers are retained.

14. Depth-limited and iterative deepening search

DLS does not expand below ℓ and distinguishes **cutoff** from definitive failure.

```
CUTOFF = object()

def dls(s, is_goal, neighbors, limit, path=frozenset()):
    if is_goal(s): return [s]
    if limit == 0: return CUTOFF
    cut = False
    for v in neighbors(s):
        if v in path: continue
        r = dls(v, is_goal, neighbors, limit - 1, path | {s})
        if r is CUTOFF: cut = True
        elif r is not None: return [s] + r
    return CUTOFF if cut else None
```

DLS costs $O(b^\ell)$ time and $O(b\ell)$ space, and is complete only with sufficient limit. IDS runs limits 0, 1, 2, ...:

$$N_{IDS} = (d+1)b^0 + db^1 + \dots + b^d = O(b^d).$$

For the slide's $b = 10, d = 5$, one DLS through depth 5 generates

$$1 + 10 + 10^2 + 10^3 + 10^4 + 10^5 = 111,111$$

nodes, while IDS generates

$$6 + 50 + 400 + 3,000 + 20,000 + 100,000 = 123,456.$$

The slide prints `123,450`; that is an arithmetic typo. Its printed BFS value `1,111,100` is also inconsistent with the geometric sum: BFS through depth 6 generates 1,111,111 nodes including the root, or 1,111,110 excluding it. IDS is complete for finite b , optimal for unit/increasing-by-depth cost, $O(b^d)$ time, and $O(bd)$ space.

Excerpt source: `09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md`, original lines 225–257. Original section numbers are retained.

15. Bidirectional search

Search from start and explicit goal until frontiers meet. Balanced BFS costs $O(b^{d/2})$ time and space. It requires computable predecessors/reverse actions, correct intersection checks, and manageable explicit goals. Multiple/implicit goals, directed/weighted transitions, and predecessors of checkmate complicate it; two naive BFSs do not solve a weighted problem optimally.

Excerpt source: `09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md`, original lines 258–261. Original section numbers are retained.

16. Comparison

Strategy	Selection	Complete?	Optimal?	Time	Space
BFS	shallowest	finite b	equal costs	$O(b^{d+1})$	$O(b^{d+1})$

Strategy	Selection	Complete?	Optimal?	Time	Space
UCS	minimum g	$c \geq \epsilon$	nonnegative costs	cost-contour bound	same order
DFS	deepest	no in general	no	$O(b^m)$	$O(bm)$
DLS	depth up to ℓ	if sufficient	no generally	$O(b^\ell)$	$O(b\ell)$
IDS	increasing limits	yes	equal costs	$O(b^d)$	$O(bd)$
bidirectional BFS	two frontiers	conditional	unit costs	$O(b^{d/2})$	$O(b^{d/2})$

Excerpt source: *09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md*, original lines 262–272. Original section numbers are retained.

17. Greedy and A*

A heuristic $h(n)$ estimates cheapest remaining goal cost. Greedy uses $f = h$, UCS $f = g$, and A* $f = g + h$. Greedy ignores money already spent, so it is not generally optimal or complete in unbounded spaces.

Romania trace

Greedy follows Arad→Sibiu ($h = 253$)→Făgăraş (176)→Bucharest, cost 450. A* orders:

Candidate	g	h	f
Arad	0	366	366
Sibiu	140	253	393
Rîmnicu via Sibiu	220	193	413
Făgăraş via Sibiu	239	176	415
Piteşti via Rîmnicu	317	100	417
Bucharest via Piteşti	418	0	418
Bucharest via Făgăraş	450	0	450

It returns Arad–Sibiu–Rîmnicu–Piteşti–Bucharest at 418.

Excerpt source: *09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md*, original lines 275–294. Original section numbers are retained.

18. Admissibility, consistency, reopening

Admissible means

$$0 \leq h(n) \leq h^*(n).$$

Consistent means for every edge

$$h(n) \leq c(n, a, n') + h(n').$$

Then $f(n') = g(n) + c + h(n') \geq g(n) + h(n) = f(n)$, so f is nondecreasing on a path. A* tree search is optimal with admissible h under standard conditions. Graph A* can permanently close a popped node with consistent h ;

with admissible but inconsistent h , reopen a state when a better g appears. Consistency normally implies admissibility when $h(goal) = 0$ and goals are reachable.

Example: edges $S-A:2, S-B:2, A-C:2, B-C:1, C-G:2$, with $h(A) = 3, h(B) = 3, h(C) = 0, h(G) = 0$. These are admissible, but $h(B) > 1 + h(C)$. A tie can expand A then C at $g = 4$; B later reveals C at $g = 3$. Refusing to reopen can retain goal cost 6 instead of optimum 5.

Dominance: if admissible $h_2(n) \geq h_1(n)$ everywhere, h_2 dominates h_1 and under comparable conditions expands no more nodes. A costly heuristic may still increase total runtime.

Excerpt source: `09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md`, original lines 295–310. Original section numbers are retained.

19. A* optimality proof and the reported $h = 0$ question

Suppose A* is about to pop suboptimal goal G_2 , so $f(G_2) = g(G_2) > C^*$. Some frontier node n remains on an optimal route. By admissibility,

$$f(n) = g(n) + h(n) \leq g(n) + h^*(n) = C^* < f(G_2),$$

a contradiction. For graph search add consistency/permanent closing or correct reopening. Completeness also needs finite branching/positive-cost assumptions so only finitely many nodes lie in the relevant f contour.

If $h(n) = 0$, then $f = g$: A* becomes UCS, equivalent to Dijkstra ordering for nonnegative graph edges. It remains optimal under UCS assumptions but loses heuristic direction. It is not universally “inefficient”; it simply cannot exploit heuristic information.

Excerpt source: `09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md`, original lines 311–320. Original section numbers are retained.

20. Building heuristics

For 8-puzzle, misplaced tiles h_1 and Manhattan distance h_2 are admissible relaxed-problem costs; Manhattan dominates misplaced tiles. Other methods: abstractions, pattern databases, and $\max(h_1, \dots, h_k)$ (admissible if components are). Arbitrary sums can overestimate unless costs are safely partitioned/additive.

Effective branching factor solves

$$N + 1 = 1 + b^* + (b^*)^2 + \dots + (b^*)^d.$$

Lower b^* means stronger guidance. A* remains exponential in worst-case time and especially memory.

Excerpt source: `09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md`, original lines 321–330. Original section numbers are retained.

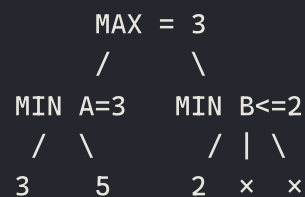
25. Game model and minimax

Main assumptions: two-player, alternating, deterministic, zero-sum, finite, perfect information. Define initial state/player, actions/result, terminal test, and terminal utility. MAX wants high utility; MIN low.

$$V(s) = \begin{cases} \text{Utility}(s), & \text{terminal,} \\ \max_a V(\text{Result}(s, a)), & \text{MAX,} \\ \min_a V(\text{Result}(s, a)), & \text{MIN.} \end{cases}$$


```
def minimax_decision(state, actions, result, terminal, utility):
    def value(s, maximizing):
        if terminal(s): return utility(s)
        vals = (value(result(s,a), not maximizing) for a in actions(s))
        return max(vals) if maximizing else min(vals)
    return max(actions(state), key=lambda a: value(result(state,a), False))
```

If root MIN children have leaves $A = [3, 12, 8]$, $B = [2, 4, 6]$, $C = [14, 5, 2]$, their values are 3, 2, 2 and MAX chooses A—not visible leaf 14. Full minimax is complete for a finite tree, optimal against optimal play, $O(b^m)$ time and depth-first $O(bm)$ space.



Search A first: root alpha becomes 3.

At B, the first leaf makes beta=2. Since beta <= alpha, the remaining B leaves cannot make MAX prefer B, so alpha-beta prunes them.

A minimax strategy guarantees the game value against every legal response. A weak opponent may allow better, but pure minimax does not explicitly model or maximally exploit their mistake pattern.

Excerpt source: `09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md`, original lines 462–492. Original section numbers are retained.

26. Reported infinite-depth question

1. Finite completely generated tree with correct utilities: minimax computes game-theoretic value. A human cannot force better than that value. If agent can force win it wins; if it can force draw, human cannot force win; if its starting position is theoretically lost, perfect human can beat it because no saving move exists.
2. Literal infinite generation cannot finish. A nonterminating/cyclic game needs draws, limits, discounting, or mathematical fixed-point/value conditions; ordinary terminal backup may be undefined.
3. Real cutoff depth, imperfect evaluation, time/memory, hidden information, stochasticity, or bugs can be exploited.

Short answer: exact minimax cannot make a game-theoretic mistake, but the outcome is bounded by the position's value; literal infinite search is not computable.

Excerpt source: `09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md`, original lines 493–500. Original section numbers are retained.

27. Alpha–beta pruning

α is MAX's best guaranteed ancestor value; β is MIN's best. Prune when $\alpha \geq \beta$.

```
from math import inf

def alpha_beta(s, maximizing, alpha, beta, actions, result, terminal, utility):
```

```

if terminal(s): return utility(s)
if maximizing:
    v = -inf
    for a in actions(s):
        v = max(v, alpha_beta(result(s,a), False, alpha, beta,
                               actions,result,terminal,utility))
        if v >= beta: return v
    alpha = max(alpha,v)
    return v
v = inf
for a in actions(s):
    v = min(v, alpha_beta(result(s,a), True, alpha, beta,
                           actions,result,terminal,utility))
    if v <= alpha: return v
    beta = min(beta,v)
return v

```

Trace: first MIN child $[3, 5]$ returns 3, so root $\alpha = 3$. Next MIN child sees 2, so $\beta = 2 \leq \alpha$ and its remaining leaves are irrelevant. Alpha-beta returns identical minimax choice. Worst $O(b^m)$; ideal ordering $O(b^{m/2})$, roughly double depth. Ordering changes work, not correctness.

Excerpt source: `09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md`, original lines 501–528. Original section numbers are retained.

28. Cutoff, evaluation, horizon, chance

Practical search replaces terminal utility at a cutoff with

$$Eval(s) = w_1 f_1(s) + \dots + w_k f_k(s).$$

Non-quiescence means cutoff in an unstable tactic; quiescence search extends volatile positions. Horizon effect postpones unavoidable harm beyond cutoff. Iterative deepening always retains the last completed move and improves ordering.

Chance node:

$$V(s) = \sum_i P(i \mid s) V(Result(s, i)).$$

Expectiminimax maximizes at MAX, minimizes at MIN, averages at chance. Multiplayer games can back up a utility vector and let player i maximize component i . Hidden-information/stochastic games do not satisfy ordinary minimax assumptions.

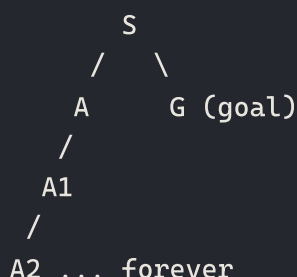
Excerpt source: `09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md`, original lines 529–544. Original section numbers are retained.

Workbook supplement — completeness, rationality, and game assumptions

Source: AI B4–B34 and DSA B15/B17 in the seniors' workbook. The spreadsheet is a list of rehearsal prompts, not an authoritative statement of these bounds.

A1. Completeness is not the same as termination on every possible input

In search, **complete** normally means guaranteed to find a solution when one exists, under stated assumptions. Terminating and reporting failure when none exists is an additional issue in an infinite search space. A decider in TOC has the stronger every-input halting requirement. Always specify the model.



Left-first DFS can follow the infinite branch and never visit G. BFS visits S, then A, then G with finite branching. But on a **finite graph with a visited set**, both BFS and DFS terminate after exploring reachable states and are complete. “DFS is never complete” is therefore wrong.

Claim	Conditions that must be said
BFS is complete	finite branching and a finite-depth reachable goal
DFS is complete	finite reachable graph plus duplicate/cycle handling
DLS is complete for a goal	chosen depth limit reaches some solution
IDS finds a shallowest goal	finite branching; increasing finite depth limits
BFS/IDS are cost-optimal	equal edge costs, or an appropriate cost-by-depth condition
UCS is complete in an infinite tree	finite branching and a positive lower bound on step costs

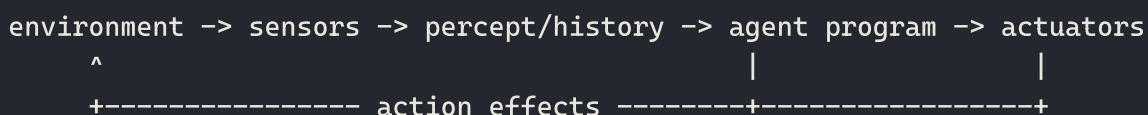
A limit of 2 cannot find a goal at depth 3. Infinite branching can prevent BFS from finishing one level. DFS is not inherently faster: it may find a deep leftmost goal quickly or waste all its work on the wrong branch. In a finite adjacency-list traversal, both BFS and DFS are $O(V + E)$: for $E = O(V)$ this is $O(V)$; for simple dense graphs $E = \Theta(V^2)$ it becomes $O(V^2)$.

Admissibility here is a property of a **heuristic**, $h(n) \leq h^*(n)$, not an unqualified synonym for algorithm optimality. An admissible heuristic must still be used by a correctly implemented search algorithm with the required duplicate/reopening rules (§18).

A2. Rational, intelligent, ideal and autonomous agent

An ideal rational agent selects an action maximizing **expected** performance given its percept sequence and available knowledge. It is not omniscient and cannot guarantee a good realized outcome in a stochastic world. Looking before crossing a road can be rational even if a rare unobservable event causes harm.

An autonomous agent uses its own experience to improve beyond designer-supplied knowledge. “Intelligent” is a broad description, not a disjoint formal category. “Discrete agent” should be clarified: usually the state, action, percept or time space is discrete. It is not a standard counterpart to “rational agent.”



For a vacuum world, PEAS could be cleanliness/energy/time; rooms/dirt; move/suck; location/dirt sensors. Whether it is fully observable or stochastic depends on the specified sensors and transition model, not the word "vacuum."

A3. Minimax extensions are not the same algorithm under new assumptions

Ordinary minimax assumes two-player zero-sum perfect-information deterministic play. Chance requires expectation nodes with known probabilities (expectiminimax). Three or more players may require utility-vector methods such as max-n or a different strategic model. A Nash equilibrium is a solution concept, not a universal replacement search algorithm for multiplayer games. Infinite-depth trees cannot be fully generated by a terminating program; finite solved games and mathematical infinite-horizon models are different claims. Use the worked minimax/alpha-beta trees in §§25–28 for the board.

Excerpt source: 09_ARTIFICIAL_INTELLIGENCE_SLIDE_COMPLETE.md, original lines 1200–1280. Original section numbers are retained.

10. Machine learning

Workbook coverage: Entire ML tab; classification/regression/ensemble screenshots; BRAC bias–variance, k-means, MLE/EM, XGBoost, AdaBoost, GD, LSTM, Transformers and LM/LLM.

1. What is machine learning?

Tom Mitchell’s operational definition is useful in a viva:

A computer program learns from **experience E**, with respect to a class of **tasks T** and **performance measure P**, if its performance at tasks in T, as measured by P, improves with experience E.

Example: spam filtering. **T** = classify email, **E** = labeled historical emails, **P** = F1/precision-recall or cost-weighted error. The definition forces us to say what improvement means.

AI, ML, and deep learning

- **Artificial intelligence** is the broad aim of making systems perform tasks associated with intelligent behavior.
- **Machine learning** learns behavior or predictive structure from experience/data instead of specifying every task rule.
- **Deep learning** is ML using multilayer representation-learning models. It is a subset of ML, not a synonym for AI.

Learning settings

Setting	Signal	Typical task	Example
Supervised	(x, y) labels	classification/regression	disease class, house price
Unsupervised	no task label	clustering/density/structure	customer segments
Semi-supervised	few labeled + many unlabeled	prediction	medical images with scarce labels
Self-supervised	targets constructed from the data	representation/pretraining	mask a token; predict next token
Reinforcement learning	reward after actions	sequential decision	game playing, control

Generative vs discriminative: a generative model learns a joint/data mechanism such as $p(x, y) = p(y)p(x|y)$ or $p(x)$ and can often sample; a discriminative model directly learns $p(y|x)$ or a decision boundary. Naive Bayes is generative; logistic regression is discriminative.

Hypothesis space and inductive bias

A hypothesis space **H** is the set of functions the learner can choose from. Learning chooses $h \in H$ that minimizes an empirical objective, often with regularization:

$$\hat{\theta} = \operatorname{argmin}_{\theta} \left[\left(\frac{1}{n} \right) \sum_i L(y_i, f_{\theta}(x_i)) + \lambda \Omega(\theta) \right].$$

An **inductive bias** is an assumption that lets the learner generalize beyond observed examples. A linear model assumes a linear boundary in its feature space; a CNN assumes locality and shared patterns; regularization

prefers certain parameter values. “Bias” here is a broad preference and must not be confused automatically with the statistical bias in the bias–variance decomposition.

Parametric and non-parametric models

- **Parametric:** fixed-form, finite parameter vector independent of training-set size, e.g. linear/logistic regression.
- **Non-parametric:** effective complexity may grow with data, e.g. k-NN, an unconstrained decision tree. It does not mean “has no parameters.”

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 30–68. Original section numbers are retained.

2. End-to-end ML workflow

1. Define the prediction unit, population, target, inference-time information, metric, and error costs.
2. Collect and audit representative data; document how labels were produced.
3. Split by the **independence unit** before data-dependent preprocessing.
4. Fit preprocessing only on training data/fold.
5. Establish a simple baseline.
6. Train candidate models on train data; choose hyperparameters on validation data.
7. Freeze all decisions; evaluate once on the locked test set.
8. Perform error analysis, robustness/fairness checks, deployment monitoring, and retraining policy.

Train, validation, and test

- **Training set:** estimates parameters.
- **Validation set:** chooses model, hyperparameters, threshold, features, and stopping time.
- **Test set:** estimates generalization after all choices are frozen.

If you repeatedly choose based on test results, the test set has become a validation set. With limited data, use **k**-fold cross-validation: divide into **k** folds, train on **k–1**, validate on the remaining fold, rotate, and average. For unbiased outer performance estimation while tuning, use **nested CV**. Stratification preserves class proportions; group/time-series splits preserve the real independence/deployment structure.

Data leakage

Leakage means training or selection uses information that would be unavailable at real inference or improperly crosses data partitions. Examples:

- normalizing or imputing using the full dataset;
- using a post-outcome/future feature;
- oversampling before splitting;
- the same patient, user, recording, near-duplicate image, or graph template in train and test;
- target encoding without out-of-fold construction;
- choosing checkpoints repeatedly from test performance.

Fit preprocessing in a training-only pipeline, split by group/time, deduplicate, and lock the final test set.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 69–100. Original section numbers are retained.

3. Preprocessing and feature engineering

Missing data

- Remove rows/columns only when justified; deletion can bias the sample.
- Numeric imputation: mean (sensitive to outliers), median (robust), model-based imputation.
- Categorical imputation: mode or an explicit "missing" category.
- Add a missingness indicator when absence itself may carry signal.
- Fit imputation statistics on training data only.

Missing completely at random, missing at random conditional on observed variables, and not missing at random lead to different statistical risks; imputation is not proof that bias vanished.

Scaling

- Min-max: $x' = (x - x_{\min}) / (x_{\max} - x_{\min})$ maps training range to $[0, 1]$.
- Standardization: $z = (x - \mu) / \sigma$ gives training mean zero and standard deviation one.
- Robust scaling uses median and interquartile range.

Scale-sensitive models include k-NN, k-means, SVM, PCA, and gradient-based linear/neural models. Ordinary trees generally do not need scaling because order-preserving transforms do not change split order.

Categorical encoding

- **Label/ordinal encoding:** integers; appropriate only when order is real or the model safely treats values categorically.
- **One-hot:** one indicator per category; avoids false numeric order but can be high-dimensional.
- **Target encoding:** category statistics derived from target; powerful but leakage-prone, so compute out-of-fold with smoothing.

Feature engineering and EDA

Inspect distributions, duplicates, outliers, correlations, class imbalance, label consistency, subgroup coverage, and train/deployment mismatch. Transformations should be justified by domain and invariance, not applied mechanically.

Excerpt source: 11_MACHINE_LEARNING_SLIDE_COMPLETE.md, original lines 101–130. Original section numbers are retained.

4. Loss, risk, and model selection

The **empirical risk** is training loss:

$$\hat{R}_D(h) = (1/n) \sum_i L(y_i, h(x_i)) .$$

The **generalization risk** is population loss:

$$R(h) = E_{(X, Y) \sim P} [L(Y, h(X))] .$$

The generalization gap is $R(h) - \hat{R}_D(h)$ (estimated using unseen data). Sources of error discussed in the slides include noise, sampling variance, model unrealizability/misspecification, and optimization/computational limitations.

Regularized empirical risk minimization uses $\hat{R}_D(h) + \lambda \Omega(h)$. Hyperparameter/model selection is different from parameter optimization: gradient descent may optimize weights for one model, while validation chooses depth, λ , learning rate, or architecture.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 131–144. Original section numbers are retained.

5. Classification metrics

For positive class `1` :

	Predicted +	Predicted -
Actual +	TP	FN
Actual -	FP	TN

$\text{accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$

$\text{precision} = \text{TP} / (\text{TP} + \text{FP})$ — of predicted positives, how many were correct?

$\text{recall/sensitivity/TPR} = \text{TP} / (\text{TP} + \text{FN})$ — of actual positives, how many did we find?

$\text{specificity/TNR} = \text{TN} / (\text{TN} + \text{FP})$

$\text{FPR} = \text{FP} / (\text{FP} + \text{TN}) = 1 - \text{specificity}$

$\text{F1} = 2\text{PR} / (\text{P} + \text{R}) = 2\text{TP} / (2\text{TP} + \text{FP} + \text{FN})$

$\text{balanced accuracy} = (\text{sensitivity} + \text{specificity}) / 2$

Matthews correlation coefficient:

$\text{MCC} = (\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN}) / \sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}$.

MCC uses all four cells and remains informative under imbalance; it ranges from `-1` (opposite), through `0` (chance-like), to `1` (perfect), when denominators are defined.

Worked example: TP=40, FP=10, FN=20, TN=130. Accuracy= $170/200 = .85$; precision= $40/50 = .80$; recall= $40/60 = .667$; F1= $80/(80+10+20) = .727$; specificity= $130/140 = .929$. The example shows why accuracy alone hides missed positives.

Multiclass averaging

- **Macro:** compute each class metric then average equally; highlights minority classes.
- **Micro:** pool all class decisions first; weights individual examples.
- **Weighted macro:** class metric weighted by support; can hide minority weakness.

ROC, PR, threshold, and calibration

ROC plots TPR against FPR over thresholds; ROC-AUC measures ranking. PR plots precision against recall; PR-AUC is usually more revealing when positives are rare. Neither chooses the deployment threshold nor proves probabilities are calibrated.

A calibrated model's predictions near `0.8` should be correct about 80% of the time. Assess with reliability diagrams, Brier score, or log loss. Select thresholds from validation data using costs/constraints, not automatically `0.5` .

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 145–187. Original section numbers are retained.

6. Regression metrics

- $MSE = (1/n) \sum (y - \hat{y})^2$: penalizes large errors; differentiable.
- $RMSE = \sqrt{MSE}$: same units as target.
- $MAE = (1/n) \sum |y - \hat{y}|$: more robust to outliers.
- $R^2 = 1 - \sum (y - \hat{y})^2 / \sum (y - \bar{y})^2$: improvement over predicting the mean. Test R^2 can be negative.

Squared loss makes the population-optimal prediction $E[Y|X=x]$; absolute loss makes a conditional median optimal. A metric must reflect the real cost of errors.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md` , original lines 188–198. Original section numbers are retained.

7. Bias–variance: the precise answer

The viva answer

Bias and variance are statistical components of expected predictive error. Underfitting and overfitting describe fitting/generalization behavior. High bias often manifests as underfitting, and high variance often manifests as overfitting, but the pairs are not definitions or exact synonyms.

What is random, and which expectation is taken?

Assume squared-error regression:

$Y = f(x) + \varepsilon$, where $f(x) = E[Y|X=x]$, $E[\varepsilon|X=x] = 0$, and $\text{Var}(\varepsilon|X=x) = \sigma^2(x)$.

Draw a training dataset $D \sim P^n$; the learning algorithm produces h_D . At one fixed test input x , define:

- Mean learned prediction: $\bar{h}(x) = E_D[h_D(x)]$.
- Bias: $\text{Bias}(x) = E_D[h_D(x)] - f(x) = \bar{h}(x) - f(x)$.
- Squared bias: $\text{Bias}^2(x) = (\bar{h}(x) - f(x))^2$.
- Variance: $\text{Var}(x) = E_D[(h_D(x) - \bar{h}(x))^2]$.
- Irreducible noise: $\sigma^2(x) = \text{Var}(Y|X=x) = E_{\varepsilon}(Y - f(x))^2$.

The expectation for bias and variance is over repeated training datasets. It may also include algorithm randomness such as initialization, shuffling, augmentation, and minibatch sampling. The target Y is a fresh outcome at the fixed x ; its randomness is separate. To get overall risk, integrate the final expression over new inputs $X \sim P_X$.

Decomposition and derivation

For a fresh target Y independent of training set D given fixed x :

$$E_D E_{\varepsilon}(Y|x) [(Y - h_D(x))^2]$$

Add and subtract $f(x)$ and $\bar{h}(x)$:

$$Y - h_D = (Y - f) + (f - \bar{h}) + (\bar{h} - h_D) .$$

Square and take expectations. The cross terms are zero because:

- $E_{\varepsilon}(Y|x)[Y - f] = 0$;

- $E_D[\bar{h}-h_D]=0$;
- fresh label noise and training-set randomness are independent under the setup.

Therefore:

$$E_D E_{(Y|x)}[(Y-h_D(x))^2] = \sigma^2(x) + \text{Bias}^2(x) + \text{Var}(x) .$$

And global expected test MSE is:

$$E_X[\sigma^2(X)+\text{Bias}^2(X)+\text{Var}(X)] .$$

This simple additive form is for squared-error regression. Bias–variance ideas extend more broadly, but do **not** quote this identical algebra as the exact decomposition for 0–1 classification error or every loss.

Numerical example

At a fixed x , suppose the true conditional mean is $f(x)=10$, label-noise variance is 4 , and models trained on five independent datasets predict $[7,9,8,8,8]$.

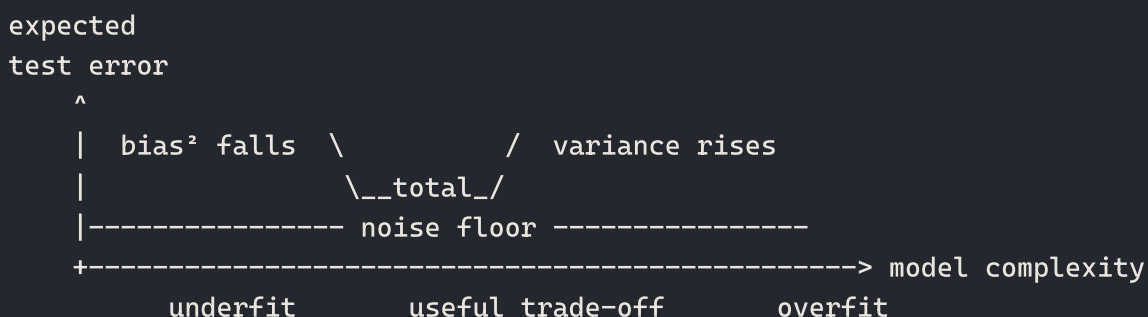
- $\bar{h}=(7+9+8+8+8)/5=8$.
- $\text{Bias}=8-10=-2$; $\text{bias}^2=4$.
- $\text{Variance}= [(7-8)^2+(9-8)^2+0+0+0]/5=0.4$.
- Expected prediction MSE= $\text{noise}+\text{bias}^2+\text{variance}=4+4+0.4=8.4$.

Bias is signed; the error contribution is squared bias. Variance is not “how noisy the labels are”—that is the irreducible-noise term.

Why underfitting/overfitting are different

- **Underfitting:** the fitted model fails to capture useful structure, commonly producing high training and validation error. Excessive statistical bias, insufficient features/capacity, too much regularization, or failed optimization can all produce this behavior.
- **Overfitting:** the model fits training idiosyncrasies/noise so training performance is much better than unseen performance. High sampling variance is a common cause, but leakage, split shift, and validation over-tuning can create a similar observed gap.

The classical picture is a memory aid, not the definition:



Bias and variance in the equation require repeated training sets; the plotted train/validation behavior is only a practical diagnostic. The curves are schematic, not a law of every modern model—optimization effects and double descent can change their shape.

Thus a train/validation plot is evidence, not a mathematical measurement of the decomposition. Bias and variance ideally concern repetition over many possible training sets, which we rarely observe directly.

Diagnostic patterns and remedies

Train behavior	Validation behavior	Common diagnosis	Check first	Typical remedies
poor	poor, small gap	underfitting/high-bias-like	optimization, features, labels	richer features/model, less regularization, train properly
excellent	much poorer, large gap	overfitting/high-variance-like	leakage/split mismatch	more independent data, regularization, augmentation, simpler model, early stopping, bagging
good	good	appropriate fit	subgroup/shift/calibration	preserve; validate robustness
poor	erratic/poor	not automatically "bias"	bad learning rate, bugs, noisy labels	debug optimization/data first

Validation loss can be lower than training loss when training uses dropout, augmentation, label smoothing, or an explicit regularization term. Deep networks may exhibit **double descent**, so validation error need not always be the textbook single U-shaped curve.

Complexity, regularization, and data

- Increasing capacity often reduces bias but may increase variance.
- Stronger regularization often increases bias but reduces variance.
- More representative independent data mainly reduces estimation variance; it cannot fix a fundamentally wrong target/feature space by itself.
- Bagging reduces variance by averaging unstable learners. If M model errors each have variance s^2 and pairwise correlation ρ , average variance is approximately $s^2 [\rho + (1-\rho)/M]$. Diversity matters; perfectly correlated models do not gain from averaging.
- Irreducible noise is irreducible only relative to the current features and label process. Better sensors/features/labels can change it.

Real-world example

Suppose we predict house price from floor area alone. A straight line misses locality, age, and nonlinear effects: stable across samples but systematically wrong—high-bias-like underfitting. A depth-unlimited tree on 100 houses memorizes rare details: nearly zero train error but changes sharply when a few houses are replaced—high-variance-like overfitting. A regularized ensemble with more representative houses can balance both; unpredictable seller urgency remains noise unless new information captures it.

Follow-up traps

- "Is bias an error?" — It is signed systematic deviation of the *mean learned predictor*; squared bias contributes to expected MSE.
- "Can a model have low bias and low variance?" — Yes, with appropriate representation, data, and signal; noise can still keep total error nonzero.
- "Does more data reduce bias?" — Usually it directly reduces variance; it may indirectly help if training/model selection changes, but it does not enlarge a misspecified hypothesis class.

- “Does high training error prove high bias?” — No. It could be optimization failure, corrupted labels, or an unsuitable metric.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 201–311. Original section numbers are retained.

8. Regularization

Lasso regression uses an L1 penalty: $\Omega(w) = \|w\|_1 = \sum |w_j|$. It encourages exact zeros/sparsity but is nondifferentiable at zero; a subgradient, coordinate-descent, or proximal method handles it. With strongly correlated features, it may select one and suppress another unstably.

Ridge regression uses an L2 penalty: $\Omega(w) = \|w\|_2^2 = \sum w_j^2$. It smoothly shrinks weights but ordinarily does not make them exactly zero. For objective $L + \lambda \|w\|^2$, the gradient adds $2\lambda w$ (or λw if the penalty is $\lambda \|w\|^2/2$). Correlated features can share distributed L2 weights.

Elastic Net combines them, for example

$$L(w) + \lambda [\alpha \|w\|_1 + (1-\alpha) \|w\|_2^2/2], \text{ with } 0 \leq \alpha \leq 1.$$

At $\alpha=1$ this parameterization is Lasso; at $\alpha=0$ it is Ridge. Intermediate values retain sparsity while often stabilizing groups of correlated features. Libraries use differing λ/α conventions, so state the implemented objective. In linear regression, the intercept is commonly left unpenalized after centering/scaling.

Other regularizers include limited tree depth/pruning, dropout, early stopping, data augmentation, label smoothing, architectural constraints/weight sharing, and ensembling. λ is selected using validation, never the test set.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 312–327. Original section numbers are retained.

9. Linear regression

Model and objective

Univariate: $\hat{y} = w_0 + w_1 x$. Multivariate with intercept column: $\hat{y} = Xw$.

Ordinary least squares minimizes

$$J(w) = (1/2n) \|Xw - y\|^2.$$

Gradient:

$$\nabla J(w) = (1/n) X^T (Xw - y).$$

Closed-form derivation

Set the gradient to zero:

$$X^T (Xw - y) = 0 \rightarrow X^T Xw = X^T y \rightarrow \hat{w} = (X^T X)^{-1} X^T y, \text{ if } X^T X \text{ is invertible.}$$

In practice use QR/SVD or pseudoinverse instead of explicitly forming an inverse; they are numerically safer and handle rank deficiency. With ridge regression:

$$\hat{w} = (X^T X + \lambda I)^{-1} X^T y \text{ (usually do not penalize the intercept).}$$

For one feature:

$w_1 = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / \Sigma(x_i - \bar{x})^2 = \text{Cov}(x, y) / \text{Var}(x)$, and $w_0 = \bar{y} - w_1 \bar{x}$.

Gradient descent

```
import numpy as np

def linear_regression_gd(X, y, lr=0.05, epochs=2000):
    X = np.c_[np.ones(len(X)), X]      # intercept
    w = np.zeros(X.shape[1])
    for _ in range(epochs):
        error = X @ w - y
        grad = X.T @ error / len(X)
        w -= lr * grad
    return w
```

Complexity per full-batch step is $O(nd)$ for $n \times d$ data; storing X is $O(nd)$. Solving normal equations naïvely is roughly $O(nd^2 + d^3)$ and can be unsuitable for high dimension.

Assumptions and traps

For prediction, a linear conditional-mean approximation may be useful without perfect classical assumptions. For unbiased/valid classical coefficient inference, discuss linear specification, independent zero-mean errors conditional on features, no perfect multicollinearity, and often homoscedasticity/normality for particular standard-error tests. Correlation or a fitted coefficient does not prove causation.

Excerpt source: 11_MACHINE_LEARNING_SLIDE_COMPLETE.md, original lines 330–378. Original section numbers are retained.

10. Perceptron

Binary labels $y \in \{-1, +1\}$; score $s = w^T x + b$; predict $\text{sign}(s)$. For a misclassified example ($y(w^T x + b) \leq 0$):

$w \leftarrow w + \eta y x$, $b \leftarrow b + \eta y$.

```
def perceptron(X, y, lr=1.0, epochs=20):
    w = [0.0] * len(X[0]); b = 0.0
    for _ in range(epochs):
        for xi, yi in zip(X, y):
            if yi * (sum(a*z for a, z in zip(w, xi)) + b) <= 0:
                w = [a + lr*yi*z for a, z in zip(w, xi)]
                b += lr*yi
    return w, b
```

Perceptron convergence is guaranteed in finitely many mistakes if data are linearly separable with a positive margin under the theorem's assumptions. It does not output calibrated probabilities and may cycle on nonseparable data.

Excerpt source: 11_MACHINE_LEARNING_SLIDE_COMPLETE.md, original lines 379–397. Original section numbers are retained.

11. Logistic regression

$z = w^T x + b$, $\sigma(z) = 1/(1 + e^{-z})$, $p(y=1|x) = \sigma(z)$.

Taking odds gives $p/(1-p) = e^z$; therefore log-odds $\log[p/(1-p)] = w^T x + b$. The decision boundary at threshold 0.5 is $w^T x + b = 0$.

Bernoulli likelihood is $\prod_i p_i^{y_i} (1-p_i)^{1-y_i}$. Negative log-likelihood/binary cross-entropy:

$J = -\sum_i [y_i \log p_i + (1-y_i) \log(1-p_i)]$.

Because $d\sigma/dz = \sigma(1-\sigma)$, the derivative with respect to the logit simplifies to $p-y$; hence:

$\nabla_w J = X^T(p-y)$ (divide by n if using mean loss).

```
import numpy as np

def sigmoid(z):
    z = np.clip(z, -40, 40)
    return 1 / (1 + np.exp(-z))

def logistic_regression_gd(X, y, lr=0.1, epochs=2000, l2=0.0):
    X = np.c_[np.ones(len(X)), X]
    w = np.zeros(X.shape[1])
    for _ in range(epochs):
        p = sigmoid(X @ w)
        reg = np.r_[0.0, w[1:]] # no intercept penalty
        w -= lr * ((X.T @ (p-y))/len(X) + l2*reg)
    return w
```

Despite its name, it is a classification model. Its boundary is linear in the supplied features but nonlinear feature maps can create nonlinear boundaries. Choose the probability threshold based on validation costs; 0.5 is not sacred.

Perceptron vs logistic regression

Perceptron	Logistic regression
mistake-driven update	smooth likelihood/log-loss optimization
hard class score	probability model
separability convergence theorem	convex objective under standard setup
no calibrated probability	probabilities can be calibrated but must be checked

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 398–439. Original section numbers are retained.

12. Multiclass softmax and cross-entropy

For logits z_k ,

$p_k = \exp(z_k) / \sum_j \exp(z_j)$.

For numerical stability compute $\exp(z_k - \max(z))$. With one-hot target y , categorical cross-entropy is $L = -\sum_k y_k \log p_k = -\log p_{\text{true}}$. The crucial derivative is:

$$\partial L / \partial z_k = p_k - y_k.$$

Softmax is invariant to adding the same constant to every logit. For mutually exclusive classes use one softmax; for independent multilabel targets use one sigmoid+BCE per label.

Excerpt source: 11_MACHINE_LEARNING_SLIDE_COMPLETE.md, original lines 440–451. Original section numbers are retained.

15. Decision trees

A tree recursively chooses a feature/test that most reduces impurity. It produces interpretable axis-aligned regions, captures nonlinear interactions, and needs little scaling; deep trees are unstable/high variance.

Entropy and information gain

For class proportions p_k :

$$H(S) = -\sum_k p_k \log_2 p_k.$$

For Boolean proportion q , $B(q) = -q \log_2 q - (1-q) \log_2 (1-q)$, with $0 \log 0 = 0$. Entropy is 0 for a pure node and 1 for a 50/50 binary node.

For split A with children S_v :

$$\text{Remainder}(A) = \sum_v |S_v| / |S| \cdot H(S_v)$$

$$\text{Gain}(S, A) = H(S) - \text{Remainder}(A).$$

Worked split: parent has 6 positive, 4 negative: $H \approx 0.971$. A split gives left $(4+, 0-)$ of size 4 and right $(2+, 4-)$ of size 6. Weighted child entropy = $0.4 \cdot 0 + 0.6 \cdot H(1/3, 2/3) \approx 0.6 \cdot 0.918 = 0.551$; gain = 0.420 bits.

Gini impurity is $1 - \sum p_k^2$; CART commonly chooses binary splits using Gini for classification and squared-error reduction for regression.

Continuous feature

Sort unique feature values; candidate thresholds lie between adjacent values (often only where labels change). Evaluate impurity decrease and choose the best. A straightforward scan after sorting is $O(n \log n)$ per feature; cumulative counts make threshold evaluation linear after sorting.

Stopping and pruning

Stop by max depth, min samples, min impurity decrease, or purity. **Pre-pruning** stops early. **Post-pruning** first grows then removes branches whose validation/cost-complexity benefit is insufficient. Pruning trades a little training fit for lower variance/generalization.

```
from math import log2

def entropy(labels):
    n = len(labels)
    counts = {c: labels.count(c) for c in set(labels)}
    return -sum((m/n)*log2(m/n) for m in counts.values())
```

```
def information_gain(parent, children):
    n = len(parent)
    return entropy(parent) - sum(len(c)/n * entropy(c) for c in children if c)
```

ID3/C4.5/CART recall

- ID3: categorical splits, information gain.
- C4.5: gain ratio, continuous/missing handling, pruning.
- CART: binary tree; Gini classification or squared-error regression; cost-complexity pruning.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 470–518. Original section numbers are retained.

16. Bagging and random forest

Bagging: draw bootstrap samples, train learners independently, average/vote. Bootstrap means sampling n points with replacement; roughly $1 - e^{-1} \approx 63.2\%$ unique examples appear in one sample. Out-of-bag examples can estimate error.

Random forest: bagged decision trees plus a random subset of features considered at each split. Feature randomness decorrelates trees, making averaging reduce variance more effectively. Trees can train in parallel. Classification votes; regression averages.

Viva contrast: bagging mainly stabilizes an unstable learner and reduces variance; it does not sequentially correct earlier learners.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 519–526. Original section numbers are retained.

17. AdaBoost

For binary labels $y_i \in \{-1, +1\}$:

1. Initialize sample weights $w_i = 1/n$.
2. Train weak learner h_t using weights.
3. Weighted error $\epsilon_t = \sum_i w_i [h_t(x_i) \neq y_i]$.
4. Learner weight $\alpha_t = 1/2 \ln((1 - \epsilon_t)/\epsilon_t)$.
5. Update $w_i \leftarrow w_i \exp(-\alpha_t y_i h_t(x_i))$; normalize.
6. Predict $\text{sign}(\sum_t \alpha_t h_t(x))$.

Misclassified examples have $y_i h_t = -1$, so their weights multiply by $e^{-\alpha_t}$; correctly classified weights multiply by e^{α_t} . If $\epsilon_t = 0.5$, $\alpha_t = 0$ (no value); if error exceeds 0.5 , the learner is worse than random unless reversed. AdaBoost is sensitive to mislabeled/outlier points because their weights may grow.

```
AdaBoost(D, T):
    w[i] = 1/n
    for t = 1..T:
        fit h_t using w
        err = sum_i w[i] * I(h_t(x_i) != y_i)
        alpha[t] = 0.5 * ln((1-err)/err)
```



```

w[i] *= exp(-alpha[t] * y_i * h_t(x_i))
normalize w
return sign(sum_t alpha[t] h_t(x))

```

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 527–551. Original section numbers are retained.

18. Gradient boosting

Gradient boosting builds an additive model $F_m(x) = F_{(m-1)}(x) + \eta h_m(x)$ by fitting each new weak learner to the **negative gradient** of the loss with respect to current predictions.

Squared-error regression

Negative gradient is residual $r_i = y_i - F_{(m-1)}(x_i)$. Initialize with mean target. Fit a regression tree to residuals, then add its leaf outputs multiplied by learning rate.

Slide-style worked start: for targets whose mean is `71.2`, set every initial prediction to `71.2`. Residuals are $y_i - 71.2$. A small tree groups similar residuals; if a sample's leaf predicts `18.8` and $\eta = .1$, new prediction is $71.2 + .1(18.8) = 73.08$. Repeat on new residuals.

Binary log-loss classification

Initialize log-odds $F_0 = \log(p/(1-p))$, where p is positive fraction. Current probability is $p_i = \sigma(F(x_i))$. Pseudo-residual is $y_i - p_i$. Fit a tree; a Newton-style leaf value is approximately:

$$p_{\text{leaf}} = \frac{\sum_{(i \text{ in leaf})} (y_i - p_i)}{\sum_{(i \text{ in leaf})} p_i(1 - p_i)}.$$

Update logit $F \leftarrow F + \eta p$, then convert with sigmoid. Do not average probabilities directly during additive logit boosting.

```

# Conceptual squared-error gradient boosting
F = np.full(len(y), y.mean())
models = []
for _ in range(M):
    residual = y - F
    tree = SmallRegressionTree().fit(X, residual)
    F += learning_rate * tree.predict(X)
    models.append(tree)

```

Small learning rate usually needs more trees. Depth controls interactions. Early stopping and subsampling regularize. Unlike bagging, boosting is sequential and primarily reduces residual bias, though it can overfit/noise-chase.

Where XGBoost fits

XGBoost means *eXtreme Gradient Boosting*: a highly optimized, regularized gradient-boosted decision-tree system, not a separate family unrelated to gradient boosting. Its common formulation uses first- and second-order loss derivatives to score tree splits and leaf values, adds penalties on tree complexity/leaf weights, and supports shrinkage plus row/column subsampling, missing-value routing, and efficient parallelized split search. Trees are still added sequentially because each stage depends on current predictions.

Quick contrast:

Method	Main distinction
Random forest	bootstrap/feature-randomized trees trained largely independently; average/vote; strong variance reduction
AdaBoost	reweights examples according to mistakes and combines weighted weak learners
Gradient boosting	fits new learners to the loss's negative gradient/pseudo-residual
XGBoost	engineered, regularized second-order GBDT implementation/system

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 552–595. Original section numbers are retained.

19. Stacking

Train diverse base learners, then train a meta-learner on their predictions. To avoid leakage, the meta-learner must see **out-of-fold** base predictions for training examples. At inference, fit base models on all training data, obtain their predictions, and feed them to the meta-model. Simple voting/averaging has no learned meta-model.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 596–601. Original section numbers are retained.

24. Optimization algorithms

Let $\mathbf{g}_t = \nabla_{\boldsymbol{\theta}} L_t(\boldsymbol{\theta}_t)$.

Batch, stochastic, and mini-batch gradient descent

- Batch GD uses all examples per step: accurate/stable but costly.
- SGD uses one example: cheap, noisy, may escape shallow traps but has high variance.
- Mini-batch uses a vectorized subset: standard compromise and GPU-friendly.

Shuffle ordinary IID training data each epoch; do not violate temporal/group structure when it matters. Learning rate too high diverges/oscillates; too low is slow. Batch size changes gradient noise and effective optimization, so learning-rate retuning may be necessary.

Momentum and Nesterov

One convention:

$$\mathbf{v}_t = \beta \mathbf{v}_{(t-1)} + \mathbf{g}_t, \quad \boldsymbol{\theta}_{(t+1)} = \boldsymbol{\theta}_t - \eta \mathbf{v}_t.$$

Momentum smooths noisy gradients and accelerates persistent directions. Nesterov computes the gradient after a look-ahead step, approximately $\mathbf{g}(\boldsymbol{\theta}_t - \eta \beta \mathbf{v}_{(t-1)})$, allowing earlier correction.

AdaGrad

$$\mathbf{r}_t = \mathbf{r}_{(t-1)} + \mathbf{g}_t \odot \mathbf{g}_t$$

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \mathbf{g}_t / (\sqrt{\mathbf{r}_t} + \epsilon).$$

Rarely updated features get larger effective steps, useful for sparse data; the accumulator only grows, so learning may stop too early.

RMSProp and AdaDelta

RMSProp uses an exponential moving average:

$$r_t = \rho r_{t-1} + (1-\rho) g_t^2, \quad \theta \leftarrow \theta - \eta g_t / (\sqrt{r_t} + \epsilon).$$

AdaDelta also tracks a moving average of squared updates, reducing dependence on a manually chosen global learning-rate scale.

Adam

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2$$

Bias correction: $\hat{m}_t = m_t / (1-\beta_1^t)$, $\hat{v}_t = v_t / (1-\beta_2^t)$.

$$\theta \leftarrow \theta - \eta \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon).$$

Adam combines momentum-like first moment and RMSProp-like second moment. AdaMax uses the infinity norm; Nadam adds Nesterov-style momentum. AdamW decouples weight decay from Adam's adaptive gradient update and is usually the intended neural-network weight decay.

Practical training controls

- Learning-rate schedules: step, cosine decay, warmup, reduce-on-plateau.
- Early stopping: monitor validation metric, keep best checkpoint, use patience.
- Gradient clipping: clip norm/value to prevent explosion; it does not solve all vanishing-gradient causes.
- Curriculum learning orders easy-to-hard examples; it encodes a training assumption and is not always beneficial.
- Adding gradient noise can regularize/explore but needs tuning.

Distributed SGD ideas from the slides

- **Hogwild!:** workers asynchronously update shared parameters without locks; fast for sparse conflicts, but stale/racing updates.
- **Downpour SGD:** distributed workers compute/update parameter servers asynchronously; tolerates slow workers but gradients may be stale.
- **Synchronous data parallelism:** all-reduce/aggregate gradients then perform the same update; consistent but waits for stragglers.
- **Elastic Averaging SGD:** local workers explore while an elastic penalty pulls them toward a center parameter vector.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 744–806. Original section numbers are retained.

34. Maximum likelihood, MAP, and Bayesian learning

For parameters θ and data D :

- Likelihood: $p(D|\theta)$ viewed as a function of θ .
- MLE: $\theta_{ML} = \operatorname{argmax}_{\theta} p(D|\theta) = \operatorname{argmax}_{\theta} \log p(D|\theta)$.
- Posterior: $p(\theta|D) \propto p(D|\theta)p(\theta)$.

- MAP: $\theta_{\text{MAP}} = \text{argmax}[\log p(D|\theta) + \log p(\theta)]$.
- Full Bayesian prediction integrates uncertainty: $p(y^*|x^*, D) = \int p(y^*|x^*, \theta) p(\theta|D) d\theta$.

MLE is MAP under a uniform prior (where meaningful). A Gaussian prior on weights corresponds to an L2-style penalty; a Laplace prior corresponds to L1. MAP returns a point estimate, not the whole posterior.

Bernoulli MLE

For N coin flips, c heads:

$$L(\theta) = \theta^c (1-\theta)^{(N-c)}$$

$$\ell = c \log \theta + (N-c) \log(1-\theta); \text{ set derivative } c/\theta - (N-c)/(1-\theta) = 0, \text{ giving } \hat{\theta}_{\text{ML}} = c/N.$$

Gaussian MLE

For IID scalar data, MLE mean is $\hat{\mu} = (1/N) \sum x_i$; MLE variance is $\hat{\sigma}^2 = (1/N) \sum (x_i - \hat{\mu})^2$. The unbiased sample variance uses $1/(N-1)$ and serves a different objective.

Beta–Bernoulli conjugacy

Prior $\theta \sim \text{Beta}(a, b)$ has density proportional to $\theta^{a-1} (1-\theta)^{b-1}$. After c successes and $N-c$ failures:

$$\theta|D \sim \text{Beta}(a+c, b+N-c)$$

Posterior mean = $(a+c)/(a+b+N)$; MAP, when both posterior shape parameters exceed 1, is $(a+c-1)/(a+b+N-2)$. a, b act like prior pseudo-counts, but their interpretation should match the prior construction.

Excerpt source: 11_MACHINE_LEARNING_SLIDE_COMPLETE.md, original lines 1066–1097. Original section numbers are retained.

37. Gaussian mixture model

A K -component GMM density is

$$p(x) = \sum_{k=1}^K \pi_k N(x|\mu_k, \Sigma_k), \text{ with } \pi_k \geq 0, \sum \pi_k = 1.$$

It is soft/probabilistic clustering. A latent one-hot component z chooses a Gaussian. Covariance choice controls shape: spherical, diagonal, tied, or full. Full covariance is expressive but parameter-heavy and may become singular.

Excerpt source: 11_MACHINE_LEARNING_SLIDE_COMPLETE.md, original lines 1134–1141. Original section numbers are retained.

38. Expectation–maximization for GMM

Direct likelihood contains $\log \Sigma_k$, making component assignments coupled. EM alternates:

E-step: responsibilities

$$r_{ik} = P(z_i = k | x_i) = \pi_k N(x_i | \mu_k, \Sigma_k) / \sum_j \pi_j N(x_i | \mu_j, \Sigma_j).$$

M-step

$$N_k = \sum_i r_{ik}$$

$$\mu_k = (1/N_k) \sum_i r_{ik} x_i$$

$$\Sigma_k = (1/N_k) \sum_i r_{ik} (x_i - \mu_k)(x_i - \mu_k)^T$$

$\pi_k = N_k / N$.

```
# conceptual GMM-EM
for _ in range(max_iter):
    weighted = [pi[k] * gaussian_pdf(X, mu[k], cov[k]) for k in range(K)]
    R = np.stack(weighted, axis=1)
    R /= R.sum(axis=1, keepdims=True)          # E-step
    Nk = R.sum(axis=0)
    pi = Nk / len(X)
    mu = (R.T @ X) / Nk[:, None]
    for k in range(K):
        D = X - mu[k]
        cov[k] = (D.T * R[:, k]) @ D / Nk[k] + 1e-6*np.eye(X.shape[1])
```

EM alternates computing the expected complete-data log likelihood and maximizing it. Observed-data likelihood does not decrease, but EM can reach a local optimum/saddle and is initialization-sensitive. Regularize covariance, use multiple starts, and monitor log likelihood.

Slide coin-mixture EM calculation

Five length-10 sequences have head fractions $(0.4, 0.9, 0.8, 0.3, 0.7)$. A hidden variable says whether coin A or B generated each sequence; unknown parameters are head biases θ_A, θ_B . Begin $\theta_A = .60$, $\theta_B = .82$.

For the fifth sequence (7 heads, 3 tails), likelihoods (the common binomial coefficient cancels during normalization) are:

- A: $.60^7 \cdot .40^3 \approx .00179$;
- B: $.82^7 \cdot .18^3 \approx .00145$.

Normalize to responsibilities about $(0.55, 0.45)$. Across all five sequences, the slide E-step obtains A-responsibilities $[.97, .12, .29, .99, .55]$ and B-responsibilities $[.03, .88, .71, .01, .45]$. M-step uses expected heads divided by expected flips:

$$\theta_A = (.97 \cdot .4 + .12 \cdot .9 + .29 \cdot .8 + .99 \cdot .3 + .55 \cdot .7) / (.97 + .12 + .29 + .99 + .55) \approx .483$$

$$\theta_B = (.03 \cdot .4 + .88 \cdot .9 + .71 \cdot .8 + .01 \cdot .3 + .45 \cdot .7) / (.03 + .88 + .71 + .01 + .45) \approx .813$$

Here fractions can be used because every sequence has the same 10 flips; the factor 10 cancels. Iterate E/M until stable.

General EM pattern

When latent variables Z make complete data easier:

1. E-step: $Q(\theta | \theta_{old}) = E(Z | X, \theta_{old}) [\log p(X, Z | \theta)]$.
2. M-step: $\theta_{new} = \arg\max_{\theta} Q(\theta | \theta_{old})$.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 1142–1199. Original section numbers are retained.

39. K-means and Lloyd's algorithm

Objective:

$$J = \sum_i ||x_i - \mu_{(c_i)}||^2.$$

Algorithm:

1. initialize K centroids;
2. assign each point to nearest centroid;
3. replace each centroid by mean of assigned points;
4. repeat until assignments/objective stabilize.

Why the mean? For one cluster, differentiating $\sum_i ||x_i - \mu||^2$ with respect to μ gives $2\sum(\mu - x_i) = 0$, so μ is the sample mean.

Why Lloyd converges: assignment chooses the best centroid for fixed means, so objective cannot increase; mean update minimizes squared error for fixed assignments, so it cannot increase. There are finitely many assignments, so it reaches a fixed point, but only a local optimum.

```
import numpy as np

def kmeans(X, K, iters=100, seed=0):
    rng = np.random.default_rng(seed)
    C = X[rng.choice(len(X), K, replace=False)].copy()
    for _ in range(iters):
        labels = ((X[:, None, :] - C[None, :, :])**2).sum(2).argmin(1)
        newC = np.array([X[labels == k].mean(0) if np.any(labels == k)
                          else X[rng.integers(len(X))] for k in range(K)])
        if np.allclose(C, newC): break
        C = newC
    return C, labels
```

Each iteration costs $O(NKd)$; memory can be $O(NK)$ if all distances are stored or lower if streamed. K-means favors spherical, similarly scaled clusters; it is sensitive to scale, outliers, chosen K , empty clusters, and initialization.

k-means++

Choose the first centroid randomly. Choose each next point with probability proportional to squared distance from its nearest chosen centroid. It spreads initial centers and gives a known expected approximation guarantee; multiple restarts remain useful.

Soft k-means

Replace hard assignments with weights, e.g.

$$r_{ik} = \exp(-\beta ||x_i - \mu_k||^2) / \sum_j \exp(-\beta ||x_i - \mu_j||^2),$$

then $\mu_k = \sum_i r_{ik} x_i / \sum_i r_{ik}$. Large β approaches hard assignment. A spherical equal-covariance GMM has a closely related responsibility rule, but it is a probabilistic density model with mixture weights/variance.

The MDSR slide also motivates responsibility as normalized gravitational pull: for distance d_{ki} , $r_{ki} = (1/(d_{ki}^2 + \epsilon)) / \sum_l (1/(d_{li}^2 + \epsilon))$. This is an illustrative soft-assignment rule; the exponential/Gaussian rule comes from a different modeling choice. In either case the M-step is the responsibility-weighted center.

K-means vs GMM

K-means	GMM
hard cluster	posterior soft responsibility
squared-distance objective	likelihood objective
effectively spherical/equal scale	covariance models elliptical shapes
Lloyd updates	EM updates

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 1200–1256. Original section numbers are retained.

40. Dimensionality reduction recall

PCA

Center data matrix X . Covariance is $S = X^T X / (n-1)$. The first principal direction maximizes projected variance:

$v_1 = \operatorname{argmax}_{\|v\|=1} v^T S v$, so it is the top eigenvector. Further directions are orthogonal eigenvectors. Equivalently use SVD $X = U \Sigma V^T$; columns of V are directions and squared singular values determine explained variance.

PCA is linear, unsupervised, scale-sensitive, and high variance need not be label-relevant. Standardize when unit differences are arbitrary. It is used for compression, denoising, visualization, and decorrelation.

UMAP and HDBSCAN in the text-clustering slides

UMAP constructs a neighborhood graph and optimizes a low-dimensional embedding that preserves local structure approximately; it is nonlinear and stochastic, so global distances/visual clusters must not be overinterpreted. HDBSCAN finds density-stable clusters, can infer cluster count, and marks low-density points as noise; results depend on density parameters and embedding quality.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 1257–1272. Original section numbers are retained.

41. MDP definition

An MDP is (S, A, P, R, γ) :

- states $s \in S$;
- actions $a \in A$;
- transition $P(s' | s, a)$;
- reward, e.g. $R(s, a, s')$;
- discount $0 \leq \gamma < 1$ (or a finite episodic horizon).

The Markov property says the next-state distribution depends on the present state/action, not the full past, assuming the state representation is sufficient.

Return from time t :

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}.$$

A policy $\pi(a|s)$ chooses actions. γ trades immediate versus delayed reward and makes infinite sums/contractions well-behaved under bounded rewards.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 1275–1292. Original section numbers are retained.

42. Value functions and Bellman equations

$$V^{\pi}(s) = E_{\pi}[G_t | S_t = s] .$$

$$Q^{\pi}(s, a) = E_{\pi}[G_t | S_t = s, A_t = a] .$$

Policy Bellman expectation equation:

$$V^{\pi}(s) = \sum_a \pi(a|s) \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma V^{\pi}(s')] .$$

Optimality equations:

$$V^*(s) = \max_a \sum_{s'} P(s' | s, a) [R + \gamma V^*(s')] .$$

$$Q^*(s, a) = \sum_{s'} P(s' | s, a) [R + \gamma \max_{a'} Q^*(s', a')] .$$

$$\pi^*(s) = \operatorname{argmax}_a Q^*(s, a) .$$

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 1293–1310. Original section numbers are retained.

43. Dynamic programming: value and policy iteration

Value iteration

Initialize V . Repeatedly apply Bellman optimality backup:

$$V_{\text{new}}(s) = \max_a \sum_{s'} P(s' | s, a) [R + \gamma V(s')] .$$

Stop when changes are sufficiently small, then extract greedy policy. With finite discounted MDP, Bellman optimality operator is a contraction and converges to V^* .

Policy iteration

1. Policy evaluation: solve/iterate Bellman equations for current π .
2. Policy improvement: $\pi_{\text{new}}(s) = \operatorname{argmax}_a \sum_{s'} P[R + \gamma V^{\pi}(s')] .$
3. Repeat until unchanged.

Policy iteration makes fewer outer improvements but evaluation can be costly; modified policy iteration uses partial evaluation.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 1311–1328. Original section numbers are retained.

44. Passive learning, ADP, TD, and active learning

- **Passive RL:** fixed policy; learn its utility/model.
- **Direct utility estimation:** average observed returns for states; unbiased with enough samples but wastes Bellman structure and has high variance.
- **Adaptive dynamic programming (ADP):** estimate transition/reward model, then solve the induced MDP; sample-efficient but model errors matter.

- **Temporal difference (TD):** model-free bootstrapping:

$$V(s) \leftarrow V(s) + \alpha [R + \gamma V(s') - V(s)] .$$

The bracket is TD error. TD updates before an episode ends and combines sampling with bootstrapping.

Active RL must explore while exploiting. Greedy-only behavior can miss better actions. ϵ -greedy takes a random action with probability ϵ ; decay carefully so exploration does not disappear too early. Other approaches include optimistic initialization and upper-confidence/softmax action selection.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 1329–1341. Original section numbers are retained.

45. Q-learning

Off-policy TD control update:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [R + \gamma \max_{a'} Q(s', a') - Q(s, a)] .$$

```
from collections import defaultdict
import random

Q = defaultdict(lambda: [0.0] * n_actions)
for episode in range(num_episodes):
    s = env.reset()
    done = False
    while not done:
        a = random.randrange(n_actions) if random.random() < epsilon \
            else max(range(n_actions), key=lambda j: Q[s][j])
        s2, r, done = env.step(a)
        target = r if done else r + gamma * max(Q[s2])
        Q[s][a] += alpha * (target - Q[s][a])
        s = s2
```

Q-learning is **off-policy** because its target uses the greedy `max` action even when behavior explores. SARSA uses the actually chosen next action and is on-policy:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [R + \gamma Q(s', a') - Q(s, a)] .$$

Tabular Q-learning convergence requires finite states/actions, sufficient visitation, suitable decaying step sizes, and stationary Markov dynamics. With function approximation, bootstrapping, and off-policy data—the “deadly triad”—divergence is possible. DQN adds replay and target networks to improve stability but does not make all guarantees automatic.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 1342–1372. Original section numbers are retained.

46. Bag-of-words to contextual embeddings

Bag-of-words

Build a vocabulary, then represent a document by word counts (or TF-IDF). It is sparse, high-dimensional, and mostly ignores order/context: "dog bites man" and "man bites dog" have the same unigram counts. It remains a strong, interpretable baseline for many text tasks.

Traditional TF-IDF weight:

$$tfidf(t, d) = tf(t, d) \cdot \log(N/df(t)) \quad (\text{variants smooth/normalize}).$$

Static embeddings

Word2vec/GloVe map each vocabulary item to a dense vector. Similar distributional contexts create nearby vectors, enabling semantic similarity and analogical structure. A static embedding assigns one vector to "bank," so it cannot distinguish river bank from financial bank without contextual machinery.

Contextual embeddings

A language model transforms raw token embeddings using surrounding tokens; the final vector for "bank" changes with context. Token embeddings represent individual contextualized positions; a sentence/document embedding aggregates or learns a vector for the whole text. Pooling choices include [CLS], mean pooling with attention-mask handling, or a specifically trained sentence encoder.

Cosine similarity:

$$\cos(u, v) = u^T v / (||u|| ||v||).$$

It measures angle, not magnitude. Normalize when dot product should equal cosine.

Excerpt source: 11_MACHINE_LEARNING_SLIDE_COMPLETE.md, original lines 1375–1398. Original section numbers are retained.

51. Autoregressive language modeling

An autoregressive model factorizes a sequence:

$$p(x_1, \dots, x_T) = \prod_{t=1}^T p(x_t | x_{<t}).$$

During self-supervised next-token training, a text sequence supplies many shifted (prefix, next token) examples. Teacher forcing presents ground-truth previous tokens in parallel with a causal mask. At generation, append one sampled token and repeat; decoding is sequential even though training positions are processed in parallel.

Cross-entropy for a target token is $-\log p(\text{target} | \text{prefix})$. Perplexity is $\exp(\text{average token NLL})$; lower is better only when tokenization/data are comparable.

Excerpt source: 11_MACHINE_LEARNING_SLIDE_COMPLETE.md, original lines 1505–1514. Original section numbers are retained.

52. Transformer overview

Input token IDs index an embedding table. Add/encode position. A stack of Transformer blocks repeatedly applies attention and position-wise feedforward networks with residual connections and normalization. A language-model head maps final hidden states to vocabulary logits; weights may be tied to the input embedding table.

```

token IDs
|
token embeddings + positional information
|
+---- Transformer block repeated L times -----+
|  x -> LN -> masked/self-attention -> add residual  |
|      -> LN -> position-wise FFN/SwiGLU -> add residual  |
+-----+
|
final normalization -> task/LM head -> logits

```

Attention mixes information across permitted token positions.
The FFN transforms each position independently with shared weights.

The slide analogy: CNNs chop a signal into patches and process patches with shared operations. Transformers also process tokens identically/in parallel, but each token can depend on all other permitted tokens through attention.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 1515–1536. Original section numbers are retained.

53. Scaled dot-product self-attention

For input $X \in \mathbb{R}^{(n \times d_{\text{model}})}$:

$Q = XW_Q$, $K = XW_K$, $V = XW_V$, where $Q, K \in \mathbb{R}^{(n \times d_k)}$ and $V \in \mathbb{R}^{(n \times d_v)}$.

$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k} + M)V$.

M contains 0 for allowed positions and a very negative value for masked positions. Softmax is row-wise, so every query's attention weights sum to one.

Dictionary intuition

Query asks “what information am I looking for?”, keys describe what each token offers, dot products score compatibility, softmax turns scores into a soft lookup distribution, and values carry the retrieved information. The output for query i is $\sum_j a_{ij} v_j$.

Why divide by $\sqrt{d_k}$?

If query/key components have independent mean 0, variance 1, their dot product sums d_k products and has variance about d_k . Large logits make softmax saturate and gradients tiny. Scaling gives variance near one.

Causal, bidirectional, and cross-attention

- Encoder self-attention can see all non-padding tokens.
- Decoder causal self-attention masks future keys ($j > i$).
- Cross-attention uses queries from one sequence/modality and keys/values from another, e.g. decoder attends to encoder output or text attends to image features.

Dense attention constructs an $n \times n$ score matrix: time roughly $O(n^2 d + n d^2)$ and attention memory $O(n^2)$ per layer (implementation details change constants). This motivates sparse/linear attention and state-space alternatives.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 1537–1562. Original section numbers are retained.

54. Multi-head, multi-query, and grouped-query attention

For each head h , project independent Q_h, K_h, V_h , compute attention, concatenate, then output-project:

$MHA(X) = \text{Concat}(\text{head}_1, \dots, \text{head}_H)W_O$.

Heads can learn different relation/position subspaces. They are computed in parallel, though “each head learns a human-interpretable relation” is not guaranteed.

- **MHA**: every query head has its own K/V head; best flexibility, largest KV cache.
- **MQA**: many query heads share one K and one V head; smallest KV cache/faster decoding, possible quality loss.
- **GQA**: groups of query heads share K/V heads; compromise used in modern LLMs.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 1563–1574. Original section numbers are retained.

55. Positional information and RoPE

Attention without positions is permutation-equivariant: permuting tokens permutes outputs but supplies no sequence order. Solutions include learned absolute embeddings, sinusoidal embeddings, relative position biases, and rotary position embeddings (RoPE).

RoPE rotates pairs of query/key coordinates by an angle determined by token position/frequency. The dot product between rotated q_i and k_j depends on relative offset $i-j$; RoPE is applied to Q/K before attention scoring, not merely added once to raw token embeddings.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 1575–1580. Original section numbers are retained.

56. Transformer block details

A modern pre-normalized decoder block is approximately:

$u = x + \text{Attention}(\text{LN}(x))$

$y = u + \text{FFN}(\text{LN}(u))$.

FFN acts independently at every token with shared weights, commonly $W_2 \cdot \phi(W_1 x)$ or gated variants such as SwiGLU. Residual connections preserve an identity path; LayerNorm stabilizes feature scales. Original Transformer often used post-norm; pre-norm generally improves deep optimization.

Excerpt source: `11_MACHINE_LEARNING_SLIDE_COMPLETE.md`, original lines 1581–1590. Original section numbers are retained.

57. KV cache

At autoregressive step t , old tokens’ keys and values at each layer are unchanged. Cache them, compute only the new token’s Q/K/V, and attend its query to cached/new K/V. This avoids recomputing previous token states and does **not** change the intended mathematical output aside from floating-point/kernel details.

Why not QV or QKV cache?

- Past **queries** are not reused to generate the current token in ordinary decoding.
- Keys are required to compute current attention scores; QV without K is insufficient.
- QKV adds memory with negligible normal benefit because old Q is unused.

KV caching reduces repeated compute but cache memory grows roughly with layers × sequence length × KV heads × head dimension. It does not remove the current token's need to score against the context.

Excerpt source: 11_MACHINE_LEARNING_SLIDE_COMPLETE.md , original lines 1591–1602. Original section numbers are retained.

58. Encoder-only, decoder-only, and encoder–decoder models

Architecture	Visibility/objective	Strength	Example
encoder-only	bidirectional; masked-token/representation	classification, embeddings	BERT
decoder-only	causal next-token	open-ended generation	GPT/Llama
encoder–decoder	encoder bidirectional, decoder causal + cross-attention	conditioned sequence-to-sequence	original Transformer, T5

BERT masks some input tokens and predicts them, learning contextual representations. GPT removes encoder/cross-attention and predicts the next token. T5 expresses tasks as text-to-text instructions. “LLM” is a moving, informal size/capability label; both generative and representation models may be called language models.

RNN encoder–decoder to attention

An early seq2seq RNN compressed a whole source sequence into one context vector, creating a long-sequence bottleneck. Attention let the decoder form a weighted combination of all encoder states at every output step. The Transformer removed recurrence, making token training more parallel while retaining context-dependent interaction.

Excerpt source: 11_MACHINE_LEARNING_SLIDE_COMPLETE.md , original lines 1603–1616. Original section numbers are retained.

Workbook supplement — classical ML gaps and precise interview answers

Source: seniors' workbook ML B11–B19, B39–B43, A64, B75–B87; BRAC Ques Bank C11/C29/C41/C47; Random Screenshots images 2/9 and Runtime Analysis image10. Added September 2026; original slide sections are retained.

W1. Hypothesis, hypothesis space, consistency, and Occam's razor

A hypothesis h is one candidate predictor; H is the allowed family. A hypothesis is **consistent with a labeled training set** when it predicts every training label correctly. The version space is

$$VS_{H,D} = \{h \in H: \forall (x_i, y_i) \in D, h(x_i) = y_i\}.$$

Example: $h_t(x) = 1[x \geq t]$ with data $(1, 0), (3, 1)$. Exactly the thresholds $1 < t \leq 3$ are consistent: $t = 2$ works, $t = 4$ fails. Add $(2, 0)$: the version space narrows to $2 < t \leq 3$. Two identical inputs with contradictory labels leave no consistent deterministic hypothesis.

```

all allowed thresholds H
-> enforce (1,0),(3,1): 1 < t <= 3
-> enforce (2,0):      2 < t <= 3
-> contradictory label for x=2: empty version space

```

Consistency means zero training classification error, not guaranteed good test performance. It is also different from **statistical consistency** of an estimator as sample size tends to infinity. With noise, forcing interpolation may be undesirable; minimize an appropriate loss and validate instead.

Occam/Ockham's razor prefers a simpler explanation when evidence is otherwise comparable. Simplicity may mean a smaller tree, stronger regularization, shorter description, or a prior preference—not "always choose a straight line." Simpler models can underfit. A useful formal connection is regularized risk $\widehat{R}_D(h) + \lambda\Omega(h)$, or MAP with a simplicity-favoring prior. Always separate inductive bias, the intercept/bias parameter, statistical bias, and societal bias: these are four different uses of the same word.

W2. PCA versus regression, with an actual projection

Question	PCA	Ordinary least-squares regression
Inputs to fitting	feature matrix X	X and target y
Goal	preserve feature variance / minimize reconstruction error	minimize target prediction squared error
Output	orthogonal directions and transformed features	a function predicting y
Geometric error	orthogonal to retained subspace	vertical/residual error in the designated target
Is low variance unimportant?	may be discarded by PCA	may carry the entire predictive signal

For points $(1, 1), (2, 2), (3, 3)$, mean is $(2, 2)$ and centered rows are $(-1, -1), (0, 0), (1, 1)$. The sample covariance is

$$S = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \lambda_1 = 2, \quad v_1 = \frac{1}{\sqrt{2}}(1, 1)^T, \quad \lambda_2 = 0.$$

Projection scores are $(-\sqrt{2}, 0, \sqrt{2})$. Keeping one component reconstructs these points exactly because all lie on the same line.

```

x2
3 |          * (3,3)
2 |      * (2,2)    principal direction: diagonal
1 |  * (1,1)
  +----- x1

```

Counterexample to "PCA always prevents overfitting": if X_1 has variance 100 but is irrelevant, X_2 has variance .01 and $y=100 \cdot X_2$, keeping the largest variance direction can discard the target signal. PCA can reduce noise and computation, but its dimension must be validated for the downstream task. Fit centering/scaling/PCA **inside each training fold**, not on all data. [Official PCA/regression comparison](#).

```
import numpy as np

def pca_fit(X, components):
    X = np.asarray(X, dtype=float)
    if X.ndim != 2 or not 1 <= components <= min(X.shape):
        raise ValueError("invalid matrix or component count")
    mean = X.mean(axis=0)
    _, singular, Vt = np.linalg.svd(X - mean, full_matrices=False)
    directions = Vt[:components].T
    scores = (X - mean) @ directions
    return mean, directions, scores, singular

# new/test data: (X_new - training_mean) @ training_directions
```

Full dense SVD costs roughly $O(nd\min(n, d))$ for n rows, d features; truncated/randomized methods can reduce work when retaining few directions. Eigenvector sign is arbitrary: flipping both a direction and its scores does not change reconstruction. PCA components are orthogonal in feature space; uncorrelated component scores are not automatically statistically independent.

Curse of dimensionality: a grid with ten positions along each of d features has 10^d cells. Maintaining the same coverage becomes exponentially more data-hungry; distances/neighborhoods can become less discriminative. Irrelevant dimensions, scaling, and intrinsic structure matter. This is not a theorem that every high-dimensional model overfits.

W3. Clustering metrics — do not use raw class-label accuracy

Internal metric, silhouette: for point i , let a_i be mean distance to other points in its cluster, and b_i the smallest mean distance to any other cluster. Then

$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}.$$

Near +1 means good separation, near 0 a boundary, and negative means the point is closer on average to another cluster. Average over points; singleton clusters are conventionally assigned 0. At least two clusters and fewer clusters than samples are needed. If $a_i = 2, b_i = 6$, then $s_i = 2/3$. The value depends on the distance and feature scaling and can favor compact convex clusters. [Silhouette documentation](#).

External metric, adjusted Rand index (ARI): compare whether pairs are in the same/different clusters under predicted versus reference partitions, then correct agreement for chance. $ARI=1$ for identical partitions up to renaming; chance-like expected agreement is about 0, and worse agreement can be negative. For true labels $[0, 0, 1, 1]$ and predicted $[7, 7, 3, 3]$, $ARI=1$ even though literal label equality reports zero accuracy. Cluster IDs have no intrinsic identity. [ARI documentation](#).

Other recall: inertia is within-cluster squared distance and always decreases when more clusters are allowed, so it alone cannot choose k ; Davies–Bouldin prefers lower inter-cluster similarity, Calinski–Harabasz higher between/within dispersion, and NMI compares information shared by two partitions. Validate stability and domain usefulness as well as a scalar score.

W4. Metric screenshot corrections and a threshold simulation

Assume positive means “spam” or “disease present,” and specify which before counting. High precision means a high **fraction** of predictions called positive are correct; it does not alone prove a smaller absolute FP count across datasets of different sizes. Recall concerns the actual positive set.

Example	Scores, highest first	Threshold	TP	FP	FN	Precision	Recall
four cases, truth 1,0,1,0	.9, .8, .6, .2	.85	1	0	1	1	.5
same cases	same	.5	2	1	0	2/3	1

Lowering this threshold catches the missed positive but adds a false alarm. On a fixed ranked dataset, recall does not decrease as the threshold lowers; precision is not guaranteed to change monotonically at every individual step.

ROC-AUC=.5 is chance-level ranking, not “any low AUC.” AUC=0 is perfectly reversed ranking for a nondegenerate binary sample; reversing the scores gives AUC=1. High AUC does not ensure calibration. F1 ignores true negatives. R^2 can be negative and is undefined in its basic formula for a constant target; RMSE/MAE have units while R^2 does not. See the exact formulas in §§5–7 rather than memorizing a high/low-value screenshot.

W5. Cross-validation and stacking: what happens to the final model?

```
locked TEST (never used for tuning)
remaining development data:
  fold A validate | folds B,C train
  fold B validate | folds A,C train
  fold C validate | folds A,B train
  -> compare average validation scores for each setting
  -> choose setting -> refit on all development data -> test once
```

CV estimates performance and supports model selection; it does not mechanically prevent overfitting or make a repeatedly inspected test set clean. Choose group-aware/time-aware splits when appropriate. Nested CV separates outer evaluation from inner hyperparameter selection. In stacking, concatenate **out-of-fold** predictions to train the meta-model; in-sample base predictions leak training performance. Ensemble learning combines predictors and is not a fourth mutually exclusive category beside supervised/unsupervised/RL.

W6. Value-based, policy-based, actor–critic, and the horizon effect

Value-based control learns $Q(s, a)$ and derives actions (often an exploratory version of argmax). Policy-based learning directly changes $\pi_\theta(a | s)$. An actor–critic learns a policy (actor) and a value estimate (critic), often to reduce the variance of the actor’s learning signal.

For an undiscounted finite episode, a REINFORCE estimator is

$$\nabla_{\theta} J(\theta) = \mathbb{E} \left[\sum_t G_t \nabla_{\theta} \log \pi_{\theta}(A_t | S_t) \right].$$


```
repeat:
    sample a complete episode using current policy
    compute reward-to-go G[t] backward
    gradient = sum_t G[t] * grad log pi_theta(action[t] | state[t])
    theta += learning_rate * gradient
```

A two-action bandit uses $\pi(A) = \sigma(\theta)$, initially .5. If A is sampled and reward is 2, gradient is $2(1 - .5) = 1$. With step .1, theta becomes .1, raising $\pi(A)$ to about .525. This is a sampled update, not a guarantee every step improves true return. Baselines can reduce variance; exploration remains necessary. Discounted policy-gradient formulas require matching the return and state-weighting convention, not silently reusing an undiscounted equation.

“Horizon effect” normally refers to a cutoff in **game/search lookahead**, not to a decision tree’s sensitivity to one training record. Example: action A earns +1 now but an unavoidable -100 just beyond a one-step cutoff; action B returns 0. A shallow evaluation chooses A. Deeper/quiescence search can reveal the delayed loss (AI §28). Greedy decision-tree splitting also has limited lookahead, but instability from resampling is a variance issue. Name the setting instead of interchanging these meanings.

W7. LSTM — gates, memory path, and one numerical step

The workbook names LSTM explicitly (BRAC Ques Bank C41). A standard LSTM maintains hidden state h_t and cell state c_t . With $z_t = [h_{t-1}; x_t]$:

$$f_t = \sigma(W_f z_t + b_f), \quad i_t = \sigma(W_i z_t + b_i), \quad o_t = \sigma(W_o z_t + b_o), \quad \tilde{c}_t = \tanh(W_c z_t + b_c), \\ c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad h_t = o_t \odot \tanh(c_t).$$

```
old cell c[t-1] -- multiply forget f --+
                                     +-- add --> new cell c[t]
new candidate  -- multiply input i --+ |
                                     |
                                     tanh, then * output o
                                     |
                                     hidden state h[t]
```

For scalar $c_{t-1} = 2, f_t = .9, i_t = .2, \tilde{c}_t = .5, o_t = .8$: $c_t = 1.8 + .1 = 1.9$, and $h_t = .8 \tanh(1.9) \approx .765$. Forget controls retained memory, input controls writing, output controls what the current hidden state exposes. These are learned continuous gates, not literal if-statements. Gates have their own weight matrices and biases.

```
def lstm_cell(x, h, c, weights, biases): # educational NumPy cell
    z = np.concatenate([h, x])
    sigmoid = lambda v: 1 / (1 + np.exp(-np.clip(v, -60, 60)))
    f = sigmoid(weights['f'] @ z + biases['f'])
    i = sigmoid(weights['i'] @ z + biases['i'])
    o = sigmoid(weights['o'] @ z + biases['o'])
    candidate = np.tanh(weights['c'] @ z + biases['c'])
    c_new = f*c + i*candidate
    return o*np.tanh(c_new), c_new
```

The direct cell path has derivative f_t with respect to its previous cell value when the gate computations are held fixed; products near 1 can preserve gradients better than a plain RNN’s repeated nonlinear transformation. This does not make vanishing/exploding gradients impossible. Training uses BPTT; the recurrent dependency limits

parallelism across time. A standard cell with input width d and hidden width h has $4h(d + h + 1)$ parameters and roughly $O(h(d + h))$ work per step. Attention instead directly mixes accessible token representations, with its own sequence-length and memory costs.

W8. Classical ML follow-up ladder: MLE, EM, XGBoost, and language models

Use §§34–38 for the full likelihood derivation and EM responsibilities. On coin outcomes H,H,T, $L(p) = p^2(1 - p)$; maximizing $2\log p + \log(1 - p)$ gives $p = 2/3$. MLE estimates parameters, not “the probability that the parameter is true”; MAP includes a prior.

EM is for latent-variable likelihoods: compute expected latent assignments under the current parameters (E), then maximize the expected complete-data log likelihood (M). For mixture components equally likely a priori and likelihoods .8 and .2 for one observation, responsibilities are .8 and .2, not an unconditional hard assignment to the first component. Standard exact EM does not decrease observed-data likelihood but may converge to a local optimum; singular Gaussian mixtures require constraints/regularization.

For XGBoost with leaf gradient sum G , Hessian sum H , L2 coefficient λ , the quadratic leaf objective is $Gw + \frac{1}{2}(H + \lambda)w^2$. Thus $w^* = -G/(H + \lambda)$. If $G = -6, H = 3, \lambda = 1$, the leaf weight is 1.5; shrinkage .1 adds .15 to current scores. XGBoost is a regularized implementation of gradient boosting; its trees are added sequentially even when split search is parallelized. It is not guaranteed to beat every alternative on every dataset. [XGBoost’s model derivation](#).

A language model assigns probabilities to token sequences, commonly $P(w_1, \dots, w_T) = \prod_t P(w_t \mid w_{<t})$. An n-gram model is already a language model. An LLM is a large learned language model, generally neural, with no universal parameter threshold defining “large.” “Transformer” names an architecture, “LLM” a model category, and “chatbot” an application. Review the RNN/LSTM and Transformer details in the existing sections before a board comparison; do not define language model as “a small LLM.”

Small-scale versus large-scale learning is relative to sample count, feature dimension, parameter count, storage, compute and latency. The workbook’s thousands/millions cutoffs are informal examples, not universal definitions.

Excerpt source: 11_MACHINE_LEARNING_SLIDE_COMPLETE.md, original lines 2397–2689. Original section numbers are retained.